

APPENDIX A

8515982



THE UNITED STATES OF AMERICA

TO ALL TO WHOM THESE PRESENTS SHALL COME:

UNITED STATES DEPARTMENT OF COMMERCE

United States Patent and Trademark Office

August 14, 2024

**THIS IS TO CERTIFY THAT ANNEXED IS A TRUE COPY FROM THE
RECORDS OF THIS OFFICE OF THE FILE WRAPPER AND CONTENTS
OF:**

APPLICATION NUMBER: 16/267,752

FILING DATE: February 05, 2019

PATENT NUMBER: 10642359

ISSUE DATE: May 05, 2020



Certified by

Katherine Kelly Vidal

Performing the Functions and Duties of the
Under Secretary of Commerce
for Intellectual Property
and Director of the United States
Patent and Trademark Office

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UTILITY PATENT APPLICATION TRANSMITTAL <i>(Only for new nonprovisional applications under 37 CFR 1.53(b))</i>	Attorney Docket No.	012052.0799C1
	First Named Inventor	Chang Mok CHOI
	Title	WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF
	Priority Mail Express® Label No.	

APPLICATION ELEMENTS <i>See MPEP chapter 600 concerning utility patent application contents.</i>	Commissioner for Patents P.O. Box 1450 Alexandria, VA 22313-1450
1. ☒ Fee Transmittal Form (PTO/SB/17 or equivalent) 2. ☐ Applicant asserts small entity status. See 37 CFR 1.27 3. ☐ Applicant certifies micro entity status. See 37 CFR 1.29. Applicant must attach form PTO/SB/15A or B or equivalent. 4. ☒ Specification [Total Pages <u>26</u>] Both the claims and abstract must start on a new page. (See MPEP § 608.01(a) for information on the preferred arrangement) 5. ☒ Drawing(s) (35 U.S.C. 113) [Total Sheets <u>5</u>] 6. Inventor's Oath or Declaration [Total Pages <u>3</u>] (including substitute statements under 37 CFR 1.64 and assignments serving as an oath or declaration under 37 CFR 1.63(e)) a. ☐ Newly executed (original or copy) b. ☒ A copy from a prior application (37 CFR 1.63(d)) 7. ☒ Application Data Sheet * See note below. See 37 CFR 1.76 (PTO/AIA/14 or equivalent) 8. CD-ROM or CD-R in duplicate, large table, or Computer Program (Appendix) ☐ Landscape Table on CD 9. Nucleotide and/or Amino Acid Sequence Submission (if applicable, items a. – c. are required) a. ☐ Computer Readable Form (CRF) b. ☐ Specification Sequence Listing on: i. ☐ CD-ROM or CD-R (2 copies); or ii. ☐ Paper c. ☐ Statements verifying identity of above copies	ACCOMPANYING APPLICATION PAPERS 10. ☐ Assignment Papers (cover sheet & document(s)) Name of Assignee _____ 11. ☐ 37 CFR 3.73(c) Statement ☒ Power of Attorney (when there is an assignee) 12. ☐ English Translation Document (if applicable) 13. ☒ Information Disclosure Statement (PTO/SB/08 or PTO-1449) ☐ Copies of citations attached 14. ☐ Preliminary Amendment 15. ☐ Return Receipt Postcard (MPEP § 503) (Should be specifically itemized) 16. ☐ Certified Copy of Priority Document(s) (if foreign priority is claimed) 17. ☐ Nonpublication Request Under 35 U.S.C. 122(b)(2)(B)(i). Applicant must attach form PTO/SB/35 or equivalent. 18. ☐ Other: _____ _____ _____ _____ _____

***Note:** (1) Benefit claims under 37 CFR 1.78 and foreign priority claims under 1.55 **must** be included in an Application Data Sheet (ADS).
(2) For applications filed under 35 U.S.C. 111, the application must contain an ADS specifying the applicant if the applicant is an assignee, person to whom the inventor is under an obligation to assign, or person who otherwise shows sufficient proprietary interest in the matter. See 37 CFR 1.46(b).

19. CORRESPONDENCE ADDRESS					
☒ The address associated with Customer Number: <u>89980</u> OR ☐ Correspondence address below					
Name					
Address					
City	State	Zip Code			
Country	Telephone	Email			
Signature	/Thomas J. Matikainen/		Date	February 5, 2019	
Name (Print/Type)	Thomas J. Matikainen		Registration No. (Attorney/Agent)	73,291	

This collection of information is required by 37 CFR 1.53(b). The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	012052.0799C1
		Application Number	
Title of Invention	WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐	Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
--------------------------	---

Inventor Information:

Inventor	1				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Chang Mok		CHOI		
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service					
City	Suwon-si		Country of Residence ⁱ	KR	
Mailing Address of Inventor:					
Address 1		Samsung Advanced Institute of Technology			
Address 2		130 Samsung-ro, Youngtong-gu, Gyeonggi-do			
City	Suwon-si		State/Province		
Postal Code		16678	Country ⁱ	KR	
Inventor	2				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Sang Joon		KIM		
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service					
City	Suwon-si		Country of Residence ⁱ	KR	
Mailing Address of Inventor:					
Address 1		Samsung Advanced Institute of Technology			
Address 2		130 Samsung-ro, Youngtong-gu, Gyeonggi-do			
City	Suwon-si		State/Province		
Postal Code		16678	Country ⁱ	KR	
Inventor	3				Remove
Legal Name					

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	012052.0799C1
		Application Number	
Title of Invention	WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF		

Prefix	Given Name	Middle Name	Family Name	Suffix
	Seung Keun		YOON	
Residence Information (Select One) US Residency ☒ Non US Residency Active US Military Service				
City	Suwon-si	Country of Residence ⁱ	KR	

Mailing Address of Inventor:				
Address 1	Samsung Advanced Institute of Technology			
Address 2	130 Samsung-ro, Youngtong-gu, Gyeonggi-do			
City	Suwon-si	State/Province		
Postal Code	16678	Country ⁱ	KR	
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button. Add				

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).			
☐ An Address is being provided for the correspondence information of this application.			
Customer Number	89980		
Email Address	pto@nsiplaw.com	Add Email	Remove Email

Application Information:

Title of the Invention	WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF		
Attorney Docket Number	012052.0799C1	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Total Number of Drawing Sheets (if any)	5	Suggested Figure for Publication (if any)	

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country ⁱ

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	012052.0799C1
		Application Number	
Title of Invention	WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF		

Publication Information:

☐	Request Early Publication (Fee required at time of Request 37 CFR 1.219)
☐	Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application has not and will not be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.			
Please Select One:	☒ Customer Number	US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	89980		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78. When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
	Continuation of	14913085	2016-02-19
Prior Application Status	Expired	Remove	
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
14913085	a 371 of international	PCT/KR2014/004073	2014-05-08
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button.			Add

Foreign Priority Information:

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	012052.0799C1
		Application Number	
Title of Invention	WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF		

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX)ⁱ the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ⁱ (if applicable)	Remove
10-2013-0098476	KR	2013-08-20	EAAE	

Additional Foreign Priority Data may be generated within this form by selecting the **Add** button.

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013. NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.
--

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	012052.0799C1
		Application Number	
Title of Invention	WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF		

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	012052.0799C1
		Application Number	
Title of Invention	WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF		

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Applicant	1	Remove
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section. Clear		
☒ Assignee	Legal Representative under 35 U.S.C. 117	Joint Inventor
Person to whom the inventor is obligated to assign.		Person who shows sufficient proprietary interest
If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:		
Name of the Deceased or Legally Incapacitated Inventor:		
If the Applicant is an Organization check here. ☒		
Organization Name	Samsung Electronics Co., Ltd.	
Mailing Address Information For Applicant:		
Address 1	129, Samsung-ro, Yeongtong-gu, Gyeonggi-do	
Address 2		
City	Suwon-si	State/Province
Country	KR	Postal Code
Phone Number		Fax Number
Email Address		
Additional Applicant Data may be generated within this form by selecting the Add button. Add		

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	012052.0799C1
		Application Number	
Title of Invention	WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF		

Assignee	1		
Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.			
Remove			
If the Assignee or Non-Applicant Assignee is an Organization check here. ☒			
Organization Name	Samsung Electronics Co., Ltd.		
Mailing Address Information For Assignee including Non-Applicant Assignee:			
Address 1	129, Samsung-ro, Yeongtong-gu, Gyeonggi-do		
Address 2			
City	Suwon-si	State/Province	
Country i	KR	Postal Code	16677
Phone Number		Fax Number	
Email Address			
Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button. Add			

Signature:[Remove](#)

NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). However, if this Application Data Sheet is submitted with the **INITIAL** filing of the application and either box A or B is **not** checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c).

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Signature	/Thomas J. Matikainen/		Date (YYYY-MM-DD)	2019-02-04	
First Name	Thomas	Last Name	Matikainen	Registration Number	73,291
Additional Signature may be generated within this form by selecting the Add button. Add					

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	012052.0799C1
		Application Number	
Title of Invention	WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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The information provided by you in this form will be subject to the following routine uses:

- 1 The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
- 2 A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
- 3 A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
- 4 A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
- 5 A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
- 6 A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
- 7 A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
- 8 A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
- 9 A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF

CROSS-REFERENCE TO RELATED APPLICATION(S)

This application is a Continuation Application of U.S. Application No. 14/913,085, filed
5 on February 19, 2016, which is a U.S. National Stage Application of International Application No.
PCT/KR2014/004073, filed on May 8, 2014, which claims the benefit under 35 USC 119(a) of
Korean Patent Application No. 10-2013-0098476 filed on August 20, 2013, in the Korean
Intellectual Property Office, the entire disclosure of which is incorporated herein by reference for
all purposes.

10

BACKGROUND

1. Field

The following description relates to a wearable biosignal interface and an operation
method thereof, and to an apparatus for controlling an external device and a method of controlling
15 an external device using a wearable apparatus.

2. Description of Related Art

Recently, wearable-type mobile devices are emerging at an accelerated rate amid the
proliferation of smartphones. A wearable-type mobile device may include a separate screen for
20 displaying information obtained by a smartphone through a wireless data network, allowing its
user to process data without directly viewing the screen of the smartphone.

A gesture recognized by an accelerometer or a biosignal sensor may be used to operate
the wearable mobile device. However, operating the wearable mobile device based only on the
gesture may increase a probability of misrecognizing the intention of the user. For example, in a
25 wearable mobile device designed to be operated by recognizing a hand gesture, a movement of the

arm connected to the hand may be recognized as a hand gesture, misinterpreting the intention of the user in operating the wearable mobile device.

Accordingly, a wearable mobile device model that recognizes an intention of a user with a greater precision is desirable to improve user experience.

5

SUMMARY

In one general aspect, there is provided a wearable biosignal interface including a motion sensor configured to obtain a first signal based on a motion of a target, a biosignal sensor disposed on the target and configured to obtain a second signal from the target, and a determination
10 controller configured to determine a validity of the second signal based on the first signal.

The second signal may include biological information obtained by one or more biosensor in contact with skin of the target.

The determination controller may be configured to determine the second signal to be valid in response to a magnitude of the first signal satisfying a first threshold value.

15 The determination controller may be configured to identify an interval during which a magnitude of the first signal is less than or equal to a first threshold value and to determine a signal, among a plurality of second signals obtained by the biosignal sensor in the identified interval, to be a valid second signal.

The determination controller may be configured to determine the first threshold value
20 based on a magnitude of the second signal.

The determination controller may be configured to determine the second signal to be invalid, irrespective of the first signal, in response to the magnitude of the second signal not satisfying a second threshold value.

In response to a determination that the second signal is valid, the determination controller
25 may be configured to operate in an active mode and to control an external device based on the

second signal.

In another general aspect, an operation method of a wearable biosignal interface may involve using a motion sensor to obtain a first signal based on a motion of a target, using a biosignal sensor disposed on the target to obtain a second signal, and determining a validity of the second
5 signal based on the first signal.

The second signal may include biological information obtained by one or more biosensor in contact with skin of the target.

The determining of the validity may involve determining the second signal to be valid in response to a magnitude of the first signal satisfying a first threshold value.

10 The determining of the validity may involve identifying an interval during which a magnitude of the first signal is less than or equal to a first threshold value, and determining a signal, among second signals obtained by the biosignal sensor in the identified interval, to be a valid second signal.

The determining of the validity may further include determining a first threshold value
15 based on a magnitude of the second signal.

The determining of the validity may involve determining the second signal to be invalid, irrespective of the first signal, in response to a magnitude of the second signal not satisfying a second threshold value.

The general aspect of the method may further involve operating in an active mode and
20 controlling an external device based on the second signal, in response to the second signal being determined to be valid.

In another general aspect, an apparatus including a motion sensor configured to detect movement of a target, a second sensor configured to detect biological information of the target, and a controller configured to determine an interval of stability from the movement of the target
25 detected by the motion sensor, to determine whether a magnitude of a biological signal obtained

during the interval is less than or equal to a first threshold value, and to identify a command for controlling another device in response to the magnitude being less than or equal to the first threshold value.

The interval may be determined by extracting a first processed signal from the detected movement of the target, and the biological signal may be obtained by extracting a second processed signal from the biological information obtained by the second sensor, the second sensor comprising one or more biosensor.

The interval may correspond to a time duration in which a magnitude of the first processed signal is less than or equal to a second threshold value.

The general aspect of the apparatus may further include a transmission unit configured to transmit the command to the another device, and the determination controller may be configured to identify the command based on the second processed signal.

In response to a determination that the magnitude of the biological signal is greater than the first threshold value, the apparatus may be configured not to transmit a command based on the biological information obtained during the interval.

Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram illustrating an example of a wearable biosignal interface.

FIG. 2A through FIG. 2D are diagrams illustrating an example of a method of determining the validity of a signal based on a first signal and a second signal.

FIG. 3 is a diagram illustrating an example of a method of performing a command given to an external device, using a magnitude of a first signal and a magnitude of a second signal.

FIG. 4 is a flowchart illustrating an example of a method of operating a wearable biosignal

interface.

Throughout the drawings and the detailed description, unless otherwise described, the same drawing reference numerals will be understood to refer to the same elements, features, and structures. The relative magnitude and depiction of these elements may be exaggerated for clarity, illustration, and convenience.

DETAILED DESCRIPTION

The following detailed description is provided to assist the reader in gaining a comprehensive understanding of the methods, apparatuses, and/or systems described herein.

Accordingly, various changes, modifications, and equivalents of the methods, apparatuses, and/or systems described herein will be suggested to those of ordinary skill in the art. The progression of processing steps and/or operations described is an example; however, the sequence of and/or operations is not limited to that set forth herein and may be changed as is known in the art, with the exception of steps and/or operations necessarily occurring in a certain order. Also, description of well-known functions and constructions may be omitted for increased clarity and conciseness.

The features described herein may be embodied in different forms, and are not to be construed as being limited to the examples described herein. Rather, the examples described herein have been provided so that this disclosure will be thorough and complete, and will convey the full scope of the disclosure to one of ordinary skill in the art.

As described herein, a sensor may refer to a sensor or a plurality of sensors that may be attached to a target to be in contact therewith or disposed thereon so as to obtain a signal generated from the target or to detect a motion of the target.

For example, a sensor may include a biosensor that may be attached to a body part of a living being or living organism such as, for example, a wrist or a forearm of a person, in order to

obtain a biological or physiological signal. For example, a biosensor may be able to detect signals generated by muscle contraction or muscle relaxation of a body part, such as a wrist or a heart. Herein, such a sensor may refer to as a biosignal sensor. A biosignal sensor may detect biological information from a target and may convert the biological information into a signal waveform to generate a biosignal. Such a biosignal sensor may include a photo sensor, a piezoelectric sensor, a force sensor, an electromyogram (EMG) sensor, and the like.

Herein, a muscle may refer to a striated muscle, a cardiac muscle or a smooth muscle, and may refer to a muscle group, a number of muscle groups in a body region, bundles of muscle fibers and the like. For instance, an arm may include several different muscle groups, including a tricep, a bicep and a brachioradialis.

A sensor may include an acceleration sensor, a gyro sensor, a geomagnetic sensor, and a Global Positioning System (GPS) sensor. Such a sensor may directly or indirectly monitor a motion performed by the living target and may obtain a signal associated with the motion performed by the living target. Hereinafter, the sensor described above will be referred to as a motion sensor.

To control an external device using a biosignal obtained by a biosignal sensor, an example of a wearable biosignal interface may ignore a biosignal in the event that a magnitude of a signal associated with a motion of a target exceeds a threshold value. Conversely, the wearable biosignal interface may receive the biosignal in the event that the magnitude of the signal associated with the motion of the target is less than or equal to the threshold value.

The wearable biosignal interface may make a determination as to whether a motion is unintentionally or spontaneously made by a user, and may ignore a biosignal obtained as a result of the unintentional or spontaneous motion. Using this approach, the accuracy of determination regarding the intention of the user may be improved.

FIG. 1 is a diagram illustrating an example of a wearable biosignal interface 100.

Referring to FIG. 1, the example of the wearable biosignal interface 100 include a motion sensor 110, a biosignal sensor 120, a determination controller 130, and a transmission unit 140.

The motion sensor 110 may detect a motion of a target 105 and may obtain a first signal. For instance, the first signal may include an acceleration signal, a rotation signal, a bearing signal, and a coordinate signal. These signals may be associated with velocity variance based on a motion of the target 105.

The motion sensor 110 may monitor a motion of the target 105 and may receive the first signal recognizing a movement of the wearable biosignal interface 100 associated with the motion. Also, the motion sensor 110 may identify a gesture performed by the target 105 by obtaining a second signal after obtaining a first signal associated with an identified gesture.

The motion sensor 110 may include, for example, an acceleration sensor. Such a motion sensor 110 may convert an acceleration signal corresponding to the obtained first signal into a velocity component by calculating an integral of the acceleration signal. The motion sensor 110 may determine one or more numerical values indicating a magnitude of a motion of the target 105.

In the event that the target 105 is an arm that is flexed in any direction, the motion sensor 110 may obtain a first signal associated an "arm motion" based on the flexing. Also, the motion sensor 110 may measure a magnitude of the obtained first signal and may measure a magnitude of a velocity of the "arm motion."

The motion sensor 110 may include a gyrosensor. Using a gyrosensor, such a motion sensor 110 may detect a rotation signal corresponding to the first signal.

The target 105 may be an arm. In the event that the arm performs a large rotation motion, the motion sensor 110 may obtain a first signal associated with the "arm rotation" based on the rotating motion of the arm. In addition, the motion sensor 110 may measure a magnitude of the obtained first signal and may measure a magnitude of a rotational velocity of the "arm rotation."

The motion sensor 110 may include a geomagnetic sensor or a GPS sensor. Using the

geomagnetic sensor or the GPS sensor, such a motion sensor 110 may detect a bearing signal or a coordinate signal corresponding to the first signal.

The motion sensor 110 may be fixed to a form of a bracelet fastened onto the target 105, along with the biosignal sensor 120.

5 The biosignal sensor 120 may be in contact with the target 105 and may obtain a second signal from the target 105. In the illustrated example, the biosignal sensor 120 directly contact the skin of the target 105. However, in another example, the biosignal sensor 120 may be disposed on the target 105 indirectly, such as through a clothing that covers the skin. The second signal may be a biosignal associated with biological information such as, for example, muscle
10 contraction and muscle relaxation occurring in the target 105. The second signal may be a bio-electric/magnetic signal, a bio-impedance signal, and a bio-mechanical signal generated in association with a muscle.

The biosignal sensor 120 may be in contact with the target 105 and may obtain the second signal generated by at least one of muscle contraction and muscle relaxation. In the event that
15 the target 105 is a wrist, the biosignal sensor 120 may detect the second signal generated by the muscle contraction or muscle relaxation of the wrist.

The target 105 may be a wrist, a forearm, a face, a neck, or the like of a living being, and a muscle that generates a second signal may be located within the portion of body.

The biosignal sensor 120 may include at least one biosensor that may detect the second
20 signal from a muscle of the target 105, and may monitor the contraction or relaxation of the muscle using the at least one biosensor. The biosensor may be a photo sensor, a piezoelectric sensor, a force sensor, an electromyogram (EMG) sensor, and the like.

For example, the biosignal sensor 120 may include the photo sensor. Using the photo sensor, such a biosignal sensor 120 may obtain the second signal associated with a status of the
25 target 105, based on light scattering.

The biosignal sensor 120 may include a piezoelectric sensor or a force sensor. Using a piezoelectric sensor or a force sensor, such a biosignal sensor 120 may obtain a second signal associated with an extension and/or contraction of a muscle of the target 105.

5 The biosignal sensor 120 may include an EMG sensor. Using an EMG sensor, a biosignal sensor may obtain a second signal associated with the muscle contraction or relaxation of a heart muscle of a target 105.

The biosignal sensor 120 may be fixed to the bracelet fastened onto the target 105, along with the motion sensor 110. A sensing surface of the biosignal sensor 120 may be fixed to the bracelet to be in a direct contact with the target 105. On the sensing surface, the at least one
10 biosensor may be arranged to detect the second signal generated from a target 105. Using the biosignal sensor 120, the bracelet may broadly detect a second signal such as the bio-electric/magnetic signal, the bio-impedance signal, and the bio-mechanical signal over an area of the body of the target 105.

The bracelet may be fastened onto the target 105 by being wrapped around a wrist muscle
15 over the skin, for example. The bracelet may be fastened onto the target 105 by covering the wrist muscle with an area of the bracelet, for example, the sensing surface, in which one or more biosignal sensor 120 is arranged. Using one or more biosensors, the biosignal sensor 120 may collect a second signal from throughout an entire area of the wrist muscle of the target 105.

The determination controller 130 may determine a validity of the second signal based on
20 the first signal. The determination controller 130 may determine whether the second signal obtained using the biosignal sensor 120 is generated from the target 105 due to an intentional motion or a motion incidentally generated from the target 105 due to an unintentional or spontaneous motion made irrespective of an intention of a user to operate an external device.

The determination controller 130 may determine the second signal to be valid when a
25 magnitude of the first signal satisfies a first threshold value.

The first threshold value may be a reference value used to determine the intentional motion of the target 105. The first threshold value may be determined as an average value of magnitudes of first signals obtained when the user moves a muscle for an actual operation.

5 The determination controller 130 may determine the first threshold value based on a magnitude of the second signal to be obtained. The determination controller 130 may determine the first threshold value for an operation involving a large motion of the target 105 to be greater in proportion to the magnitude of the second signal. Accordingly, the second signal generated by an intentional motion may not be ignored due to a low threshold value.

10 For example, the determination controller 130 may determine the first threshold value for a quick display shift operation of a display apparatus for which a relatively greater second signal is generated to be greater than the first threshold value for an ON/OFF operation of an illumination apparatus for which a relatively smaller second signal is generated.

To determine a validity of a second signal, the determination controller 130 may identify an interval during which the magnitude of the first signal is less than or equal to the first threshold value and may determine a signal, among the second signals obtained during the identified interval, 15 to be a valid second signal. The determination controller 130 may determine, to be valid, the second signal obtained in the interval during which the magnitude of the first signal is less than or equal to the first threshold value and a small motion of the target 105 is performed.

In the event that the magnitude of the second signal does not satisfy a second threshold 20 value, the determination controller 130 may determine the second signal to be invalid, irrespective of the first signal. The second threshold value may be a reference value used to determine a second signal that is valuable as information. The second threshold value may be set as an averaged minimum value obtained based on data obtained by simulations in which users moved their muscles to initiate an actual operation.

25 In response to the magnitude of the second signal being less than or equal to the

determined second threshold value, the determination controller 130 may determine whether the target 105 moves intentionally or unintentionally to be indefinite and invalidate the second signal. Conversely, in response to the magnitude of the second signal being greater than the determined second threshold value, the determination controller 130 may further determine whether the magnitude of the first signal satisfies the first threshold value and may determine the validity of the second signal.

Based on the determination, the determination controller 130 may operate in an active mode in response to the second signal being determined to be valid and may allow an external device to be controlled based on the second signal.

An external device may refer to a device controlled by a command signal, such as a mobile device, a television (TV) screen, a digital versatile disc (DVD) player, a radio, a heating thermostat, a vehicle door, and the like. The command signal may be generated in association with a valid second signal of the wearable biosignal interface 100.

In the event that the magnitude of the first signal is less than or equal to the first threshold value and the magnitude of the second signal is greater than the second threshold value, the determination controller 130 may shift a mode of the wearable biosignal interface 100 to the active mode and may allow the external device to be controlled based on the second signal. Thus, the accuracy of determining an intention of a user to operate the device may be improved.

To control the external device after the shift to the active mode, the wearable biosignal interface 100 may transmit the command signal based on the second signal determined to be valid to an appliance or a device in order to enable the external device to perform a control operation, using the transmission unit 140. For example, the wearable biosignal interface 100 may transmit a command signal to an appliance such as a TV, an air conditioner, a computer, a refrigerator, and the like, or to a living environment controlling device such as a light, a curtain, a window, a door lock device, and the like. The signal transmission may enable the external device to perform a

control operation of the appliance or device such as an ON/OFF operation, an open/close operation, and the like, in accordance with the command signal.

In another example, the wearable biosignal interface 100 may use a valid second signal (or a second signal determined to be valid) as a signal of a remote controller of a vehicle to start
5 the vehicle or as an ID signal to make a payment.

In yet another example, the wearable biosignal interface 100 may use the valid second signal for an interaction control with a computer, or a signal to edit a file stored on the computer or to move a page for a presentation.

The wearable biosignal interface 100 may detect and invalidate a biosignal generated by
10 the unintentional motion of the target 105; thus, the wearable biosignal interface 100 may optimally avoid a control error for the external device.

The wearable biosignal interface 100 may identify a biosignal generated only when the target 105 is under a certain level of stability, and may use the biosignal obtained during the stable condition to control an external device; thus, the wearable biosignal interface 100 may provide a
15 precise determination of the user's intention for operating the external device.

FIG. 2A through FIG. 2D are diagrams illustrating examples of methods of determining a second signal as a valid signal based on a first signal and a second signal.

FIG. 2A is a diagram illustrating an example of a first signal obtained using a motion sensor 110 of FIG. 1 of a wearable biosignal interface 100 of FIG. 1. When a target 105 of FIG. 1 performs a rapid motion, such as a motion of flexing an arm, in a interval from t_2 to t_3 , the motion
20 sensor 110 may obtain the first signal having a magnitude measured to be great in the interval from t_2 to t_3 . The first signal obtained in the interval from t_2 to t_3 may be greater than a first threshold value used as a reference value to determine an intentional motion of the target 105.

FIG. 2B is a diagram illustrating an example of a second signal obtained by a biosignal
25 sensor 120 of FIG. 1 of a wearable biosignal interface 100 of FIG. 1. As illustrated in FIG. 2B,

the obtained second signal may be in a form of an analog sine waveform and may have a greater waveform in a interval from t_1 to t_3 during which muscles of an arm are tense due to the flexing of the arm.

FIG. 2C is a graph illustrating an example in which a determination controller 130 of FIG. 1 of a wearable biosignal interface 100 of FIG. 1 processes magnitudes of the obtained second signals and compares the magnitudes. The determination controller 130 may plot, based on the second signal of FIG. 2B, a parabola in which an inclination increases drastically after a point t_1 at which a waveform becomes larger and the inclination decreases drastically after a point t_3 at which the waveform becomes smaller.

The second signal in the interval from t_1 to t_3 may be obtained when satisfying a second threshold value used as a reference value to determine a second signal that is valuable as information.

FIG. 2D illustrates an example of a method of determining, by a determination controller 130 of FIG. 1, a validity of a second signal based on a first signal and a second signal.

The determination controller 130 may invalidate the second signal obtained in the interval from t_2 to t_3 in which the first signal exceeding the first threshold value is obtained. Also, the determination controller 130 may invalidate the second signal obtained during an interval before t_1 and after t_3 during which the second signal less than or equal to the second threshold value is obtained.

As shown in FIG. 2D, the determination controller 130 may determine the second signal obtained in of the interval from t_1 to t_2 to be a valid signal and may enable a command signal to be generated to control an external device during the interval.

The wearable biosignal interface 100 may operate, using the second signal, an external device such as a mobile device, a TV screen, a DVD, a radio, a heating thermostat, a vehicle door, and the like. The wearable biosignal interface 100 may allow the operation of the external device,

using the second signal, only when the magnitude of the first signal decreases to be less than or equal to the first threshold value due to a small motion performed by a user in order to improve the accuracy of the wearable biosignal interface 100 in determining an intention of a user for operation.

5 The wearable biosignal interface 100 may also allow the operation of the external device in response to the magnitude of the second signal satisfying the second threshold value, indicating that the second signal includes valuable information.

FIG. 3 is a diagram illustrating an example of a method of performing a command given to an external device, using a magnitude of a first signal and a magnitude of a second signal.

10 Referring to FIG. 3, in 311, a motion sensor of a wearable biosignal interface may detect a motion of a target, and the motion sensor may obtain a first signal. For example, the motion sensor may detect and obtain a first signal associated with the flexing of an arm. In this example, the arm is the target from which signals can be detected.

In 312, the wearable biosignal interface may process the first signal obtained by the motion sensor and may extract a characteristic. For example, wearable biosignal interface may extract the magnitude of the first signal corresponding to a characteristic associated with a speed at which an arm is flexed.

In 313, the wearable biosignal interface may determine whether the magnitude of the first signal is greater than a first threshold value. For example, the wearable biosignal interface may determine a magnitude of a first signal obtained in response to a user moving a muscle to actually operate an external device to be the first threshold value, and may compare the first threshold value to magnitudes of the obtained first signals.

Also, in 313, the wearable biosignal interface may identify an interval during which the magnitude of the first signal is less than or equal to the first threshold value.

25 In response to a determination that the magnitude of the first signal is greater than the first

threshold value, for example, in a 'Yes' direction from 313, the wearable biosignal interface may ignore the second signal obtained in the same interval during which the first signal is obtained. Also, the wearable biosignal interface may re-obtain the first signal in 311.

Conversely, in response to a determination that the magnitude of the first signal is less than or equal to the first threshold value, for example, in a 'No' direction of 313, the wearable biosignal interface may proceed to 323, as described later. The wearable biosignal interface may determine whether the second signal is valid based on a result of comparing the magnitude of the second signal obtained in the same interval during which the first signal is obtained to the second threshold value.

If the magnitude of the second signal is less than the second threshold value, in 321, a biosignal sensor of the wearable biosignal interface may detect and may obtain a second signal from the target. For example, the biosignal sensor may obtain the second signal in response to detecting a tenseness of the muscles of the arm due to a flexing of the arm. The second signal may be obtained from the biological information detected by the biosignal sensor in 320.

In 322, the wearable biosignal interface may process the second signal obtained by the biosignal sensor 320 and may extract a characteristic of the second signal. For example, the wearable biosignal interface may extract a magnitude of the second signal corresponding to a characteristic associated with the tension in the flexed arm.

In 323, the wearable biosignal interface may determine whether the magnitude of the second signal is greater than a second threshold value. For example, the wearable biosignal interface may determine a reference value of the magnitude of the second signal, which may be a valuable information for determining the second threshold value, and may compare the second threshold value to magnitudes of the obtained second signals. In 323, the wearable biosignal interface may determine a signal obtained in the identified interval during which the magnitude of the first signal is less than or equal to the first threshold value to be a valid second signal.

In the event that the magnitude of the second signal is less than or equal to the second threshold value, for example, a 'No' direction of 323, the wearable biosignal interface may ignore the obtained second signal irrespective of a determination made in 313. Also, the wearable biosignal interface may re-obtain the second signal in 321.

5 Conversely, in the event that the magnitude of the second signal is greater than the second threshold value, for example, in a 'Yes' direction of 323, the wearable biosignal interface may ultimately determine the second signal to be valid based on a result of determination in 313 and may proceed to 330 to execute the predetermined command.

Operations described in the foregoing include using a motion sensor to obtain a first signal
10 in 311, and using a biosignal sensor to obtain a second signal in 321. However, it may be possible to perform 321 prior to or concurrent with 311.

In 330, the wearable biosignal interface may execute a command based on the second signal determined to be valid. In 330, the wearable biosignal interface may operate in an active mode and may allow an external device to be controlled based on the second signal. Using the
15 second signal that is determined to be valid, the wearable biosignal interface may operate an external device such as a mobile device, a TV screen, a DVD player, a radio, a heating thermostat, a vehicle door and the like.

Hereinafter, operations of a wearable biosignal interface 100 according to another example will be described with reference to FIG. 4.

20 FIG. 4 is a flowchart illustrating an example of an operation method of a wearable biosignal interface 100.

In 410, the wearable biosignal interface 100 may detect a motion of a target 105 of FIG. 1 and may obtain a first signal. The first signal may be, for example, an acceleration signal, a rotation signal, a bearing signal, a coordinate signal, and the like, that is associated with a velocity
25 variance based on the motion of the target 105. In 410, the wearable biosignal interface 100 may

monitor, using a motion sensor 110 of FIG. 1, the motion of the target 105 and may receive the first signal to perceive a movement involved with the motion. For example, the wearable biosignal interface 100 may identify a gesture performed by the target 105 in response to obtaining a biosignal, and may obtain the first signal associated with the identified gesture.

5 For example, the motion sensor 110 may include an acceleration sensor. By applying integral calculus, the motion sensor 110 may convert an acceleration signal obtained by the acceleration sensor as the first signal into its velocity component. Based on the calculation, the wearable biosignal interface 100 may numerically indicate a magnitude of a motion of the target.

For example, in the event that the target 105 is an arm that is flexed, the wearable biosignal
10 interface 100 may obtain the first signal associated with an "arm motion" based on the flexing of the arm. Also, the wearable biosignal interface 100 may measure the magnitude of the obtained first signal and may measure a velocity magnitude of the "arm motion."

The motion sensor 110 may include a gyrosensor. For example, using the gyrosensor, the wearable biosignal interface 100 may measure a rotation signal corresponding to the first signal.

15 In one example, target 105 may be an arm, and the arm may be performing a large rotation. The wearable biosignal interface 100 may obtain the first signal associated with the arm rotation based on the detected movement of the arm. Also, the wearable biosignal interface 100 may measure the magnitude of the obtained first signal and/or may measure the magnitude of the angular velocity of the arm rotation.

20 The motion sensor 110 may include a geomagnetic sensor or a GPS sensor. Using the geomagnetic sensor or the GPS sensor, the motion sensor 110 may measure a bearing signal or a coordinate signal corresponding to the first signal.

According to an example, the motion sensor 110 may be fixed to a form of a bracelet fastened onto the target 105, along with a biosignal sensor 120.

25 The wearable biosignal interface 100 may be in contact with the target 105 and may obtain

a second signal from the target 105. In this example, the second signal may be a biosignal associated with biological information such as muscle contraction and muscle relaxation that is occurring in the target 105. The second signal may be a bio-electric/magnetic signal, a bio-impedance signal, and/or a bio-mechanical signal generated in association with a muscle movement or muscle tone.

In 420, the wearable biosignal interface 100 may be in contact with the target 105 through the biosignal sensor 120 and may obtain the second signal generated by at least one of muscle contraction and muscle relaxation. In an example in which the target 105 is a wrist, the wearable biosignal interface 100 may detect the second signal generated by the muscle contraction or muscle relaxation of the wrist.

The target 105 may be, for example, a wrist, a forearm, a face, a neck and the like of a living being. The body part may include a muscle that generates the second signal.

The biosignal sensor 120 may include at least one biosensor detecting the second signal from a muscle of the target 105 and may monitor contraction or relaxation of the muscle. The biosensor may be a photo sensor, a piezoelectric sensor, a force sensor, an EMG sensor, and the like by using the at least one biosensor.

For example, the biosignal sensor 120 may include the photo sensor. Using the photo sensor, the wearable biosignal interface 100 may obtain the second signal associated with a state of the target 105, based on light scattering.

The biosignal sensor 120 may include a piezoelectric sensor or a force sensor. Using the piezoelectric sensor or the force sensor, the wearable biosignal interface 100 may obtain the second signal associated with an extension and/or contraction state of the target 105.

The biosignal sensor 120 may include an EMG sensor. Using the EMG sensor, the wearable biosignal interface 100 may obtain the second signal associated with a muscle contraction or relaxation of the target 105.

The biosignal sensor 120 may be affixed to a bracelet fastened onto the target 105, along with a motion sensor 110, as illustrated in FIG. 1. In this example, a sensing surface of the biosignal sensor 120 may be affixed to the bracelet to be in a direct contact with the skin of the target 105. On the sensing surface, one or more biosensor may be arranged to detect a second
 5 signal generated from the target 105. For example, the bracelet may detect, using the biosignal sensor 120, various forms of second signals such as a bio-electric/magnetic signal, a bio-impedance signal, and a bio-mechanical signal associated with the target 105.

The bracelet may be fastened onto the target 105, wrapping around the skin that covers a wrist muscle. For example, the bracelet may be fastened onto the target 105 by covering the wrist
 10 muscle with a specialized area such as the sensing surface of the bracelet on which the one or more biosignal sensor 120 is disposed. Using the one or more biosensor, the biosignal sensor 120 may collect the second signal throughout an entire area of the wrist muscle of the target 105.

In 430, the wearable biosignal interface 100 may determine a validity of the second signal based on the first signal. In 430, the wearable biosignal interface 100 may determine whether the
 15 second signal is generated by an intentional motion of the target 105 or incidentally generated by an unintentional or spontaneous motion of the target 105.

The wearable biosignal interface 100 may determine the second signal to be valid in the event that the magnitude of the first signal satisfies a first threshold value.

Here, the first threshold value may be a reference value used to determine the intentional
 20 motion of the target 105. The first threshold value may be determined as an average value of magnitudes of the first signals obtained while a user moves a muscle for an actual operation.

The wearable biosignal interface 100 may determine the first threshold value based on a magnitude of the second signal to be obtained. For example, the wearable biosignal interface
 100 may determine the first threshold value for operation involving a large motion of the target
 25 105 to be greater in proportion to the magnitude of the second signal; thus, the second signal

generated by the intentional motion of the target 105 may not be ignored due to a threshold value determined to be small.

The wearable biosignal interface 100 may determine the first threshold value for a quick display shift operation of a display apparatus for which a relatively greater second signal is generated to be greater than the first threshold value for an ON/OFF operation of an illumination apparatus for which a relatively smaller second signal is generated.

To determine a validity of the second signal, the wearable biosignal interface 100 may identify interval during which the magnitude of the first signal is less than or equal to the first threshold value and may determine a signal, among the second signals, obtained in the identified interval to be a valid second signal. The wearable biosignal interface 100 may determine the second signal obtained in the interval during which the magnitude of the first signal is less than or equal to the first threshold value and a small motion of the target 105 is determined to be valid.

The wearable biosignal interface 100 may determine the second signal to be invalid, irrespective of the first signal, in the event that the magnitude of the obtained second signal does not satisfy a second threshold value. The second threshold value may be a reference value used to determine the second signal that is valuable as information. The second threshold value may be determined as an averaged minimum value obtained a user moves a muscle for an actual operation. For instance, the second threshold value may be determined from historic values obtained based on many actual operations performed by a target.

In the event that the magnitude of the obtained second signal is less than or equal to the second threshold value, the wearable biosignal interface 100 may determine whether the target 105 moved intentionally for operation to be indefinite and accordingly, invalidate the second signal. Conversely, in the event that the magnitude of the second signal is greater than the second threshold value, the wearable biosignal interface 100 may further determine whether the magnitude of the first signal satisfies the first threshold value and may determine the validity of the second signal.

Based on a result of the determination, in the event that the second signal is determined to be valid, the wearable biosignal interface 100 may operate in an active mode and may allow an external device to be controlled by the second signal. The external device may be a general term for devices controlled by a command signal, such as a mobile device, a TV screen, a DVD player, a radio, a heating thermostat, a vehicle door, and the like. Using the wearable biosignal interface 100, the command signal may be generated based on the valid second signal.

For example, in the event that the magnitude of the first signal is less than or equal to the first threshold value, and the magnitude of the second signal exceeds the second threshold value, the wearable biosignal interface 100 may shift a mode of the wearable biosignal interface 100 to the active mode and may allow the external device to be controlled based on the second signal; thus, the accuracy in determining an intention of a user may be improved.

To control the external device after the shift to the active mode, the wearable biosignal interface 100 may transmit the command signal based on the valid second signal to an appliance such as a TV, an air conditioner, a computer, a refrigerator, and the like, or to a living environment controlling device such as a light, a curtain, a window, a door lock device, and the like. The transmission of the command signal enables the external device to perform a control operation such as an ON/OFF operation and an open/close operation, requested by the command signal.

The wearable biosignal interface 100 may use the valid second signal as a signal for a remote controller of a vehicle to start the vehicle, or as an identification (ID) signal to make a payment.

The wearable biosignal interface 100 may use the valid second signal as a signal for interactive control of a computer, or to edit a file stored on the computer or move a page for a presentation.

The wearable biosignal interface 100 may invalidate a biosignal generated by the unintentional or spontaneous motion of the target 105 to optimally avoid a control error for the

external device.

The wearable biosignal interface 100 may identify a biosignal exclusively generated during the state in which the target 105 is in a certain level of stability and may use the biosignal to control the external device. By using the biosignals generated during a stable state, it is possible to determine an intention of a user with improved precision.

Program instructions to perform a method described herein, or one or more operations thereof, may be recorded, stored, or fixed in one or more computer-readable storage media. The program instructions may be implemented by a computer. For example, the computer may cause a processor to execute the program instructions. The media may include, alone or in combination with the program instructions, data files, data structures, and the like. Examples of non-transitory computer-readable storage media include magnetic media, such as hard disks, floppy disks, and magnetic tape; optical media such as CD ROM discs and DVDs; magneto-optical media, such as optical discs; and hardware devices that are specially configured to store and perform program instructions, such as read-only memory (ROM), random access memory (RAM), flash memory, and the like. Examples of program instructions include machine code, such as produced by a compiler, and files including higher level code that may be executed by the computer using an interpreter. The program instructions, that is, software, may be distributed over network coupled computer systems so that the software is stored and executed in a distributed fashion. For example, the software and data may be stored by one or more computer readable storage mediums. Also, functional programs, codes, and code segments that accomplish the examples disclosed herein can be easily construed by programmers skilled in the art to which the examples pertain based on and using the flow diagrams and block diagrams of the figures and their corresponding descriptions as provided herein. Also, the described unit to perform an operation or a method may be hardware, software, or some combination of hardware and software. For example, the unit may be a software package running on a computer or the computer on which that software is

running.

While this disclosure includes specific examples, it will be apparent to one of ordinary skill in the art that various changes in form and details may be made in these examples without departing from the spirit and scope of the claims and their equivalents. The examples described
5 herein are to be considered in a descriptive sense only, and not for purposes of limitation. Descriptions of features or aspects in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if the described techniques are performed in a different order, and/or if components in a described system, architecture, device, or circuit are combined in a different manner and/or replaced or supplemented
10 by other components or their equivalents. Therefore, the scope of the disclosure is defined not by the detailed description, but by the claims and their equivalents, and all variations within the scope of the claims and their equivalents are to be construed as being included in the disclosure.

CLAIMS

1. A wearable device comprising:

a motion sensor;

a biological sensor; and

a controller configured to obtain a movement of an object using the motion sensor, and a biological signal of the object using the biological sensor,

wherein the controller is configured to perform a predetermined function using at least portion of the biological signal obtained during a period that the movement corresponds to a specified condition.
2. The wearable device of claim 1,

wherein the controller is configured to perform, in response to the movement being less than or equal to a first threshold value, the wearable device using the biological signal.
3. The wearable device of claim 2, wherein the controller is configured to determine the first threshold value based on a magnitude of the biological signal.
4. The wearable device of claim 1, wherein the controller is configured to determine the biological signal to be invalid, irrespective of the movement, in response to the magnitude of the biological signal not satisfying a second threshold value.
5. The wearable device of claim 1, wherein, response to a determination that the biological signal is valid, the controller is configured to operate in an active mode and to control an external device based on the biological signal.

6. The wearable device of claim 1, wherein the controller configured to determine, based on an analysis of the movement, whether to perform the function of the wearable device using the biological signal.

ABSTRACT

A wearable biosignal interface, an apparatus for operating an external device, and an operation method of a wearable biosignal interface are provided. A wearable biosignal interface may include a motion sensor configured to obtain a first signal based on a motion of a target, a
5 biosignal sensor disposed on the target and configured to obtain a second signal from the target, and a determination controller configured to determine a validity of the second signal based on the first signal.

FIG. 1

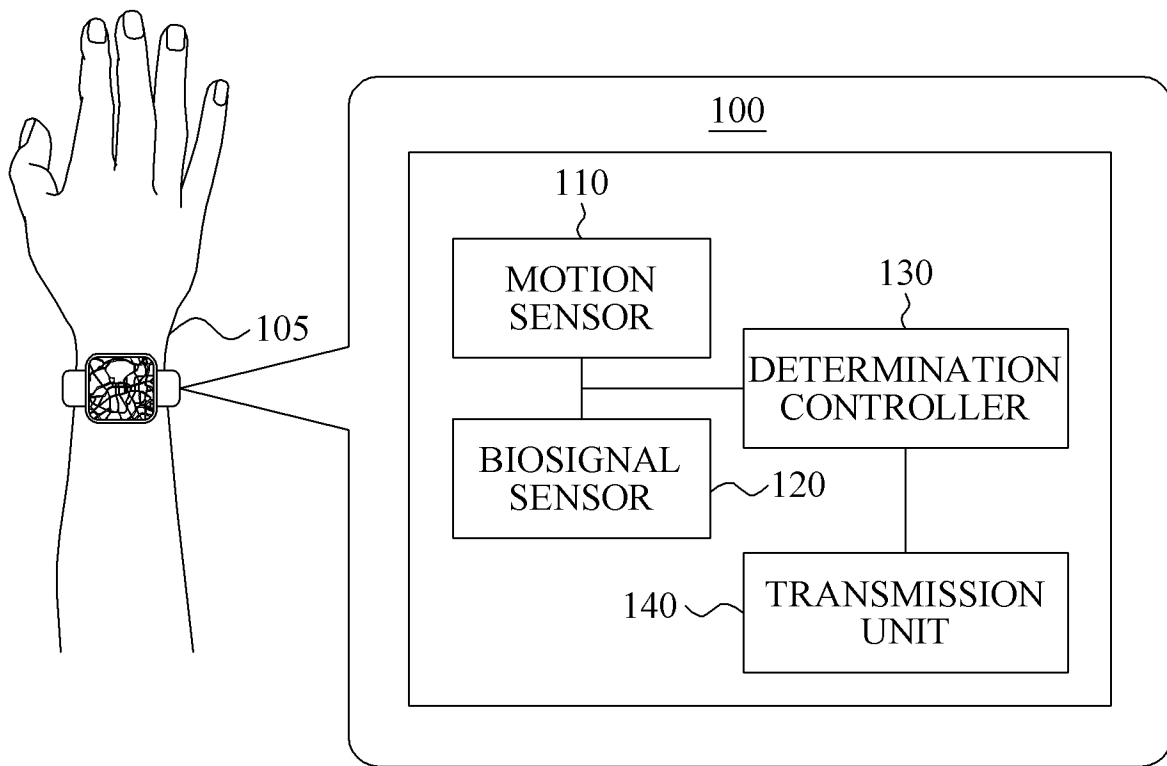


FIG. 2A

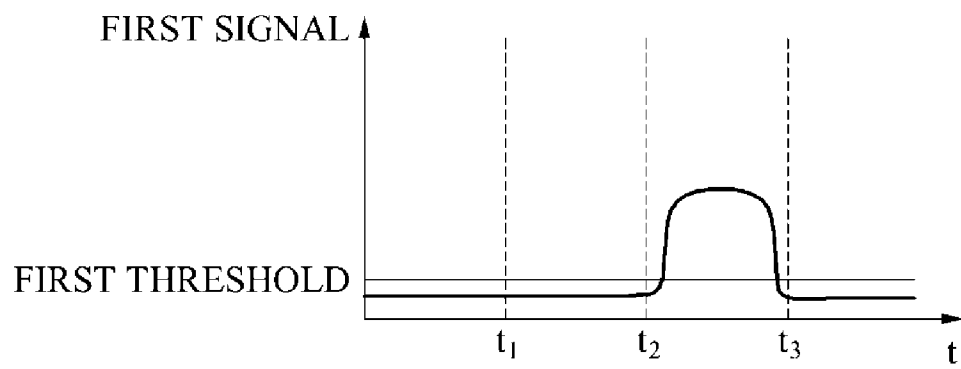


FIG. 2B

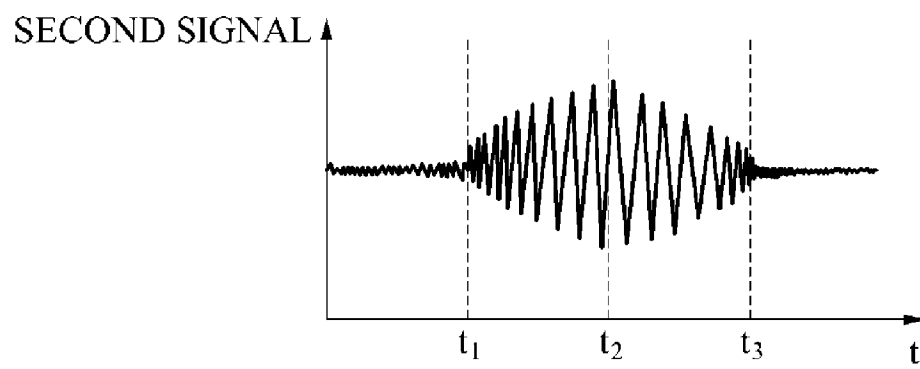


FIG. 2C

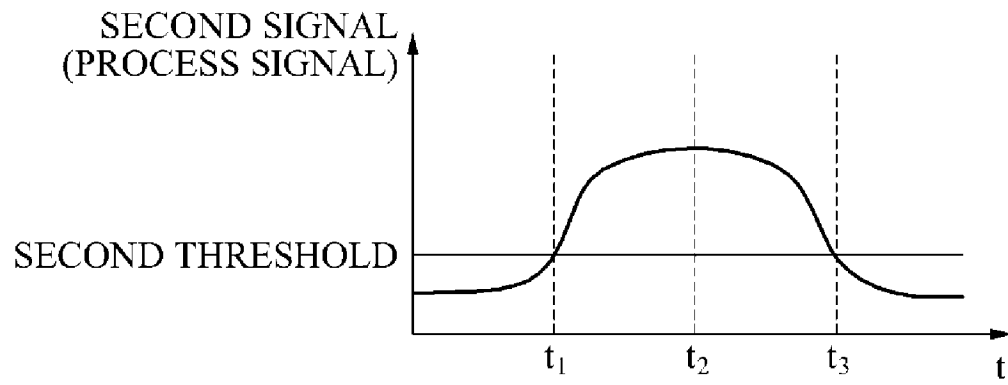


FIG. 2D

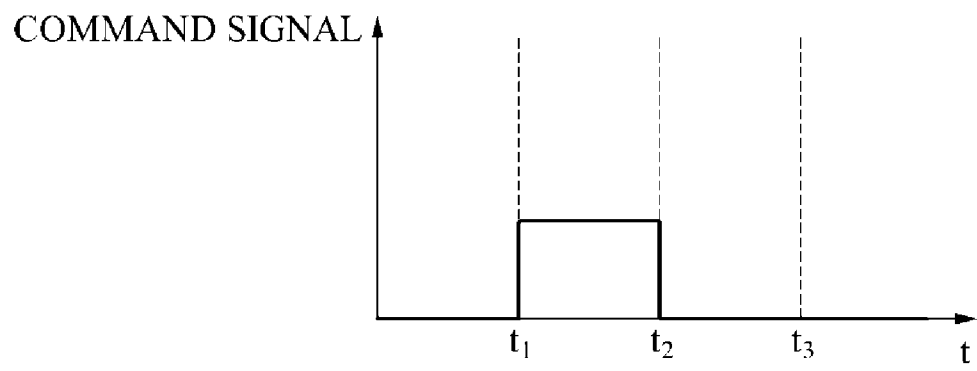


FIG. 3

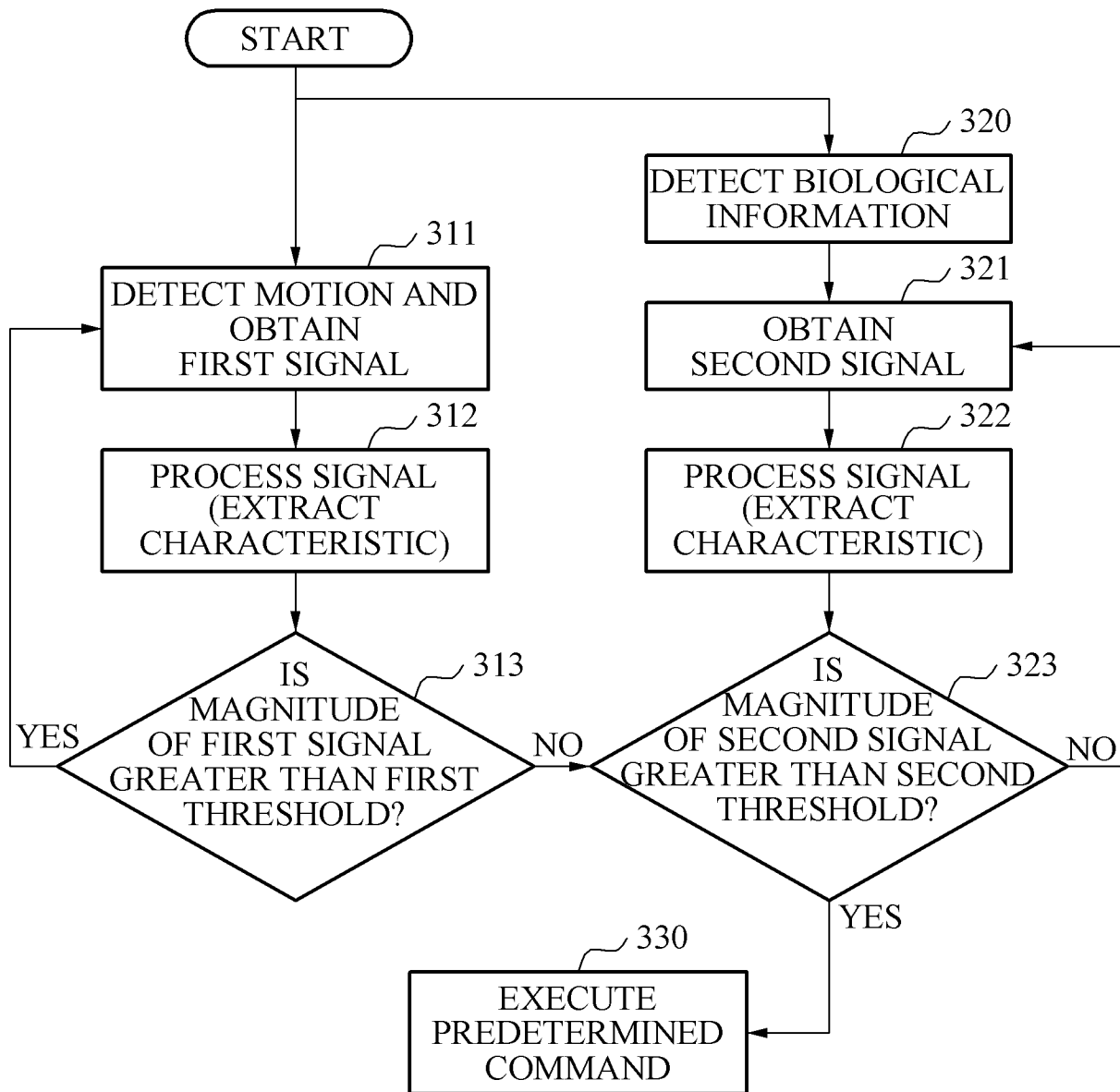
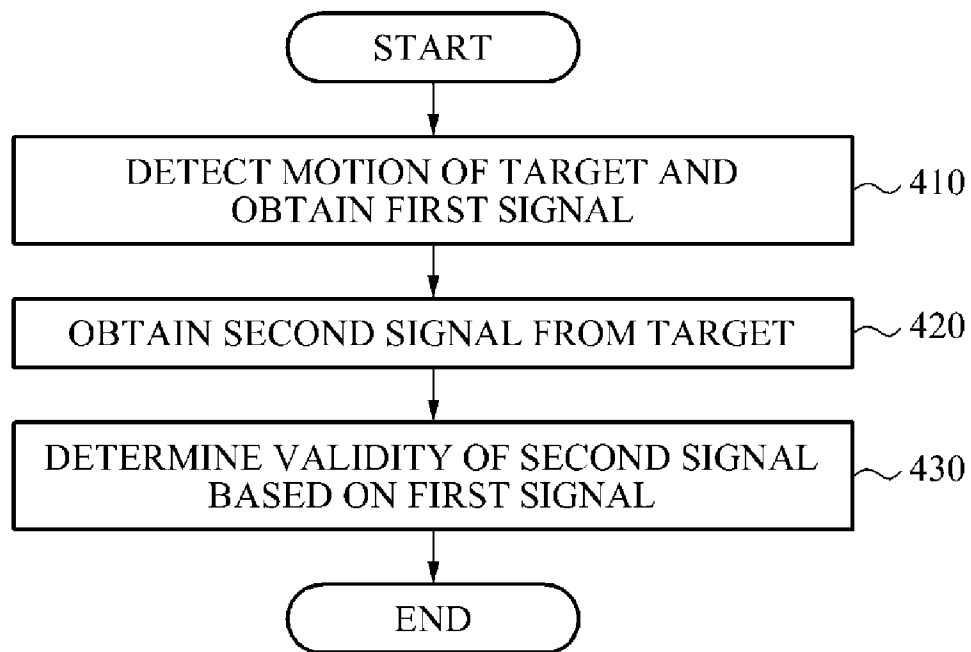


FIG. 4



COMBINED DECLARATION AND ASSIGNMENT UNDER 37 CFR 1.63(e)

DECLARATION

As a below-named inventor, I hereby declare that this declaration is directed to the application attached hereto, or to United States Application Number or PCT International Application Number PCT/KR2014/004073 filed on May 8, 2014 (if applicable), entitled:

WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF

The above-identified application was made or authorized to be made by me.

I believe I am the original inventor or an original joint inventor of a claimed invention in the above-identified application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 USC 1001 by fine or imprisonment of not more than five (5) years, or both.

I have reviewed and understand the contents of the above-identified application, including the claims.

I am aware of the duty to disclose to the United States Patent and Trademark Office all information known to me to be material to patentability as defined in 37 CFR 1.56, including for continuation-in-part applications, material information that became available between the filing date of the prior application and the national or PCT international filing date of the continuation-in-part application.

ASSIGNMENT

For valuable consideration, I, as a below-named assignor, hereby assign to:

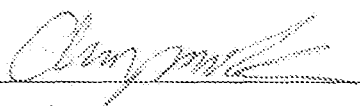
Samsung Electronics Co., Ltd.
129, Samsung-ro, Yeongtong-gu
Suwon-si, Gyeonggi-do, 443-742
Republic of Korea

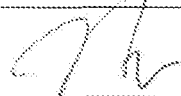
and its successors and assigns (collectively hereinafter called "the Assignee"), the entire right, title, and interest throughout the world in the inventions and improvements that are the subject of the application identified in the above declaration, which is United States Application Number or PCT International Application Number PCT/KR2014/004073 filed on May 8, 2014.

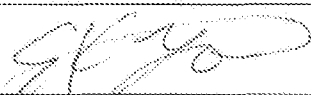
I authorize and request the attorneys appointed in the above-identified application to hereafter complete this Assignment by inserting above the application number and the filing date of the above-identified application when known, and to correct any typographical errors that may be discovered in this Assignment.

This Assignment includes the above-identified application, any and all United States and foreign patents, utility models, and design registrations granted for any of the inventions and improvements that are the subject of the above-identified application, and the right to claim priority based on the filing date of the above-identified application under the International Convention for the Protection of Industrial Property, the Patent Cooperation Treaty, the European Patent Convention, and all other treaties of like purposes; and I authorize the Assignee to apply in all countries in my name or in its own name for patents, utility models, design registrations, like rights of exclusion, and inventors' certificates for any of the inventions and improvements that are the subject of the above-identified application; and I agree for myself and my heirs, legal representatives, and assigns without further compensation to perform such lawful acts and to sign such further applications, Assignments, preliminary statements, and other lawful documents as the Assignee may reasonably request to effectuate fully this Assignment.

In witness whereof, I, as a below-named inventor and assignor, intending to be legally bound, have hereunto affixed my signature on the date indicated below next to my signature.

Inventor's Legal Name	Chang Mok CHOI		
Inventor's Signature		Date	27 Dec. 2015
Residence (City, Country)	Suwon-si, Republic of Korea		
Mailing Address	Samsung Advanced Institute of Technology 130, Samsung-ro, Youngtong-gu Suwon-si, Gyeonggi-do, 443-803 Republic of Korea		

Inventor's Legal Name	Sang Joon KIM		
Inventor's Signature		Date	12. 18. 2015
Residence (City, Country)	Suwon-si, Republic of Korea		
Mailing Address	Samsung Advanced Institute of Technology 130, Samsung-ro, Youngtong-gu Suwon-si, Gyeonggi-do, 443-803 Republic of Korea		

Inventor's Legal Name	Seung Keun YOON		
Inventor's Signature		Date	12. 18. 2015
Residence (City, Country)	Suwon-si, Republic of Korea		
Mailing Address	Samsung Advanced Institute of Technology 130, Samsung-ro, Youngtong-gu Suwon-si, Gyeonggi-do, 443-803 Republic of Korea		

POWER OF ATTORNEY BY APPLICANT

I hereby revoke all previous powers of attorney given in the application identified in the accompanying transmittal letter (form PTO/AIA/15, PTO/SB/16, PTO/AIA/18, PTO/AIA/19, PTO/SB/29, PTO/AIA/50, PTO/AIA/82A, or equivalent).

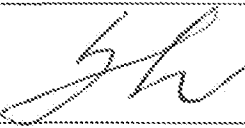
I hereby appoint the practitioners of NSIP Law associated with the following Customer Number as my attorneys or agents to transact all business in the United States Patent and Trademark Office connected with the application identified in the accompanying transmittal letter:

89980

Please recognize or change the correspondence address for the application identified in the accompanying transmittal letter to the address of NSIP Law associated with the following Customer Number:

89980

I, the below-named Assignee, am the original Applicant (if there is only one Applicant) or an original Applicant (if there is more than one Applicant) of the application identified in the accompanying transmittal letter. The undersigned, whose title is supplied below, is authorized to act on behalf of the Assignee.

Name of Assignee	Samsung Electronics Co., Ltd.		
Address of Assignee	129, Samsung-ro, Yeongtong-gu Suwon-si, Gyeonggi-do (16677) Republic of Korea		
Signature		Date	April 7, 2016
Name	Chanho CHO		
Title	Group Leader, Samsung Electronics Co. Ltd		

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

INVENTORS: Chang Mok CHOI, et al.

CONFIRMATION NO.: Not Yet Assigned

APPLICATION NO.: New Application

GROUP ART UNIT: Not Yet Assigned

FILING DATE: Concurrently Herewith

EXAMINER: Not Yet Assigned

TITLE: WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF

INFORMATION DISCLOSURE STATEMENT

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

Applicant hereby submits the following information in accordance with the duty of disclosure set forth in 37 C.F.R. § 1.56.

MPEP 609.02(A)(2) states as follows (emphasis added):

The examiner will consider information which has been considered by the Office in a parent application when examining: (A) a continuation application filed under 37 CFR 1.53(b), (B) a divisional application filed under 37 CFR 1.53(b), or (C) a continuation-in-part application filed under 37 CFR 1.53(b). A listing of the information need not be resubmitted in the continuing application unless the applicant desires the information to be printed on the patent.

If resubmitting a listing of the information, applicant should submit a new listing that complies with the format requirements in 37 CFR 1.98(a)(1).

Accordingly, attached hereto is an Information Disclosure Statement complying with the format requirements in 37 CFR 1.98(a)(1) and listing the references cited in parent application number 14/913,085. This Information Disclosure Statement is being presented concurrently with the filing of the new application, and therefore no fee is due at this time.

It is respectfully requested that an Examiner-initialed copy of the attached PTO/SB08 form be returned to the undersigned.

Dated: February 5, 2019

By: /Thomas J. Matikainen/
Thomas J. Matikainen, Reg. No. 73,291
NSIP Law
P.O. Box 65745
Washington, DC 20035
Tel: (202) 429-0020

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	New Application
				Filing Date	Concurrently Herewith
				First Named Inventor	Chang Mok CHOI
				Art Unit	Not Yet Assigned
				Examiner Name	Not Yet Assigned
Sheet	1	of	2	Attorney Docket Number	012052.0799C1

U.S. PATENT DOCUMENTS					
Examiner Initials*	Cite No. ¹	Document Number Number - Kind Code ² (if known)	Patent or Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
	A	US 5,832,296 A	11-03-1998	Wang et al.	
	B	US 2006/0149146 A1	07-06-2006	Schmidt et al.	
	C	US 2006/0281979 A1	12-14-2006	Kim et al.	
	D	US 2010/0145171 A1	06-10-2010	PARK et al.	
	E	US 2012/0172733 A1	07-05-2012	Park	
	F	US 2012/0184826 A1	07-19-2012	KEENAN et al.	
	G	US 2012/0259182 A1	10-11-2012	KIM et al.	

FOREIGN PATENT DOCUMENTS						
Examiner Initials*	Cite No. ¹	Foreign Patent Document Country Code ³ - Number ⁴ - Kind Code ⁵ (if known)	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear	T ⁶
	a	CN 1488115 A	04-07-2004	Bridicum AS	Abstract of WO02059803 A1	X
	b	CN 100440310 C	12-03-2008	Thomson Licensing	Abstract of US 2004218104 A1	X
	c	KR 10-2003-0040316 A	05-22-2003	LEE		X
	d	KR 10-2006-0069985 A	06-23-2006	ELECTRONICS AND TELECOMMUNICATIONS RESEARCH INSTITUTE		X
	e	KR 10-2007-0000848 A	01-03-2007	PANTECH CO., LTD.		X
	f	KR 10-2007-0109044 A	11-15-2007	ELECTRONICS AND TELECOMMUNICATIONS RESEARCH INSTITUTE		X
	g	KR 10-2009-0027390 A	03-17-2009	Lee et al.		X

Examiner Signature		Date Considered	
--------------------	--	-----------------	--

*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant. ¹Applicant's unique citation designation number (optional). ²See Kinds Codes of USPTO Patent Documents at www.uspto.gov or MPEP 901.04(a). ³Enter Office that issued the document, by the two-letter code (WIPO Standard ST.3). ⁴For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁶Applicant is to place a check mark here if English language Translation is attached (X indicates Abstract only provided).

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	New Application
				Filing Date	Concurrently Herewith
				First Named Inventor	Chang Mok CHOI
				Art Unit	Not Yet Assigned
				Examiner Name	Not Yet Assigned
Sheet	2	of	2	Attorney Docket Number	012052.0799C1

FOREIGN PATENT DOCUMENTS						
Examiner Initials*	Cite No. ¹	Foreign Patent Document Country Code ³ - Number ⁴ - Kind Code ⁵ (if known)	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear	T ⁶
	h	KR 10-2009-0032537 A	04-01-2009	ELECTRONICS AND TELECOMMUNICATIONS RESEARCH INSTITUTE		X
	i	KR 10-2010-0027375 A	03-11-2010	KOREA INSTITUTE OF ORIENTAL MEDICINE		X
	j	KR 10-2011-0123708 A	11-15-2011	Jawon Medical Co., Ltd.		X
	k	KR 10-2011-0062558 A	06-10-2011	SOONGSIL UNIVERSITY RESEARCH CONSORTIUM TECHNOPARK		X
	l	KR 10-2012-0014233 A	02-16-2012	QUALCOMM INCORPORATED	Abstract of WO/2010/141878	X
	m	KR 10-2012-0111030 A	10-10-2012	LAXTHA		X
	n	KR 10-2012-0133350 A	12-10-2012	NEOFACT CO., LTD.		X
	o	WO 2008/073801 A2	06-19-2008	Emotiv Systems Pty Ltd.		

NON-PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author, title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ⁶
	1	International Search Report issued on August 7, 2014 in counterpart International Application No. PCT/KR2014/004073 (2 pages in English; 2 pages in Korean).	✓
	2	European Search Report issued on July 11, 2017 in Corresponding European Patent Application No. 14838613.9 (11 pages in English).	
	3	Chinese Office Action issued on January 2, 2018 in corresponding Chinese Patent Application No. 201480044084.7 (8 pages in English and 7 pages in Chinese).	✓

Examiner Signature		Date Considered	
--------------------	--	-----------------	--

*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant. ¹Applicant's unique citation designation number (optional). ²See Kinds Codes of USPTO Patent Documents at www.uspto.gov or MPEP 901.04(a). ³Enter Office that issued the document, by the two-letter code (WIPO Standard ST.3). ⁴For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁶Applicant is to place a check mark here if English language Translation is attached (X indicates Abstract only provided).

Electronic Patent Application Fee Transmittal

Application Number:				
Filing Date:				
Title of Invention:	WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF			
First Named Inventor/Applicant Name:	Chang Mok CHOI			
Filer:	Thomas Joseph Matikainen/Brandon Bender			
Attorney Docket Number:	012052.0799C1			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
UTILITY APPLICATION FILING	1011	1	300	300
UTILITY SEARCH FEE	1111	1	660	660
UTILITY EXAMINATION FEE	1311	1	760	760
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1720

Electronic Acknowledgement Receipt

EFS ID:	35062025
Application Number:	16267752
International Application Number:	
Confirmation Number:	9126
Title of Invention:	WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF
First Named Inventor/Applicant Name:	Chang Mok CHOI
Customer Number:	89980
Filer:	Thomas Joseph Matikainen/Brandon Bender
Filer Authorized By:	Thomas Joseph Matikainen
Attorney Docket Number:	012052.0799C1
Receipt Date:	05-FEB-2019
Filing Date:	
Time Stamp:	14:04:49
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 1720
RAM confirmation Number	020619INTEFSW14070700
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Transmittal of New Application	NewContApp_0120520799C1_ UtilityTransmittal.pdf	280413	no	2
			32dc74b23f252ad2acfa45bbe2c7b17470cd36801		
Warnings:					
Information:					
2	Application Data Sheet	NewContApp_0120520799C1_ ADS.pdf	1256288	no	9
			92c9af7af042c710e25fadeb6ac3cf949bd24057		
Warnings:					
Information:					
3		NewContApp_0120520799C1_ AppAsFiled.pdf	166054	yes	26
			e985d0bfed9dceb96fcaccccd0b4238b2c8c9a17		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Specification		1	23	
	Claims		24	25	
	Abstract		26	26	
Warnings:					
Information:					
4	Drawings-only black and white line drawings	NewContApp_0120520799C1_ DrawingsAsFiled.pdf	116029	no	5
			e66d632bb2a66e8568fa475c4f43572f9f0e80cb		
Warnings:					
Information:					
5	Oath or Declaration filed	NewContApp_0120520799C1_ ExecutedDec_fromParent.pdf	3096963	no	3
			4ae184e46485f18f1620e4690997a1f48d24a3b2		

Warnings:					
Information:					
6	Power of Attorney	NewContApp_0120520799C1_POA_byApplicant_SamsungSALT.pdf	909304 fc6bc14364c776d17ea8c3ff5ca93e37ce372567	no	1
Warnings:					
Information:					
7	Transmittal Letter	NewContApp_0120520799C1_Statement.pdf	18384 3c00e084a67dd8fc5fe6f3e4699906855df33de	no	2
Warnings:					
Information:					
8	Information Disclosure Statement (IDS) Form (SB08)	NewContApp_0120520799C1_IDS.pdf	97936 0522729fc95ac6a99acacdf4b3504e59225c244b	no	2
Warnings:					
Information:					
This is not an USPTO supplied IDS fillable form					
9	Fee Worksheet (SB06)	fee-info.pdf	35083 34ac98b036d74ab086e504d4ee665e2b73388b3b	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			5976454		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
16/267,752

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	300
SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A			N/A	660
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	760
TOTAL CLAIMS (37 CFR 1.16(j))	6 minus 20 =	*			OR	x 100 =	0.00
INDEPENDENT CLAIMS (37 CFR 1.16(h))	1 minus 3 =	*				x 460 =	0.00
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						0.00
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0.00
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL	1720

APPLICATION AS AMENDED - PART II

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT A	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=		OR	x	=
Independent (37 CFR 1.16(h))	*	Minus	***	=		OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT B	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=		OR	x	=
Independent (37 CFR 1.16(h))	*	Minus	***	=		OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
16/267,752	02/05/2019		1720	012052.0799C1	6	1

CONFIRMATION NO. 9126

FILING RECEIPT



0000000106056523

89980
NSIP LAW
P.O. Box 65745
Washington, DC 20035

Date Mailed: 02/25/2019

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Chang Mok CHOI, Suwon-si, KOREA, REPUBLIC OF;
Sang Joon KIM, Suwon-si, KOREA, REPUBLIC OF;
Seung Keun YOON, Suwon-si, KOREA, REPUBLIC OF;

Applicant(s)

Samsung Electronics Co., Ltd., Suwon-si, KOREA, REPUBLIC OF;

Assignment For Published Patent Application

Samsung Electronics Co., Ltd., Suwon-si, KOREA, REPUBLIC OF

Power of Attorney: The patent practitioners associated with Customer Number 89980

Domestic Priority data as claimed by applicant

This application is a CON of 14/913,085 02/19/2016
which is a 371 of PCT/KR2014/004073 05/08/2014

Foreign Applications (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.)

REPUBLIC OF KOREA 10-2013-0098476 08/20/2013 Access Code Provided

Permission to Access Application via Priority Document Exchange: Yes

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The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 16/267,752**

Projected Publication Date: 06/06/2019

Non-Publication Request: No

Early Publication Request: No
Title

WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF

Preliminary Class

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

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Feb 25, 2019 03:40:57 AM

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Application	Document	Mailroom Date	Attorney Docket No.
16267752	APP.FILE.REC	02/25/2019	012052.0799C1

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Application number: 10-2013-0098476

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출 원 번 호 : 10-2013-0098476

Application Number

출 원 년 월 일 : 2013년 08월 20일

Filing Date Aug. 20, 2013

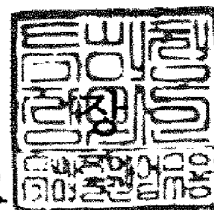
출 원 인 : 삼성전자주식회사

Applicant(s) SAMSUNG ELECTRONICS CO., LTD.

2019년 03월 04일

특 허 청

COMMISSIONER



【서지사항】

【서류명】	특허출원서
【출원구분】	특허출원
【출원인】	
【명칭】	삼성전자주식회사
【특허고객번호】	1-1998-104271-3
【대리인】	
【명칭】	특허법인 무한
【대리인번호】	9-2007-100061-4
【지정된변리사】	천성진
【포괄위임등록번호】	2007-019579-5
【발명의 국문명칭】	착용형 생체신호 인터페이스 및 착용형 생체신호 인터페이스의 동작 방법
【발명의 영문명칭】	WEARABLE BIO-SIGNAL INTERFACE AND METHOD OF OPERATION OF WEARABLE BIO-SIGNAL INTERFACE
【발명자】	
【성명】	최창목
【성명의 영문표기】	CHOI Chang Mok
【주민등록번호】	820711-1XXXXXX
【우편번호】	135-826
【주소】	서울특별시 강남구 논현동 192-15 휴먼스페이스빌 802호
【국적】	KR
【발명자】	

제출 일자 : 2013-08-20

【성명】 김상준
【성명의 영문표기】 KIM Sang Joon
【주민등록번호】 780919-1XXXXXX
【우편번호】 445-724
【주소】 경기도 화성시 반송동 시범다은마을포스코더샵아파트 321동
204호
【국적】 KR

【발명자】

【성명】 윤승근
【성명의 영문표기】 YOON Seung Keun
【주민등록번호】 780417-1XXXXXX
【우편번호】 137-779
【주소】 서울특별시 서초구 서초4동 삼풍아파트 10동 1303호
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【출원료】	0 면	38,000 원
【가산출원료】	32 면	0 원
【우선권주장료】	0 건	0 원
【심사청구료】	0 항	0 원

제출 일자 : 2013-08-20

【합계】 38,000 원

【발명의 설명】

【발명의 명칭】

착용형 생체신호 인터페이스 및 착용형 생체신호 인터페이스의 동작 방법
{WEARABLE BIO-SIGNAL INTERFACE AND METHOD OF OPERATION OF WEARABLE BIO-SIGNAL
INTERFACE}

【기술분야】

【0001】 이하의 실시예들은 착용형 생체신호 인터페이스 및 착용형 생체신호 인터페이스의 동작 방법에 관한 것이다.

【발명의 배경이 되는 기술】

【0002】 근래에는, 스마트폰의 확산과 함께, 새로운 착용형 모바일 기기의 출현이 가속화 되고 있다. 이러한 착용형 모바일 기기에서는, 무선 데이터망을 통해 스마트폰으로 얻은 정보를, 스마트폰을 직접 보지 않고도, 데이터를 가공하여 별도로 제공되는 화면을 통해 시각화하고 있다.

【0003】 상기 착용형 모바일 기기를 조작하는 데에는, 주로 가속도계나 생체신호 센서를 이용하여 인식되는 제스처를 이용할 수 있다. 하지만, 제스처만을 이용한 기기 조작은, 사용자의 조작 의도를 잘못 인식하는 확률을 증가 시킬 수 있다. 예컨대, 손의 제스처를 인식하여 조작되는 착용형 모바일 기기에서는, 상기 손과 연결된 팔이 심하게 움직여 발생하는 손의 제스처 조작, 사용자의 조작 의도로 잘못 인식할 수 있다.

【발명의 내용】

【과제의 해결 수단】

【0004】 이에 따라, 착용형 모바일 기기에서, 사용자 조작 의도를 보다 정확하게 인식하는 모델의 등장어 요구되고 있다.

【0005】 일실시예에 따른 착용형 생체신호 인터페이스는, 측정 대상의 움직임을 감지하여 제1 신호를 획득하는 모션 센서부와, 상기 측정 대상에 접촉되어 상기 측정 대상으로부터 제2 신호를 획득하는 생체신호 센서부, 및 상기 제1 신호를 고려하여, 상기 제2 신호에 대한 유효성을 판단하는 판단 제어부를 포함할 수 있다.

【0006】 상기 판단 제어부는, 상기 제1 신호의 크기가 선정된 제1 임계치를 만족하여 획득되면, 상기 제2 신호를 유효한 것으로 판단할 수 있다.

【0007】 상기 판단 제어부는, 상기 제1 신호의 크기가 선정된 제1 임계치 이하인 시간을 식별하고, 상기 제2 신호 중 상기 식별된 시간에 획득되는 신호를, 유효한 제2 신호로서 판단할 수 있다.

【0008】 상기 판단 제어부는, 획득되는 상기 제2 신호의 크기에 기초하여, 상기 제1 임계치를 선정할 수 있다.

【0009】 상기 판단 제어부는, 획득된 상기 제2 신호의 크기가 선정된 제2 임계치를 만족하지 않으면, 상기 제1 신호와 무관하게, 상기 제2 신호를 유효하지 않는 것으로 판단할 수 있다.

【0010】 상기 판단 제어부는, 상기 판단 결과, 상기 제2 신호가 유효하면, 활성모드로 동작하여, 상기 제2 신호에 의한 외부 기기의 제어를 허용할 수 있다.

【0011】 일실시예에 따른 착용형 생체신호 인터페이스의 동작 방법은, 모션 센서부에 의해, 측정 대상의 움직임을 감지하여 제1 신호를 획득하는 단계, 상기 측정 대상에 접촉되는 생체신호 센서부에 의해, 상기 측정 대상으로부터 제2 신호를 획득하는 단계, 및 상기 제1 신호를 고려하여, 상기 제2 신호에 대한 유효성을 판단하는 단계를 포함할 수 있다.

【0012】 상기 제2 신호에 대한 유효성을 판단하는 단계는, 상기 제1 신호의 크기가 선정된 제1 임계치를 만족하여 획득되면, 상기 제2 신호를 유효한 것으로 판단하는 단계를 포함할 수 있다.

【0013】 상기 제2 신호에 대한 유효성을 판단하는 단계는, 상기 제1 신호의 크기가 선정된 제1 임계치 이하인 시간을 식별하는 단계, 및 상기 제2 신호 중 상기 식별된 시간에 획득되는 신호를, 유효한 제2 신호로서 판단하는 단계를 포함할 수 있다.

【0014】 상기 제2 신호에 대한 유효성을 판단하는 단계는, 획득되는 상기 제2 신호의 크기에 기초하여, 상기 제1 임계치를 선정하는 단계를 포함할 수 있다.

【0015】 상기 제2 신호에 대한 유효성을 판단하는 단계는, 획득된 상기 제2 신호의 크기가 선정된 제2 임계치를 만족하지 않으면, 상기 제1 신호와 무관하게, 상기 제2 신호를 유효하지 않는 것으로 판단하는 단계를 포함할 수 있다.

【0016】 상기 작용형 생체신호 인터페이스의 동작 방법은, 상기 판단 결과, 상기 제2 신호가 유효하면, 활성모드로 동작하여, 상기 제2 신호에 의한 외부 기기의 제어를 허용하는 단계를 더 포함할 수 있다.

【도면의 간단한 설명】

【0017】 도 1은 일실시예에 따른 작용형 생체신호 인터페이스에 대한 구성을 설명하기 위한 도면이다.

도 2는 일실시예에 따라, 제1 신호와 제2 신호에 기초하여, 유효한 제2 신호를 판단하는 일례를 도시하는 도면이다.

도 3은 일실시예에 따라, 제1 신호의 크기와 제2 신호의 크기를 이용하여, 외부 기기에 대한 명령을 수행하는 일례를 도시하는 도면이다.

도 4는 일실시예에 따른 작용형 생체신호 인터페이스의 동작 방법을 구체적으로 도시한 작업 흐름도이다.

【발명을 실시하기 위한 구체적인 내용】

【0018】 이하에서, 실시예들을, 첨부된 도면을 참조하여 상세하게 설명한다. 그러나, 실시 가능한 예시는, 명세서에 기재된 실시예들에 의해 제한되거나 한정되는 것은 아니다. 각 도면에 제시된 동일한 참조 부호는 동일한 부재를 나타낸다.

【0019】 명세서에서 지속적으로 사용되는 센서부는, 측정 대상에 접촉되게 부착되어, 측정 대상에서 발생하는 신호를 획득하거나, 또는 측정 대상의 움직임을 감지하는 센서 또는 센서들의 집합을 지칭할 수 있다.

【0020】예컨대, 센서부는, 팔목, 팔뚝 등의 생체에 부착되어, 상기 생체의 근육 수축 및 근육 이완에 따라 발생하는 신호를 획득하는 생체 센서(biosensor)를 예시할 수 있고, 본 명세서에서는 이를 생체신호 센서부로 명명한다. 상기 생체 센서는, 측정 대상으로부터 생물학적 정보를 감지하여, 신호 파형으로 변환 함으로써 생체신호를 생성할 수 있다. 상기 생체 센서로는 광 센서, 압전(piezoelectric) 센서, 힘 센서, 근전도(EMG) 센서 등을 예시할 수 있다.

【0021】다른 예시에서, 센서부는 생체가 취하는 동작을 직간접적으로 모니터링하여, 상기 생체의 움직임에 관한 신호를 획득하는 가속도 센서, 자이로 센서, 지자계 센서, GPS 센서 등으로 예시할 수 있으며, 본 명세서에서는 이를 모션 센서부로 명명한다.

【0022】일실시예에 따른 착용형 생체신호 인터페이스는, 생체신호 센서부에서 획득한 생체신호를 이용하여 외부 기기를 제어하는 데에 있어, 측정 대상의 움직임에 따른 신호의 크기가 설정된 임계치를 넘어 크게 측정되면, 상기 생체신호를 무시하고, 반면 상기 측정 대상의 움직임에 따른 신호의 크기가 상기 임계치 이하로 작으면 상기 생체신호를 받아들일 수 있다.

【0023】이를 통해, 착용형 생체신호 인터페이스는, 사용자가 조작 의도를 갖고 있지 않고 취한 움직임을 파악하여, 그에 따라 획득한 생체신호를 무시 함으로써, 생체 기반의 사용자 기기 조작 의도를 판단하는데 있어서 정확도의 향상을 도모할 수 있다.

【0024】도 1은 일실시예에 따른 착용형 생체신호 인터페이스에 대한 구성을

설명하기 위한 도면이다.

【0025】 도 1을 참조하여, 착용형 생체신호 인터페이스(100)는 모션 센서부(110), 생체신호 센서부(120) 및 판단 제어부(130)를 포함하여 구성할 수 있다.

【0026】 우선, 모션 센서부(110)는 측정 대상(105)의 움직임에 감지하여 제1 신호를 획득한다. 여기서, 상기 제1 신호는 측정 대상(105)의 운동 상태에 따른 속도 변화에 관한 가속도 신호, 회전 신호, 방위 신호, 좌표 신호 등일 수 있다.

【0027】 모션 센서부(110)는 측정 대상(105)의 동작을 모니터링하고, 상기 동작에 따른 움직임을 인지하는 제1 신호를 수신할 수 있다. 예컨대, 모션 센서부(110)는 후술하는 제2 신호의 획득시 측정 대상(105)에 의해 취해지는 제스처를 식별하고, 상기 식별되는 제스처와 관련한 제1 신호를 획득할 수 있다.

【0028】 예를 들어, 모션 센서부(110)는 가속도 센서를 포함하여 구성할 수 있고, 이 경우, 모션 센서부(110)는 획득한 제1 신호로서의 가속도 신호를, 적분을 통해 속도 성분으로 변환할 수 있다. 모션 센서부(110)는 측정 대상(105)의 움직임 크기를 수치적으로 표시할 수 있다.

【0029】 예컨대, 측정 대상이 팔이고, 상기 팔이 정면으로 접었다 펴지는 경우, 모션 센서부(110)는 상기 움직임으로서 '팔 동작'과 관련한 제1 신호를 획득할 수 있다. 또한, 모션 센서부(110)는 상기 획득한 제1 신호의 크기를 측정하여, 상기 '팔 동작'의 속도 크기를 측정할 수 있다.

【0030】 또한, 모션 센서부(110)는 자이로 센서를 포함하여 구성할 수 있고, 이 경우, 모션 센서부(110)는 상기 자이로 센서를 통해 제1 신호로서의 회전 신호를 측정할 수 있다.

【0031】 예컨대, 측정 대상이 팔이고, 상기 팔이 크게 회전하는 경우, 모션 센서부(110)는 상기 움직임으로서 '팔 회전'과 관련한 제1 신호를 획득할 수 있다. 또한, 모션 센서부(110)는 상기 획득한 제1 신호의 크기를 측정하여, 상기 '팔 회전'의 회전속도 크기를 측정할 수 있다.

【0032】 또한, 모션 센서부(110)는 지자계 센서 또는 GPS 센서를 포함하여 구성할 수 있고, 이 경우, 모션 센서부(110)는 상기 지자계 센서 또는 상기 GPS 센서를 통해 제1 신호로서의 방위 신호 또는 좌표 신호를 측정할 수 있다.

【0033】 실시예에 따라, 모션 센서부(110)는, 후술하는 생체신호 센서부(120)와 함께, 측정 대상(105)을 체결할 수 있는 팔찌 부재에 고정될 수 있다.

【0034】 생체신호 센서부(120)는 측정 대상(105)에 접촉되어, 측정 대상(105)으로부터 제2 신호를 획득한다. 여기서, 상기 제2 신호는 측정 대상(105)에서 발생하는 근육 수축, 근육 이완 등의 생물학적 정보와 관련한 생체신호일 수 있다. 상기 제2 신호는 예컨대 근육과 연관하여 생성되는 생체 전기/자기 신호(Bio-electric/magnetic signals), 생체 임피던스 신호(Bio-impedance signals), 생체 역학 신호(Bio-mechanical signals) 등일 수 있다.

【0035】 생체신호 센서부(120)는 측정 대상(105)과 연관되어, 근육 수축 및 근육 이완 중 적어도 하나에 의해 발생하는 제2 신호를 획득할 수 있다. 예컨대, 생체신호 센서부(120)는 측정 대상(105)으로 팔목이 지정되는 경우, 상기 팔목의 근육으로부터, 팔목 근육의 수축 또는 이완에 따라 생성되는 제2 신호를 감지할 수 있다.

【0036】 여기서, 측정 대상(105)은 예컨대 상기 제2 신호를 발생시키는 근육이 위치하는, 생물체의 팔목, 팔뚝, 얼굴, 목 등을 예시할 수 있다.

【0037】 생체신호 센서부(120)는 측정 대상(105)의 근육으로부터 제2 신호를 감지하는 하나 이상의 생체 센서를 포함하고, 상기 생체 센서를 통해 상기 근육에서의 수축 또는 이완을 모니터링 할 수 있다. 상기 생체 센서로는 광 센서, 압전 (piezo-electric) 센서, 힘 센서, 근전도(EMG) 센서 등을 예시할 수 있다.

【0038】 예를 들어, 생체신호 센서부(120)는 광 센서를 포함하여 구성할 수 있고, 이 경우, 생체신호 센서부(120)는 상기 광 센서를 통해, 광산란에 따른 측정 대상(105)의 상태에 관한 제2 신호를 획득할 수 있다.

【0039】 또한, 생체신호 센서부(120)는 압전 센서 또는 힘 센서를 포함하여 구성할 수 있고, 이 경우, 생체신호 센서부(120)는 상기 압전 센서 또는 힘 센서를 통해, 측정 대상(105)의 신축 상태에 관한 제2 신호를 획득할 수 있다.

【0040】 또한, 생체신호 센서부(120)는 근전도 센서를 포함하여 구성할 수 있고, 이 경우, 생체신호 센서부(120)는 상기 근전도 센서를 통해, 측정 대상(10

5)의 근육에서의 수축 또는 이완에 관한 제2 신호를 획득할 수 있다.

【0041】 실시예에 따라, 생체신호 센서부(120)는, 측정 대상(105)을 체결할 수 있는 팔찌 부재에, 모션 센서부(110)와 함께 고정될 수 있다. 이때, 생체신호 센서부(120)의 센싱면은 측정 대상(105)과 직접 접촉하도록 상기 팔찌 부재에 고정될 수 있다. 상기 센싱면은 측정 대상(105)과 맞닿아 측정 대상(105)에서 발생하는 제2 신호를 감지하는, 하나 이상의 생체 센서가 배열되는 면일 수 있다. 예컨대 팔찌 부재는 생체신호 센서부(120)를 통해, 측정 대상(105)에 관한, 생체 전기/자기 신호, 생체 임피던스 신호, 생체 역학 신호 등의 제2 신호를 광범위하게 감지할 수 있다.

【0042】 상기 팔찌 부재는 예컨대 팔목 근육을 덮는 피부를 둘러 싸듯이 체결하되, 생체신호 센서부(120)가 배열되는 팔찌 부재의 영역(센싱면)이 상기 팔목 근육을 덮도록 하여 체결할 수 있다. 이를 통해 생체신호 센서부(120)는, 하나 이상의 생체 센서를 통해, 측정 대상(105)인 팔목 근육 전 범위에 걸쳐, 상기 제2 신호를 수집할 수 있다.

【0043】 판단 제어부(130)는 상기 제1 신호를 고려하여, 상기 제2 신호에 대한 유효성을 판단한다. 예컨대, 판단 제어부(130)는 생체신호 센서부(120)에서 획득한 상기 제2 신호가, 사용자의 조작 의도에 따라, 측정 대상(105)에서 생성된 것인지, 아니면 사용자의 조작 의도와 무관한 동작에 의해 측정 대상(105)에서 부차적으로 생성된 것인지를 결정할 수 있다.

【0044】 일례에서, 판단 제어부(130)는, 상기 제1 신호의 크기가 선정된 제1

임계치를 만족하여 획득되면, 상기 제2 신호를 유효한 것으로 판단할 수 있다.

【0045】 여기서 제1 임계치는, 조작 의도를 갖는 측정 대상(105)의 움직임을 판별하는 기준값일 수 있고, 예컨대 실제 조작을 위해 사용자가 근육을 움직일 때의, 제1 신호의 크기에 대한 평균적인 값으로 선정할 수 있다.

【0046】 실시예에 따라, 판단 제어부(130)는, 획득되는 상기 제2 신호의 크기에 기초하여, 상기 제1 임계치를 선정할 수 있다. 예컨대, 판단 제어부(130)는 측정 대상(105)에 대해 큰 동작이 수반되는 조작시의 상기 제1 임계치를, 제2 신호의 크기에 비례적으로 크게 선정 함으로써, 조작 의도를 갖고 생성된 제2 신호가, 작고 고정되는 값의 임계치에 의해 무의미하게 무시되는 문제를 보완할 수 있다.

【0047】 일례에서, 판단 제어부(130)는 비교적 큰 제2 신호가 발생되는, 디스플레이 장치에서의 빠른화면전환 조작시에 대한 제1 임계치가, 상대적으로 작은 제2 신호가 발생되는, 조명 장치에서의 On/Off 조작시에 대한 제1 임계치 보다 더 큰 값으로 선정되도록 할 수 있다.

【0048】 제2 신호의 유효성 판단에 있어서, 판단 제어부(130)는, 상기 제1 신호의 크기가 선정된 제1 임계치 이하인 시간을 식별하고, 상기 제2 신호 중 상기 식별된 시간에 획득되는 신호를, 유효한 제2 신호로서 판단할 수 있다. 예컨대, 판단 제어부(130)는 상기 제1 신호의 크기가 선정된 제1 임계치 이하로서, 측정 대상(105)의 움직임이 작은 시간대에서 획득되는 제2 신호를 유효한 것으로 판단할 수 있다.

【0049】 또한, 판단 제어부(130)는, 획득된 상기 제2 신호의 크기가 선정된 제2 임계치를 만족하지 않으면, 상기 제1 신호와 무관하게, 상기 제2 신호를 유효하지 않는 것으로 판단할 수 있다. 여기서 제2 임계치는, 정보로서 가치가 있는 제2 신호를 판별하는 기준값일 수 있고, 예컨대 실제 조작을 위해 사용자가 근육을 움직일 때의, 평균적인 최소값으로 선정할 수 있다.

【0050】 일례에서, 판단 제어부(130)는 획득된 상기 제2 신호의 크기가 선정된 제2 임계치 이하이면, 측정 대상(105)이 조작 의도를 가지고 동작했는지가 불명확하다고 판단하고, 해당 제2 신호를 무효화할 수 있다. 반면, 상기 제2 신호의 크기가 상기 제2 임계치 보다 크면, 판단 제어부(130)는 앞서 설명한 바와 같이, 상기 제1 신호의 크기가 상기 제1 임계치를 만족하는지를 더 판단하여, 해당 제2 신호에 대한 유효성을 결정할 수 있다.

【0051】 또한, 판단 제어부(130)는, 상기 판단 결과, 상기 제2 신호가 유효하면, 활성모드로 동작하여, 상기 제2 신호에 의한 외부 기기의 제어를 허용할 수 있다. 상기 외부 기기는, 예컨대 모바일 기기, TV 화면, DVD, 라디오, 난방온도기, 차량 도어 등, 명령 신호에 의해 제어되는 기기들의 총칭일 수 있다. 상기 명령 신호는, 착용형 생체신호 인터페이스(100)에서, 상기 유효한 제2 신호와 연관되어 생성될 수 있다.

【0052】 예컨대, 판단 제어부(130)는 제1 신호의 크기가 제1 임계치 이하이고, 제2 신호의 크기가 제2 임계치를 초과하여 만족하는 경우에, 착용형 생체신호 인터페이스(100)의 모드를 활성모드로 전환 함으로써, 상기 제2 신호를 이용한 외

부 기기의 조작을 허용하고, 사용자의 조작 의도의 정확성을 향상시킬 수 있게 한다.

【0053】 상기 활성모드로의 전환에 따른 외부 기기의 제어에 있어, 착용형 생체신호 인터페이스(100)는 가전기기(TV, 에어컨, 컴퓨터, 냉장고 등), 또는 생활환경기기(전등, 커튼, 창문, 문 잠금 장치 등)로, 상기 유효한 제2 신호에 의한 명령 신호를 송출하고, 상기 명령 신호에 의해 지정되는 제어 동작(예, on/off 동작, 개폐 동작 등)이 수행되도록 할 수 있다.

【0054】 또한, 착용형 생체신호 인터페이스(100)는 상기 유효한 제2 신호를, 자동차 리모컨의 신호로 활용하여 자동차를 시동시키거나, ID 신호로 활용하여 결제 처리 할 수 있다.

【0055】 또한, 착용형 생체신호 인터페이스(100)는 상기 유효한 제2 신호를, 컴퓨터와의 인터랙션(interaction) 제어에 활용하여, 컴퓨터 내 파일을 편집하거나, 프레젠테이션시의 페이지 이동 등을 수행 할 수 있다.

【0056】 이를 통해, 착용형 생체신호 인터페이스(100)는 조작 의도 없이 취해지는 측정 대상(105)의 움직임에 따라 발생된 생체신호를 무효화 함으로써, 외부 기기에 대한 제어 오류를 최소화할 수 있다.

【0057】 또한, 착용형 생체신호 인터페이스(100)는 측정 대상(105)이 일정 수준으로 안정화된 상태에서 발생하는 생체신호 만을 식별하여 외부 기기 제어에 이용 함으로써, 사용자의 조작 의도를 정확하게 판단 가능하도록 지원할 수 있다.

【0058】 도 2는 일실시예에 따라, 제1 신호와 제2 신호에 기초하여, 유효한 제2 신호를 판단하는 일례를 도시하는 도면이다.

【0059】 우선, 도 2(a)에는, 착용형 생체신호 인터페이스(100)의 모션 센서부(110)에서 획득한 제1 신호를 예시한다. 예컨대, 측정 대상(105)에서, 시간 t_2 내지 t_3 에, 급격한 움직임(예, 팔을 접었다 펴는 동작)이 있다면, 모션 센서부(110)는 상기 시간 t_2 내지 t_3 에서 크기가 크게 측정되는 제1 신호를 획득할 수 있다. 상기 시간 t_2 내지 t_3 에서 획득한 제1 신호는, 조작 의도를 갖는 측정 대상(105)의 움직임을 판별하는 기준값으로서의 제1 임계치를 초과하여 획득될 수 있다.

【0060】 도 2(b)에서는, 착용형 생체신호 인터페이스(100)의 생체신호 센서부(120)에서 획득한 제2 신호를 예시하고 있다. 도 2(b)에 도시한 바와 같이, 제2 신호는, 아날로그의 정현파 형태로 획득되며, 예컨대 사용자가 팔 자체에 힘을 강하게 주는 t_1 내지 t_3 구간에서 보다 큰 파형을 가지고 획득될 수 있다.

【0061】 도 2(c)는, 착용형 생체신호 인터페이스(100)의 판단 제어부(130)에 의해, 상기 획득된 제2 신호의 크기를 신호 처리 함으로써, 크기 비교를 가능하게 하는 그래프를 예시한다. 예컨대, 판단 제어부(130)는 도 2(b)의 제2 신호를 참조하여, 파형이 커지는 t_1 이후 기울기가 급격히 상승하고, 또한 파형이 작아지는 t_3 이후 기울기가 급격히 하강하는 2차 곡선의 그래프를 작성할 수 있다.

【0062】 상기 시간 t_1 내지 t_3 에서의 제2 신호는, 정보로서 가치가 있는 제2 신호를 판별하는 기준값으로서의 제2 임계치를 만족하여 획득될 수 있다.

【0063】 도 2(d)에서는, 판단 제어부(130)에 의해, 제1 신호와 제2 신호를 고려하여, 획득된 제2 신호에 대한 유효성을 판단하는 일례를 예시한다.

【0064】 예컨대, 판단 제어부(130)는 제1 임계치를 초과하여 제1 신호가 획득되는, 시간 t_2 내지 t_3 에서 획득되는 제2 신호를 무효화할 수 있다. 또한, 판단 제어부(130)는 제2 임계치 이하로 제2 신호가 획득되는, 시간 t_1 이하와, 시간 t_3 이상에 대해, 해당 시간 구간들에서 획득되는 제2 신호를 무효화할 수 있다.

【0065】 이에 따라, 판단 제어부(130)는 도 2(d)에 도시한 바와 같이, 시간 t_1 내지 t_2 에서 획득되는 제2 신호를, 유효한 신호로 판단하고, 해당 시간에서 외부 기기의 제어를 위한 명령 신호가 생성되도록 할 수 있다.

【0066】 이를 통해, 착용형 생체신호 인터페이스(100)는, 제2 신호로 예컨대 모바일 기기, TV 화면, DVD, 라디오, 난방온도기, 차량 도어 등의 외부기기를 조작 하되, 사용자의 모션이 작아, 제1 신호의 크기가 제1 임계치 이하로 떨어질 때에만 제2 신호를 이용한 외부 기기의 조작을 허용 함으로써, 사용자의 조작 의도의 정확성을 향상시킬 수 있게 한다.

【0067】 아울러, 착용형 생체신호 인터페이스(100)는 획득된 제2 신호의 크기가 제2 임계치를 만족하여, 상기 제2 신호가 정보로서 가치가 있을 때에, 상기 외부 기기에 대한 조작을 허용할 수 있다.

【0068】 도 3은 일실시예에 따라, 제1 신호의 크기와 제2 신호의 크기를 이용하여, 외부 기기에 대한 명령을 수행하는 일례를 도시하는 도면이다.

【0069】 도 3에 도시한 바와 같이, 착용형 생체신호 인터페이스를 구성하는 모션 센서부(310)는 측정 대상의 움직임을 감지하여 제1 신호를 획득한다(311). 예컨대, 모션 센서부(310)는 측정 대상의 팔이 접었다 펴는 동작에 따른 제1 신호를 획득할 수 있다.

【0070】 이후, 상기 착용형 생체신호 인터페이스는, 모션 센서부(310)에서 획득한 상기 제1 신호를 신호처리하여, 특징을 추출한다(312). 예컨대, 상기 착용형 생체신호 인터페이스는 팔을 접었다 펴는 동작의 속도와 연관된 제1 신호의 크기를, 상기 특징으로 추출할 수 있다.

【0071】 계속해서, 상기 착용형 생체신호 인터페이스는 제1 신호의 크기가 제1 임계치 보다 큰지를 판단한다(313). 예컨대, 상기 착용형 생체신호 인터페이스는 외부 기기의 실제 제어를 위해 사용자가 근육을 움직일 때의, 제1 신호의 크기에 관한 값을 제1 임계치로 선정하여, 상기 획득한 제1 신호의 크기와 비교할 수 있다. 단계(313)에서, 상기 착용형 생체신호 인터페이스는, 상기 제1 신호의 크기가 상기 제1 임계치 이하인 시간을 식별할 수도 있다.

【0072】 만약, 제1 신호의 크기가 제1 임계치 보다 큰 경우(단계(313)에서 Yes 방향), 상기 착용형 생체신호 인터페이스는 해당 제1 신호와 동시간에 획득되는 제2 신호를 무시할 수 있다. 또한, 상기 착용형 생체신호 인터페이스는, 단계

(311)로 리턴하여, 제1 신호를 다시 획득할 수 있다.

【0073】 반면, 제1 신호의 크기가 제1 임계치 이하인 경우(단계(313)에서 No 방향), 상기 착용형 생체신호 인터페이스는 후술하는 단계(323)로 진행하여, 해당 제1 신호와 동시간에 획득되는 제2 신호의 크기와 제2 임계치와의 비교 결과에 따라, 상기 제2 신호를 유효한 것으로 판단할 수 있다.

【0074】 상기 착용형 생체신호 인터페이스를 구성하는 생체신호 센서부(320)는 측정 대상으로부터 제2 신호를 획득한다(321). 예컨대, 생체신호 센서부(320)는 측정 대상의 팔에 강한 힘이 주어짐에 따른 제2 신호를 획득할 수 있다.

【0075】 이후, 상기 착용형 생체신호 인터페이스는, 생체신호 센서부(320)에서 획득한 상기 제2 신호를 신호처리하여, 특징을 추출한다(322). 예컨대, 상기 착용형 생체신호 인터페이스는 팔에 주어진 힘과 연관된 제2 신호의 크기를, 상기 특징으로 추출할 수 있다.

【0076】 계속해서, 상기 착용형 생체신호 인터페이스는 제2 신호의 크기가 제2 임계치 보다 큰지를 판단한다(323). 예컨대, 상기 착용형 생체신호 인터페이스는 정보로서 가치가 있는 제2 신호의 크기에 관한 기준값을 제2 임계치로 선정하여, 상기 획득한 제2 신호의 크기와 비교할 수 있다. 단계(323)에서, 상기 착용형 생체신호 인터페이스는, 앞서 단계(313)에서 식별된 시간(상기 제1 신호의 크기가 상기 제1 임계치 이하인 시간)에 획득된 신호를 유효한 제2 신호로 판단할 수 있다.

【0077】 만약, 제2 신호의 크기가 제2 임계치 이하인 경우(단계(323)에서 No 방향), 상기 작용형 생체신호 인터페이스는 단계(313)의 결과와 무관하게 상기 획득한 제2 신호를 무시할 수 있다. 또한, 상기 작용형 생체신호 인터페이스는, 단계(321)로 리턴하여, 제2 신호를 다시 획득할 수 있다.

【0078】 반면, 제2 신호의 크기가 제2 임계치 보다 큰 경우(단계(323)에서 Yes 방향), 상기 작용형 생체신호 인터페이스는 단계(313)의 결과에 따라, 상기 제2 신호를 최종적으로 유효한 것으로 판단하고, 단계(330)로 진행할 수 있다.

【0079】 상술에서, 모션 센서부(310)에 의한 제1 신호의 획득(311) 이후, 생체신호 센서부(320)에 의한 제2 신호의 획득(321)을 수행하는 것으로 설명하고 있으나, 단계 311과 단계 321은 선후가 변경되어 진행하여도 무방하고, 또한 동시에 진행하여도 무방할 수 있다.

【0080】 상기 작용형 생체신호 인터페이스는 유효한 것으로 판단된 제2 신호에 관련하여 기설정된 명령을 실행한다(330). 단계(330)에서, 상기 작용형 생체신호 인터페이스는 활성모드로 동작하여, 상기 제2 신호에 의한 외부 기기의 제어를 허용할 수 있다. 예컨대, 상기 작용형 생체신호 인터페이스는, 유효한 제2 신호를 이용하여, 모바일 기기, TV 화면, DVD, 라디오, 난방온도기, 차량 도어 등의 외부 기기를 조작할 수 있다.

【0081】 이하, 일실시예에 따른 작용형 생체신호 인터페이스(100)의 동작에 대한 흐름을 상세히 설명한다.

【0082】 도 4는 일실시예에 따른 착용형 생체신호 인터페이스의 동작 방법을 구체적으로 도시한 작업 흐름도이다.

【0083】 우선, 착용형 생체신호 인터페이스(100)는 측정 대상(105)의 움직임을 감지하여 제1 신호를 획득한다(410). 여기서, 상기 제1 신호는 측정 대상(105)의 운동 상태에 따른 속도 변화에 관한 가속도 신호, 회전 신호, 방위 신호, 좌표 신호 등일 수 있다. 단계(410)에서 착용형 생체신호 인터페이스(100)는 모션 센서부(110)를 통해 측정 대상(105)의 동작을 모니터링하고, 상기 동작에 따른 움직임을 인지하기 위한 제1 신호를 수신할 수 있다. 예컨대, 착용형 생체신호 인터페이스(100)는 생체신호의 획득시 측정 대상(105)에 의해 취해지는 제스처를 식별하고, 상기 식별되는 제스처와 관련하는 제1 신호를 획득할 수 있다.

【0084】 예를 들어, 모션 센서부(110)는 가속도 센서를 포함하여 구성할 수 있고, 이 경우, 모션 센서부(110)는 획득한 제1 신호로서의 가속도 신호를, 적분을 통해 속도 성분으로 변환할 수 있다. 착용형 생체신호 인터페이스(100)는 측정 대상(105)의 움직임 크기를 수치적으로 표시할 수 있다.

【0085】 예컨대, 측정 대상이 팔이고, 상기 팔이 정면으로 접었다 펴지는 경우, 착용형 생체신호 인터페이스(100)는 상기 움직임으로서 '팔 동작'과 관련한 제1 신호를 획득할 수 있다. 또한, 착용형 생체신호 인터페이스(100)는 상기 획득한 제1 신호의 크기를 측정하여, 상기 '팔 동작'의 속도 크기를 측정할 수 있다.

【0086】 또한, 모션 센서부(110)는 자이로 센서를 포함하여 구성할 수 있고, 이 경우, 착용형 생체신호 인터페이스(100)는 상기 자이로 센서를 통해 제1 신호로서의 회전 신호를 측정할 수 있다.

【0087】 예컨대, 측정 대상이 팔이고, 상기 팔이 크게 회전하는 경우, 착용형 생체신호 인터페이스(100)는 상기 움직임으로서 '팔 회전'과 관련한 제1 신호를 획득할 수 있다. 또한, 착용형 생체신호 인터페이스(100)는 상기 획득한 제1 신호의 크기를 측정하여, 상기 '팔 회전'의 회전속도 크기를 측정할 수 있다.

【0088】 또한, 모션 센서부(110)는 지자계 센서 또는 GPS 센서를 포함하여 구성할 수 있고, 이 경우, 착용형 생체신호 인터페이스(100)는 상기 지자계 센서 또는 상기 GPS 센서를 통해 제1 신호로서의 방위 신호 또는 좌표 신호를 측정할 수 있다.

【0089】 실시예에 따라, 모션 센서부(110)는, 생체신호 센서부(120)와 함께, 측정 대상(105)을 체결할 수 있는 팔찌 부재에 고정될 수 있다.

【0090】 또한, 착용형 생체신호 인터페이스(100)는 측정 대상(105)에 접촉되어, 측정 대상(105)으로부터 제2 신호를 획득한다(420). 여기서, 상기 제2 신호는 측정 대상(105)에서 발생하는 근육 수축, 근육 이완 등의 생물학적 정보와 관련한 생체신호일 수 있다. 상기 제2 신호는 예컨대 근육과 연관하여 생성되는 생체 전기/자기 신호(Bio-electric/magnetic signals), 생체 임피던스 신호(Bio-impedance signals), 생체 역학 신호(Bio-mechanical signals) 등일 수 있다.

【0091】 단계(420)에서 착용형 생체신호 인터페이스(100)는 생체신호 센서부(120)를 통해 측정 대상(105)과 연관되어, 근육 수축 및 근육 이완 중 적어도 하나에 의해 발생하는 제2 신호를 획득할 수 있다. 예컨대, 착용형 생체신호 인터페이스(100)는 측정 대상(105)으로 팔목이 지정되는 경우, 상기 팔목의 근육으로부터, 팔목 근육의 수축 또는 이완에 따라 생성되는 제2 신호를 감지할 수 있다.

【0092】 여기서, 측정 대상(105)은 예컨대 상기 제2 신호를 발생시키는 근육이 위치하는, 생물체의 팔목, 팔뚝, 얼굴, 목 등을 예시할 수 있다.

【0093】 생체신호 센서부(120)는 측정 대상(105)의 근육으로부터 제2 신호를 감지하는 하나 이상의 생체 센서를 포함하고, 상기 생체 센서를 통해 상기 근육에서의 수축 또는 이완을 모니터링 할 수 있다. 상기 생체 센서로는 광 센서, 압전(piezo-electric) 센서, 힘 센서, 근전도(EMG) 센서 등을 예시할 수 있다.

【0094】 예를 들어, 생체신호 센서부(120)는 광 센서를 포함하여 구성할 수 있고, 이 경우, 착용형 생체신호 인터페이스(100)는 상기 광 센서를 통해, 광산란에 따른 측정 대상(105)의 상태에 관한 제2 신호를 획득할 수 있다.

【0095】 또한, 생체신호 센서부(120)는 압전 센서 또는 힘 센서를 포함하여 구성할 수 있고, 이 경우, 착용형 생체신호 인터페이스(100)는 상기 압전 센서 또는 힘 센서를 통해, 측정 대상(105)의 신축 상태에 관한 제2 신호를 획득할 수 있다.

【0096】 또한, 생체신호 센서부(120)는 근전도 센서를 포함하여 구성할 수 있고, 이 경우, 착용형 생체신호 인터페이스(100)는 상기 근전도 센서를 통해, 측정 대상(105)의 근육에서의 수축 또는 이완에 관한 제2 신호를 획득할 수 있다.

【0097】 실시예에 따라, 생체신호 센서부(120)는, 측정 대상(105)을 체결할 수 있는 팔찌 부재에, 모션 센서부(110)와 함께 고정될 수 있다. 이때, 생체신호 센서부(120)의 센싱면은 측정 대상(105)과 직접 접촉하도록 상기 팔찌 부재에 고정될 수 있다. 상기 센싱면은 측정 대상(105)과 맞닿아 측정 대상(105)에서 발생하는 제2 신호를 감지하는, 하나 이상의 생체 센서가 배열되는 면일 수 있다. 예컨대 팔찌 부재는 생체신호 센서부(120)를 통해, 측정 대상(105)에 관한, 생체 전기/자기 신호, 생체 임피던스 신호, 생체 역학 신호 등의 제2 신호를 광범위하게 감지할 수 있다.

【0098】 상기 팔찌 부재는 예컨대 팔목 근육을 덮는 피부를 둘러 싸듯이 체결하되, 생체신호 센서부(120)가 배열되는 팔찌 부재의 영역(센싱면)이 상기 팔목 근육을 적어도 덮도록 하여 체결할 수 있다. 이를 통해 생체신호 센서부(120)는, 하나 이상의 생체 센서를 통해, 측정 대상(105)인 팔목 근육 전 범위에 걸쳐, 제2 신호를 수집할 수 있다.

【0099】 계속해서, 착용형 생체신호 인터페이스(100)는 상기 제1 신호를 고려하여, 상기 제2 신호에 대한 유효성을 판단한다(430). 단계(430)에서 착용형 생체신호 인터페이스(100)는 획득한 제2 신호가, 사용자의 조작 의도에 따라, 측정 대상(105)에서 생성된 것인지, 아니면 사용자의 조작 의도와 무관한 동작에 의해

측정 대상(105)에서 부차적으로 생성된 것인지를 결정할 수 있다.

【0100】 일례에서, 착용형 생체신호 인터페이스(100)는 상기 제1 신호의 크기가 선정된 제1 임계치를 만족하여 획득되면, 상기 제2 신호를 유효한 것으로 판단할 수 있다.

【0101】 여기서 제1 임계치는, 조작 의도를 갖는 측정 대상(105)의 움직임의 판별하는 기준값일 수 있고, 예컨대 실제 조작을 위해 사용자가 근육을 움직일 때의, 제1 신호의 크기에 대한 평균적인 값으로 선정할 수 있다.

【0102】 실시예에 따라, 착용형 생체신호 인터페이스(100)는 획득되는 상기 제2 신호의 크기에 기초하여, 상기 제1 임계치를 선정할 수 있다. 예컨대, 착용형 생체신호 인터페이스(100)는 측정 대상(105)에 대해 큰 동작이 수반되는 조작시의 상기 제1 임계치를, 제2 신호의 크기에 비례적으로 크게 선정 함으로써, 조작 의도를 갖고 생성된 제2 신호가, 작고 고정되는 값의 임계치에 의해 무의미하게 무시되는 문제를 보완할 수 있다.

【0103】 일례에서, 착용형 생체신호 인터페이스(100)는 비교적 큰 제2 신호가 발생하는, 디스플레이 장치에서의 빠른화면전환 조작시에 대한 제1 임계치가, 상대적으로 작은 제2 신호가 발생하는, 조명 장치에서의 On/Off 조작시에 대한 제1 임계치 보다 더 큰 값으로 선정되도록 할 수 있다.

【0104】 제2 신호의 유효성 판단에 있어서, 착용형 생체신호 인터페이스(100)는, 상기 제1 신호의 크기가 선정된 제1 임계치 이하인 시간을 식별하고, 상

기 제2 신호 중 상기 식별된 시간에 획득되는 신호를, 유효한 제2 신호로서 판단할 수 있다. 예컨대, 착용형 생체신호 인터페이스(100)는 상기 제1 신호의 크기가 선정된 제1 임계치 이하로서, 측정 대상(105)의 움직임이 작은 시간대에서 획득되는 제2 신호를 유효한 것으로 판단할 수 있다.

【0105】 또한, 착용형 생체신호 인터페이스(100)는 획득된 상기 제2 신호의 크기가 선정된 제2 임계치를 만족하지 않으면, 상기 제1 신호와 무관하게, 상기 제2 신호를 유효하지 않는 것으로 판단할 수 있다. 여기서 제2 임계치는, 정보로서 가치가 있는 제2 신호를 판별하는 기준값일 수 있고, 예컨대 실제 조작을 위해 사용자가 근육을 움직일 때의, 평균적인 최소값으로 선정할 수 있다.

【0106】 일례에서, 착용형 생체신호 인터페이스(100)는 획득된 상기 제2 신호의 크기가 선정된 제2 임계치 이하이면, 측정 대상(105)이 조작 의도를 가지고 동작했는지가 불명확하다고 판단하고, 해당 제2 신호를 무효화할 수 있다. 반면, 상기 제2 신호의 크기가 상기 제2 임계치 보다 크면, 착용형 생체신호 인터페이스(100)는 앞서 설명한 바와 같이, 상기 제1 신호의 크기가 상기 제1 임계치를 만족하는지를 더 판단하여, 해당 제2 신호에 대한 유효성을 결정할 수 있다.

【0107】 또한, 착용형 생체신호 인터페이스(100)는 상기 판단 결과, 상기 제2 신호가 유효하면, 활성모드로 동작하여, 상기 제2 신호에 의한 외부 기기의 제어를 허용할 수 있다. 상기 외부 기기는, 예컨대 모바일 기기, TV 화면, DVD, 라디오, 난방온도기, 차량 도어 등, 명령 신호에 의해 제어되는 기기들의 총칭일 수 있다. 상기 명령 신호는, 착용형 생체신호 인터페이스(100)에서, 상기 유효한 제2

신호와 연관되어 생성될 수 있다.

【0108】예컨대, 착용형 생체신호 인터페이스(100)는 제1 신호의 크기가 제1 임계치 이하이고, 제2 신호의 크기가 제2 임계치를 초과하여 만족하는 경우에, 착용형 생체신호 인터페이스(100)의 모드를 활성모드로 전환 함으로써, 상기 제2 신호를 이용한 외부 기기의 조작을 허용하고, 사용자의 조작 의도의 정확성을 향상시킬 수 있게 한다.

【0109】상기 활성모드로의 전환에 따른 외부 기기의 제어에 있어, 착용형 생체신호 인터페이스(100)는 가전기기(TV, 에어컨, 컴퓨터, 냉장고 등), 또는 생활환경기기(전등, 커튼, 창문, 문 잠금 장치 등)로, 상기 유효한 제2 신호에 의한 명령 신호를 송출하고, 상기 명령 신호에 의해 지정되는 제어 동작(예, on/off 동작, 개폐 동작 등)이 수행되도록 할 수 있다.

【0110】또한, 착용형 생체신호 인터페이스(100)는 상기 유효한 제2 신호를, 자동차 리모컨의 신호로 활용하여 자동차를 시동시키거나, ID 신호로 활용하여 결제 처리 할 수 있다.

【0111】또한, 착용형 생체신호 인터페이스(100)는 상기 유효한 제2 신호를, 컴퓨터와의 인터랙션(interaction) 제어에 활용하여, 컴퓨터 내 파일을 편집하거나, 프레젠테이션시의 페이지 이동 등을 수행 할 수 있다.

【0112】이를 통해, 착용형 생체신호 인터페이스(100)는 조작 의도 없이 취해지는 측정 대상(105)의 움직임에 따라 발생된 생체신호를 무효화 함으로써, 외부

기기에 대한 제어 오류를 최적하게 회피할 수 있다.

【0113】 또한, 착용형 생체신호 인터페이스(100)는 측정 대상(105)이 일정 수준으로 안정화된 상태에서 발생하는 생체신호 만을 식별하여 외부 기기 제어에 이용 함으로써, 사용자의 조작 의도를 정확하게 판단 가능하도록 지원할 수 있다.

【0114】 실시예에 따른 방법은 다양한 컴퓨터 수단을 통하여 수행될 수 있는 프로그램 명령 형태로 구현되어 컴퓨터 판독 가능 매체에 기록될 수 있다. 상기 컴퓨터 판독 가능 매체는 프로그램 명령, 데이터 파일, 데이터 구조 등을 단독으로 또는 조합하여 포함할 수 있다. 상기 매체에 기록되는 프로그램 명령은 실시예를 위하여 특별히 설계되고 구성된 것들이거나 컴퓨터 소프트웨어 당업자에게 공지되어 사용 가능한 것일 수도 있다. 컴퓨터 판독 가능 기록 매체의 예에는 하드 디스크, 플로피 디스크 및 자기 테이프와 같은 자기 매체(magnetic media), CD-ROM, DVD와 같은 광기록 매체(optical media), 플롭티컬 디스크(floptical disk)와 같은 자기-광 매체(magneto-optical media), 및 롬(ROM), 램(RAM), 플래시 메모리 등과 같은 프로그램 명령을 저장하고 수행하도록 특별히 구성된 하드웨어 장치가 포함된다. 프로그램 명령의 예에는 컴파일러에 의해 만들어지는 것과 같은 기계어 코드 뿐만 아니라 인터프리터 등을 사용해서 컴퓨터에 의해서 실행될 수 있는 고급 언어 코드를 포함한다. 상기된 하드웨어 장치는 실시예의 동작을 수행하기 위해 하나 이상의 소프트웨어 모듈로서 작동하도록 구성될 수 있으며, 그 역도 마찬가지이다.

【0115】 이상과 같이 실시예들이 비록 한정된 실시예와 도면에 의해 설명되었으나, 해당 기술분야에서 통상의 지식을 가진 자라면 상기의 기재로부터 다양한 수정 및 변형이 가능하다. 예를 들어, 설명된 기술들이 설명된 방법과 다른 순서로 수행되거나, 및/또는 설명된 시스템, 구조, 장치, 회로 등의 구성요소들이 설명된 방법과 다른 형태로 결합 또는 조합되거나, 다른 구성요소 또는 균등물에 의하여 대치되거나 치환되더라도 적절한 결과가 달성될 수 있다.

【0116】 그러므로, 다른 구현들, 다른 실시예들 및 특허청구범위와 균등한 것들도 후술하는 특허청구범위의 범위에 속한다.

【부호의 설명】

【0117】 100 : 착용형 생체신호 인터페이스

110 : 모션 센서부

120 : 생체신호 센서부

130 : 판단 제어부

【특허청구범위】

【청구항 1】

측정 대상의 움직임을 감지하여 제1 신호를 획득하는 모션 센서부;

상기 측정 대상에 접촉되어 상기 측정 대상으로부터 제2 신호를 획득하는 생체신호 센서부; 및

상기 제1 신호를 고려하여, 상기 제2 신호에 대한 유효성을 판단하는 판단 제어부

를 포함하는 착용형 생체신호 인터페이스.

【청구항 2】

제1항에 있어서,

상기 판단 제어부는,

상기 제1 신호의 크기가 선정된 제1 임계치를 만족하여 획득되면, 상기 제2 신호를 유효한 것으로 판단하는

착용형 생체신호 인터페이스.

【청구항 3】

제1항에 있어서,

상기 판단 제어부는,

상기 제1 신호의 크기가 선정된 제1 임계치 이하인 시간을 식별하고, 상기 제2 신호 중 상기 식별된 시간에 획득되는 신호를, 유효한 제2 신호로서 판단하는

착용형 생체신호 인터페이스.

【청구항 4】

제2항 또는 제3항에 있어서,

상기 판단 제어부는,

획득되는 상기 제2 신호의 크기에 기초하여, 상기 제1 임계치를 선정하는

착용형 생체신호 인터페이스.

【청구항 5】

제1항에 있어서,

상기 판단 제어부는,

획득된 상기 제2 신호의 크기가 선정된 제2 임계치를 만족하지 않으면, 상기 제1 신호와 무관하게, 상기 제2 신호를 유효하지 않는 것으로 판단하는

착용형 생체신호 인터페이스.

【청구항 6】

제1항에 있어서,

상기 판단 제어부는,

상기 판단 결과, 상기 제2 신호가 유효하면, 활성모드로 동작하여, 상기 제2 신호에 의한 외부 기기의 제어를 허용하는

착용형 생체신호 인터페이스.

【청구항 7】

모션 센서부에 의해, 측정 대상의 움직임을 감지하여 제1 신호를 획득하는 단계;

상기 측정 대상에 접촉되는 생체신호 센서부에 의해, 상기 측정 대상으로부터 제2 신호를 획득하는 단계; 및

상기 제1 신호를 고려하여, 상기 제2 신호에 대한 유효성을 판단하는 단계를 포함하는 착용형 생체신호 인터페이스의 동작 방법.

【청구항 8】

제7항에 있어서,

상기 제2 신호에 대한 유효성을 판단하는 단계는,

상기 제1 신호의 크기가 선정된 제1 임계치를 만족하여 획득되면, 상기 제2 신호를 유효한 것으로 판단하는 단계

를 포함하는 착용형 생체신호 인터페이스의 동작 방법.

【청구항 9】

제7항에 있어서,

상기 제2 신호에 대한 유효성을 판단하는 단계는,

상기 제1 신호의 크기가 선정된 제1 임계치 이하인 시간을 식별하는 단계;

및

상기 제2 신호 중 상기 식별된 시간에 획득되는 신호를, 유효한 제2 신호로서 판단하는 단계

를 포함하는 착용형 생체신호 인터페이스의 동작 방법.

【청구항 10】

제8항 또는 제9항에 있어서,

상기 제2 신호에 대한 유효성을 판단하는 단계는,

획득되는 상기 제2 신호의 크기에 기초하여, 상기 제1 임계치를 선정하는 단계

를 더 포함하는 착용형 생체신호 인터페이스의 동작 방법.

【청구항 11】

제7항에 있어서,

상기 제2 신호에 대한 유효성을 판단하는 단계는,

획득된 상기 제2 신호의 크기가 선정된 제2 임계치를 만족하지 않으면, 상기 제1 신호와 무관하게, 상기 제2 신호를 유효하지 않는 것으로 판단하는 단계

를 포함하는 착용형 생체신호 인터페이스의 동작 방법.

【청구항 12】

제7항에 있어서,

상기 판단 결과, 상기 제2 신호가 유효하면, 활성모드로 동작하여, 상기 제2 신호에 의한 외부 기기의 제어를 허용하는 단계

를 더 포함하는 착용형 생체신호 인터페이스의 동작 방법.

【요약서】

【요약】

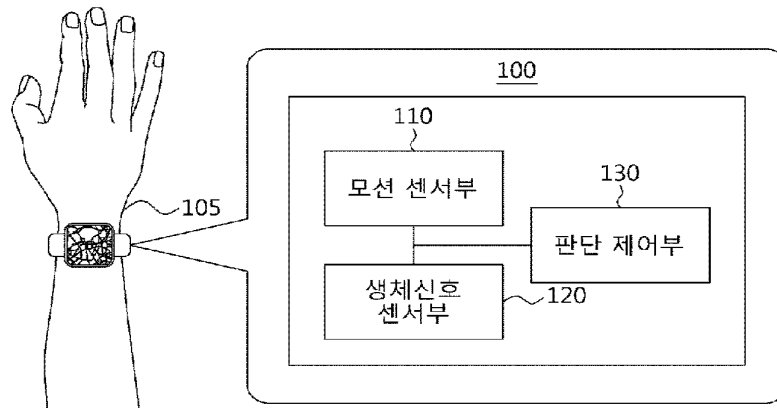
착용형 생체신호 인터페이스 및 착용형 생체신호 인터페이스의 동작 방법을 제공한다. 일실시예에 따른 착용형 생체신호 인터페이스는, 사용자가 조작 의도를 갖고 있지 않고 취한 움직임을 파악하여, 그에 따라 획득한 생체신호를 무시 함으로써, 생체 기반의 사용자 기기 조작 의도를 판단하는데 있어서 정확도의 향상을 도모할 수 있다.

【대표도】

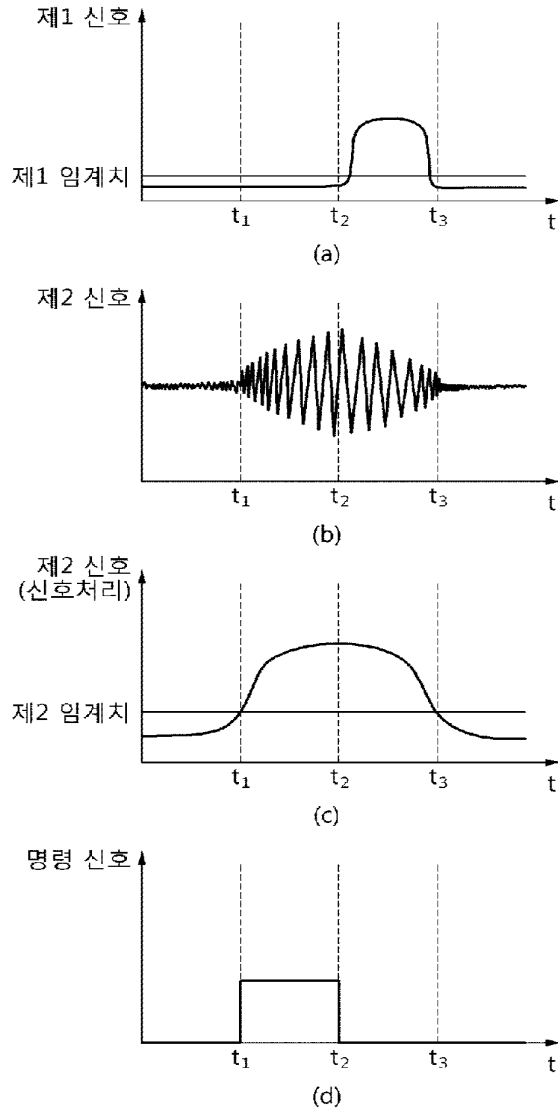
도 1

【도면】

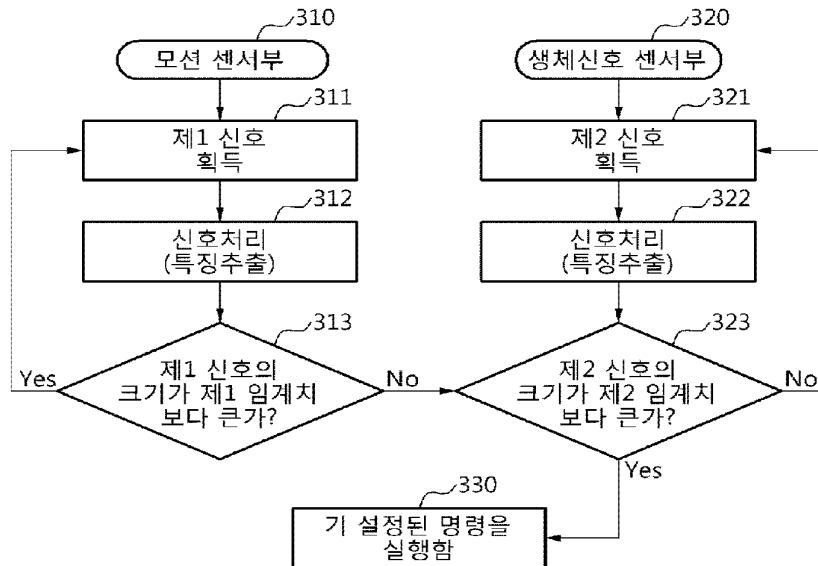
【도 1】



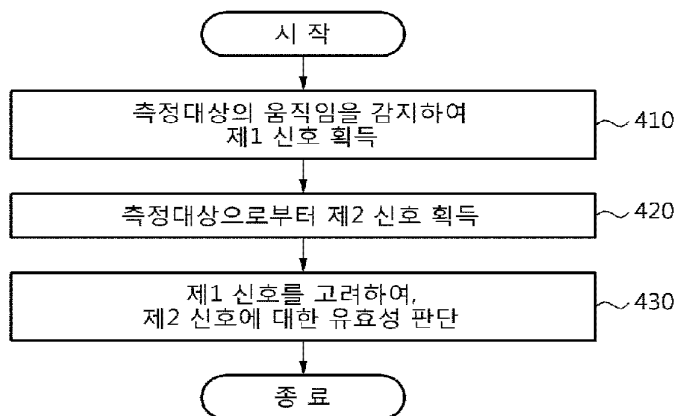
【도 2】



【도 3】



【도 4】



IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re the Application of:

Chang Mok CHOI

Application No. 16/267,752

Art Unit: 2694

Confirmation No. 9126

Filed: February 5, 2019

Examiner: Gustavo D. Polo

For: WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF

PRELIMINARY AMENDMENT PURSUANT TO 37 C.F.R. § 1.115

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

Prior to examination on the merits, please amend the above-identified application as follows:

Amendments to the claims are reflected in the listing of the claims that begins on page **2** of this Paper.

Remarks begin on page **6** of this Paper.

Electronic Acknowledgement Receipt

EFS ID:	35380202
Application Number:	16267752
International Application Number:	
Confirmation Number:	9126
Title of Invention:	WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF
First Named Inventor/Applicant Name:	Chang Mok CHOI
Customer Number:	89980
Filer:	Thomas Joseph Matikainen
Filer Authorized By:	
Attorney Docket Number:	012052.0799C1
Receipt Date:	11-MAR-2019
Filing Date:	05-FEB-2019
Time Stamp:	13:35:44
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		NewContApp_0120520799C1_PreliminaryAmendmentAsFiled.pdf	37487 7f9d0b4460d1c86651c081a226ebc7e1f356a0d0	yes	6

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Applicant Arguments/Remarks Made in an Amendment	6	6
	Claims	2	5
	Preliminary Amendment	1	1

Warnings:

Information:

Total Files Size (in bytes):	37487
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

REMARKS

In accordance with the foregoing, claim 1 has been amended and claims 7-20 have been added. Accordingly, upon entry of the amendments, claims 1-20 are pending, with claims 1, 8, and 15 being the independent claims.

No new matter is presented in the amendments and newly-added claims. Applicant has amended and added claims solely for the purpose of furthering the prosecution of a disclosed embodiment. Applicant reserves the right to pursue other claims, including broader claims or embodiments, in a continuation application.

Entry of the amendments and favorable early consideration of the present application are respectfully requested. The USPTO is directed and authorized to charge all required fees, except for the Issue Fee and the Publication Fee, to Deposit Account No. 50-5113. Please also credit any overpayments to said Deposit Account.

Respectfully submitted,

Date: March 11, 2019

/Thomas J. Matikainen/

Thomas J. Matikainen

Registration No. 73,291

NSIP Law
1120 Connecticut Avenue, NW, Suite 304
Washington, DC 20036
Telephone (202) 429-0020 x123
Facsimile (202) 315-3758
TJM

IN THE CLAIMS

This listing of the claims replaces all prior versions and listings of the claims in this application.

The text of all pending claims (including any withdrawn claims) is set forth below. Canceled and not entered claims are indicated with claim number and status only. The claims as listed below show added text with underlining and deleted text with ~~striketrough~~ or in [[double brackets]]. The status of each claim is listed with one of (Original), (Currently amended), (Canceled), (Withdrawn), (Previously presented), (New), and (Not entered).

Please **AMEND** claim 1 and **ADD** claims 7-20 in accordance with the following:

1. (Currently amended) A wearable device comprising:

a motion sensor;

a biological sensor; and

a controller configured to:

obtain a movement of an object using the motion sensor, and a biological signal of the object using the biological sensor, and

~~wherein the controller is configured to perform a predetermined~~ specified function using at least a portion of the biological signal obtained during a specified period that the movement corresponds to a specified condition.

2. (Original) The wearable device of claim 1,

wherein the controller is configured to perform, in response to the movement being less than or equal to a first threshold value, the wearable device using the biological signal.

3. (Original) The wearable device of claim 2, wherein the controller is configured to determine the first threshold value based on a magnitude of the biological signal.

4. (Original) The wearable device of claim 1, wherein the controller is configured to determine the biological signal to be invalid, irrespective of the movement, in response to the magnitude of the biological signal not satisfying a second threshold value.

5. (Original) The wearable device of claim 1, wherein, response to a determination that the biological signal is valid, the controller is configured to operate in an active mode and to control an external device based on the biological signal.

6. (Original) The wearable device of claim 1, wherein the controller configured to determine, based on an analysis of the movement, whether to perform the function of the wearable device using the biological signal.

7. (New) The wearable device of claim 1, further comprising a transmission unit, wherein the controller configured to transmit an instruction associated with an external electronic device to the external electronic device using the transmission unit as at least part of the specified function.

8. (New) A wearable device comprising:

a motion sensor;

a biological sensor; and

a controller configured to:

obtain, with respect to an object external to the wearable device, a movement of the object using the motion sensor and a biological signal of the object using the biological sensor;

identify a period that the movement is smaller than a specified condition; and

perform a specified function using at least portion of the biological signal corresponding to the period.

9. (New) The wearable device of claim 8, wherein the biological sensor comprises an optical sensor to obtain an optical signal related to the biological signal.

10. (New) The wearable device of claim 8, wherein the controller is configured to adjust the specified condition based at least in part on a magnitude of the biological signal prior to the identifying of the period.

11. (New) The wearable device of claim 8, wherein the controller is configured to:
determine the biological signal as a valid signal if that a magnitude of the biological signal falls into a first specified range; and

determine the biological signal as an invalid signal if that a magnitude of the biological signal falls into a second specified range.

12. (New) The wearable device of claim 8, wherein the controller is configured to generate an instruction using the at least portion of the biological signal as at least part of the performing.

13. (New) The wearable device of claim 12, wherein the controller is configured to control an external electronic device based at least in part on the instruction as at least part of the specified function.

14. (New) The wearable device of claim 13, further comprising a transmission unit, wherein the controller configured to transmit the instruction to the external electronic device using the transmission unit.

15. (New) A wearable device comprising:
a motion sensor;
a biological sensor; and
a controller configured to:
obtain, with respect to an object external to the wearable device, a movement of the object using the motion sensor and a biological signal of the object using the biological sensor;
determine, at least a portion of the biometric signal corresponding to a period that the movement is smaller than a specified condition as a valid signal; and
perform a specified function using the valid signal.

16. (New) The wearable device of claim 15, wherein the biological sensor comprises an optical sensor to obtain an optical signal related to the biological signal.

17. (New) The wearable device of claim 15, wherein the controller is configured to adjust the specified condition based at least in part on a magnitude of the biological signal prior to the identifying of the period.

18. (New) The wearable device of claim 15, wherein the controller configured to generate an instruction using the at least portion of the valid signal as at least part of the performing.

19. (New) The wearable device of claim 18, wherein the controller configured to control an external electronic device based at least in part on the instruction as at least part of the specified function.

20. (New) The wearable device of claim 19, further comprising a transmission unit, wherein the controller configured to transmit the instruction to the external device using the transmission unit.

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 16/267,752		Filing Date 02/05/2019		☐ To be Mailed					
ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO													
APPLICATION AS FILED - PART I													
		(Column 1)	(Column 2)										
FOR		NUMBER FILED	NUMBER EXTRA		RATE (\$)		FEE (\$)						
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A	N/A		N/A								
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))		N/A	N/A		N/A								
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A	N/A		N/A								
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 = *		x \$100 =									
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 = *		x \$460 =									
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).											
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))													
* If the difference in column 1 is less than zero, enter "0" in column 2.					TOTAL								
APPLICATION AS AMENDED - PART II													
		(Column 1)	(Column 2)		(Column 3)								
AMENDMENT	03/11/2019	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)						
	Total (37 CFR 1.16(i))	* 20	Minus	** 20	= 0	x \$100 =	0						
	Independent (37 CFR 1.16(h))	* 2	Minus	*** 3	= 0	x \$460 =	0						
	☐ Application Size Fee (37 CFR 1.16(s))												
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))												
							TOTAL ADD'L FEE		0				
		(Column 1)	(Column 2)		(Column 3)								
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)						
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$0 =							
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$0 =							
	☐ Application Size Fee (37 CFR 1.16(s))												
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))												
							TOTAL ADD'L FEE						
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.							SLIE						
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".							/APRIL L. WALKER/						
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".													
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.													

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
16/267,752	02/05/2019	Chang Mok CHOI	012052.0799C1

CONFIRMATION NO. 9126

89980
NSIP LAW
P.O. Box 65745
Washington, DC 20035

PUBLICATION NOTICE



OC000000108726837

Title:WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF

Publication No.US-2019-0171288-A1

Publication Date:06/06/2019

NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

The publication may be accessed through the USPTO's publically available Searchable Databases via the Internet at www.uspto.gov. The direct link to access the publication is currently <http://www.uspto.gov/patft/>.

The publication process established by the Office does not provide for mailing a copy of the publication to applicant. A copy of the publication may be obtained from the Office upon payment of the appropriate fee set forth in 37 CFR 1.19(a)(1). Orders for copies of patent application publications are handled by the USPTO's Public Records Division. The Public Records Division can be reached by telephone at (571) 272-3150 or (800) 972-6382, by facsimile at (571) 273-3250, by mail addressed to the United States Patent and Trademark Office, Public Records Division, Alexandria, VA 22313-1450 or via the Internet.

In addition, information on the status of the application, including the mailing date of Office actions and the dates of receipt of correspondence filed in the Office, may also be accessed via the Internet through the Patent Electronic Business Center at www.uspto.gov using the public side of the Patent Application Information and Retrieval (PAIR) system. The direct link to access this status information is currently <https://portal.uspto.gov/pair/PublicPair>. Prior to publication, such status information is confidential and may only be obtained by applicant using the private side of PAIR.

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Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

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Application	Document	Mailroom Date	Attorney Docket No.
16267752	NTC.PUB	06/06/2019	012052.0799C1

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/267,752	02/05/2019	Chang Mok CHOI	012052.0799C1	9126
89980	7590	06/21/2019	EXAMINER	
NSIP LAW			POLO, GUSTAVO D	
P.O. Box 65745				
Washington, DC 20035				
			ART UNIT	PAPER NUMBER
			2694	
			NOTIFICATION DATE	DELIVERY MODE
			06/21/2019	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

pto.nsip@gmail.com
pto@nsiplaw.com

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Claim Rejections - 35 USC § 103

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

Claims 1-20 is/are rejected under 35 U.S.C. 103 as being unpatentable over Park et al. Pub. No. US 2010/0145171 A1 [Park] in view of Kangas et al. Pub. No. US 2012/0157789 A1 [Kangas].

1. Park discloses a wearable device [Fig. 1] comprising: a motion sensor [Fig. 2, 150]; a biological sensor [130]; and a controller [110] configured to; obtain a movement of an object using the motion sensor [¶ 50], and a biological signal of the object using the biological sensor [¶ 47]. Park is silent on perform a specified function using at least a portion of the biological signal obtained during a specified period that the movement corresponds to a specified condition. Kangas teaches perform a specified function using at least a portion of the biological signal obtained during a specified period [¶

96 bio-signal information which is stored in the bio-signal profile 25 may comprise data which has been collected, over an extended period of time] that the movement corresponds to a specified condition [¶ 110 by collecting data over a period of time and storing it in a bio-signal profile 25 the outputs provided are personal to the user of the apparatus and may be optimized to make the apparatus as convenient as possible for the user]. Therefore it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify Park with Kangas as required by this claim, since such a modification improves the performance of the device.

2. Park in view of Kangas teaches wherein the controller is configured to perform, in response to the movement being less than or equal to a first threshold value, the wearable device using the biological signal [Kangas ¶ 74 upper or lower limit].

3. Park in view of Kangas teaches wherein the controller is configured to determine the first threshold value based on a magnitude of the biological signal [Kangas ¶ 74 mean or median value].

4. Park in view of Kangas teaches wherein the controller is configured to determine the biological signal to be invalid, irrespective of the movement, in response to the magnitude of the biological signal not satisfying a second threshold value [Kangas ¶ 102 different threshold values].

5. Park in view of Kangas teaches wherein, response to a determination that the biological signal is valid, the controller is configured to operate in an active mode and to control an external device based on the biological signal [Kangas ¶ 102 gaming mode].

6. Park in view of Kangas teaches wherein the controller configured to determine, based on an analysis of the movement, whether to perform the function of the wearable device using the biological signal [Kangas ¶ 102].

7. Park in view of Kangas teaches a transmission unit, wherein the controller configured to transmit an instruction associated with an external electronic device to the external electronic device using the transmission unit as at least part of the specified function [Park Fig. 2, 160].

8. Park teaches a wearable device [Fig. 1] comprising: a motion sensor [Fig. 2, 150]; a biological sensor [130]; and a controller [110] configured to: obtain, with respect to an object external to the wearable device [¶ 50], a movement of the object using the motion sensor and a biological signal of the object using the biological sensor [¶ 47]. Park is silent on identify a period that the movement is smaller than a specified condition; and perform a specified function using at least portion of the biological signal corresponding to the period. Kangas teaches identify a period that the movement is smaller than a specified condition [¶ 96 where the movement is any movement that is readable]; and perform a specified function using at least portion of the biological signal corresponding to the period [¶ 110 by collecting data over a period of time and storing it in a bio-signal profile 25 the outputs provided are personal to the user of the apparatus and may be optimized to make the apparatus as convenient as possible for the user]. Therefore it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify Park with Kangas as required by this claim, since such a modification improves the performance of the device.

9. Park in view of Kangas teaches wherein the biological sensor comprises an optical sensor to obtain an optical signal related to the biological signal [Park ¶ 21].

10. Park in view of Kangas teaches wherein the controller is configured to adjust the specified condition based at least in part on a magnitude of the biological signal prior to the identifying of the period [Kangas ¶ 101].

11. Park in view of Kangas teaches wherein the controller is configured to: determine the biological signal as a valid signal if that a magnitude of the biological signal falls into a first specified range [Kangas ¶ 101 average value]; and determine the biological signal as an invalid signal if that a magnitude of the biological signal falls into a second specified range [¶ 102 different threshold values].

12. Park in view of Kangas teaches wherein the controller is configured to generate an instruction using the at least portion of the biological signal as at least part of the performing [Kangas ¶ 96].

13. Park in view of Kangas teaches wherein the controller is configured to control an external electronic device based at least in part on the instruction as at least part of the specified function [Kangas ¶ 96].

14. Park in view of Kangas teaches a transmission unit, wherein the controller configured to transmit the instruction to the external electronic device using the transmission unit [Park Fig. 2, 160].

15. Park discloses a wearable device [Fig. 1] comprising: a motion sensor [Fig. 2, 150]; a biological sensor [130]; and a controller [110] configured to: obtain, with respect to an object external to the wearable device, a movement of the object using the motion sensor [¶ 50] and a biological signal of the object using the biological sensor [¶ 47]. Park is silent on determine, at least a portion of the biometric signal corresponding to a period that the movement is smaller than a specified condition as a valid signal; and perform a specified function using the valid signal. Kangas teaches determine, at least a portion of the biometric signal corresponding to a period that the movement is smaller than a specified condition as a valid signal [¶ 96]; and perform a specified function using the valid signal [¶ 110].

Therefore it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify Park with Kangas as required by this claim, since such a modification improves the performance of the device.

16. Park in view of Kangas teaches wherein the biological sensor comprises an optical sensor to obtain an optical signal related to the biological signal [Park ¶ 21].

17. Park in view of Kangas teaches wherein the controller is configured to adjust the specified condition based at least in part on a magnitude of the biological signal prior to the identifying of the period [Kangas ¶ 101].

18. Park in view of Kangas teaches wherein the controller configured to generate an instruction using the at least portion of the valid signal as at least part of the performing [Kangas ¶ 96].

19. Park in view of Kangas teaches wherein the controller configured to control an external electronic device based at least in part on the instruction as at least part of the specified function [Kangas ¶ 96].

20. Park in view of Kangas teaches a transmission unit, wherein the controller configured to transmit the instruction to the external device using the transmission unit [Park Fig. 2, 160].

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to GUSTAVO POLO whose telephone number is (571)270-7613. The examiner can normally be reached on Mon-Fri 9am-5pm PT.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Patrick Edouard can be reached on (571) 272-7603. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Art Unit: 2694

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Gustavo Polo/
Primary Examiner, Art Unit 2694

<i>Notice of References Cited</i>	Application/Control No. 16/267,752		Applicant(s)/Patent Under Reexamination CHOI et al.	
	Examiner GUSTAVO POLO		Art Unit 2694	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-20120157789-A1	06-2012	Kangas; Jari	A61B5/7228	600/300
	B					
	C					
	D					
	E					
	F					
	G					
	H					
	I					
	J					
	K					
	L					
	M					

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
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Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

<i>Search Notes</i> 	Application/Control No. 16/267,752	Applicant(s)/Patent Under Reexamination CHOI et al.
	Examiner GUSTAVO POLO	Art Unit 2694

CPC - Searched*		
Symbol	Date	Examiner
g06f	06/17/2019	gp

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
inventor/text	06/17/2019	gp

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner

/Gustavo Polo/ Primary Examiner, Art Unit 2694	
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Bibliographic Data

Application No: 16/267,752

Foreign Priority claimed: ☒ Yes ☐ No

35 USC 119 (a-d) conditions met: ☒ Yes ☐ No ☐ Met After Allowance

Verified and Acknowledged:

/Gustavo Polo/

Examiner's Signature

Initials

Title:

WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
02/05/2019	345	2694	012052.0799C1
RULE			

APPLICANTS

Samsung Electronics Co., Ltd., Suwon-si, KOREA, REPUBLIC OF

INVENTORS

Chang Mok CHOI Suwon-si, KOREA, REPUBLIC OF

Sang Joon KIM Suwon-si, KOREA, REPUBLIC OF

Seung Keun YOON Suwon-si, KOREA, REPUBLIC OF

CONTINUING DATA

This application is a CON of 14913085 02/19/2016 PAT 10241574

14913085 is a 371 of PCT/KR2014/004073 05/08/2014

FOREIGN APPLICATIONS

KOREA, REPUBLIC OF 10-2013-0098476 08/20/2013

IF REQUIRED, FOREIGN LICENSE GRANTED**

02/21/2019

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P.O. Box 65745

Washington, DC 20035

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FILING FEE RECEIVED

\$1,720

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	761	(bio-signal\$4 biosignal\$4 (biologic\$5 near2 (signal data))) with period	US-PGPUB; USPAT	OR	ON	2019/06/17 11:33
L2	1219871	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2019/06/17 11:33
L3	125	L1 and L2	US-PGPUB; USPAT	OR	ON	2019/06/17 11:33
L4	73	(wearable watch) and L3	US-PGPUB; USPAT	OR	ON	2019/06/17 11:33
L6	5	(US-20030078505-\$ or US-20090326398-\$ or US-20120157789-\$ or US-20130096397-\$ or US-20130109997-\$).did.	US-PGPUB	OR	ON	2019/06/17 12:05
L7	5	4 and 6	US-PGPUB; USPAT	OR	ON	2019/06/17 12:05
L8	15	(US-20100145171-\$ or US-20130328761-\$ or US-20140015750-\$ or US-20140258208-\$ or US-20140365979-\$ or US-20040042637-\$ or US-20130151730-\$ or US-20140050354-\$ or US-20030078505-\$ or US-20090326398-\$ or US-20120157789-\$ or US-20130096397-\$ or US-20130109997-\$).did. or (US-5954662-\$ or US-5832296-\$).did.	US-PGPUB; USPAT	OR	ON	2019/06/17 12:35
L9	10	8 and threshold	US-PGPUB; USPAT	OR	ON	2019/06/17 12:35
L10	7	((("5832296") or ("20060149146") or ("20060281979") or ("20100145171") or ("20120172733") or ("20120184826") or ("20120259182")).PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2019/06/17 13:23
L19	25913	L15 L16 L17 L18	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
L20	139	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
L21	21	L19 and L20	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
L22	1219871	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
L23	249799	(acceleration motion) near2 (sensor	US-	OR	ON	2019/06/17

		detector)	PGPUB; USPAT			13:25
L24	64689	(watch wearable attachable) and L23	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L25	1689	(bio-signal biosignal) and L24	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L26	1218439	(second two) near2 (signal wave)	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L27	241	L25 and L26	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L28	28769	(wearable watch) with interfac\$3	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L38	25913	L34 L35 L36 L37	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L39	139	((bio-signal biosignal) and motion).clm.	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L40	21	L38 and L39	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L41	1219871	g06f\$.cpc.	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L42	249799	(acceleration motion) near2 (sensor detector)	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L43	64689	(watch wearable attachable) and L42	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L44	1689	(bio-signal biosignal) and L43	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L45	1218439	(second two) near2 (signal wave)	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L46	241	L44 and L45	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L47	28769	(wearable watch) with interfac\$3	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L58	25913	L54 L55 L56 L57	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L59	139	((bio-signal biosignal) and motion).clm.	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L60	21	L58 and L59	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L61	1219871	g06f\$.cpc.	US-	OR	ON	2019/06/17

			PGPUB; USPAT			13:25
L62	249799	(acceleration motion) near2 (sensor detector)	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L63	64689	(watch wearable attachable) and L62	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L64	1689	(bio-signal biosignal) and L63	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L65	1218439	(second two) near2 (signal wave)	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L66	241	L64 and L65	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L67	28769	(wearable watch) with interfac\$3	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L77	25913	L73 L74 L75 L76	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L78	139	((bio-signal biosignal) and motion).clm.	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L79	21	L77 and L78	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L80	1219871	g06f\$.cpc.	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L81	249799	(acceleration motion) near2 (sensor detector)	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L82	64689	(watch wearable attachable) and L81	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L83	1689	(bio-signal biosignal) and L82	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L84	1218439	(second two) near2 (signal wave)	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L85	241	L83 and L84	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L86	28769	(wearable watch) with interfac\$3	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L98	25913	L94 L95 L96 L97	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L99	139	((bio-signal biosignal) and motion).clm.	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L100	21	L98 and L99	US-	OR	ON	2019/06/17

			PGPUB; USPAT			13:25
L101	1219871	g06f\$.cpc.	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L102	249799	(acceleration motion) near2 (sensor detector)	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L103	64689	(watch wearable attachable) and L102	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
L104	1689	(bio-signal biosignal) and L103	US- PGPUB; USPAT	OR	ON	2019/06/17 13:25
S1	1	("20160202759").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2017/07/27 19:18
S2	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US- PGPUB; USPAT	OR	ON	2017/07/28 12:11
S3	1	(bio-signal biosignal) and motion and S2	US- PGPUB; USPAT	OR	ON	2017/07/28 12:51
S4	1	("20160202759").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2017/07/28 18:22
S5	14137	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD"))).AANM.	USPAT	OR	ON	2017/07/29 17:00
S6	40	((("CHOI") near3 ("Chang") near3 ("Mok"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2017/07/29 17:00
S7	256	((("KIM") near3 ("Sang") near3 ("Joon"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2017/07/29 17:00
S8	37	((("YOON") near3 ("Seung") near3 ("Keun"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2017/07/29 17:00
S9	14382	S5 S6 S7 S8	US- PGPUB; USPAT	OR	ON	2017/07/29 17:03
S10	97	((bio-signal biosignal) and motion).clm.	US- PGPUB; USPAT	OR	ON	2017/07/29 17:03
S11	6	S9 and S10	US- PGPUB; USPAT	OR	ON	2017/07/29 17:04
S12	1029344	g06f\$.cpc.	US- PGPUB; USPAT	OR	ON	2017/07/29 17:04
S13	188878	(acceleration motion) near2 (sensor detector)	US- PGPUB; USPAT	OR	ON	2017/07/29 17:05

S14	38789	(watch wearable attachable) and S13	US-PGPUB; USPAT	OR	ON	2017/07/29 17:06
S15	602	(bio-signal biosignal) and S14	US-PGPUB; USPAT	OR	ON	2017/07/29 17:06
S16	1081387	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2017/07/29 17:07
S17	126	S15 and S16	US-PGPUB; USPAT	OR	ON	2017/07/29 17:08
S18	18967	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2017/07/29 17:08
S19	29	S17 and S18	US-PGPUB; USPAT	OR	ON	2017/07/29 17:08
S20	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2017/07/29 17:09
S21	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US-PGPUB; USPAT	OR	ON	2017/07/29 17:09
S22	1	(bio-signal biosignal) and motion and S21	US-PGPUB; USPAT	OR	ON	2017/07/29 17:09
S23	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2017/07/29 17:09
S24	14137	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD"))).AANM.	USPAT	OR	ON	2017/07/29 17:09
S25	40	((("CHOI") near3 ("Chang") near3 ("Mok"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2017/07/29 17:09
S26	256	((("KIM") near3 ("Sang") near3 ("Joon"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2017/07/29 17:09
S27	37	((("YOON") near3 ("Seung") near3 ("Keun"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2017/07/29 17:09
S28	14382	S24 S25 S26 S27	US-PGPUB; USPAT	OR	ON	2017/07/29 17:09
S29	97	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2017/07/29 17:09
S30	6	S28 and S29	US-PGPUB; USPAT	OR	ON	2017/07/29 17:09
S31	1029344	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2017/07/29 17:09

S32	188878	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2017/07/29 17:09
S33	38789	(watch wearable attachable) and S32	US-PGPUB; USPAT	OR	ON	2017/07/29 17:09
S34	602	(bio-signal biosignal) and S33	US-PGPUB; USPAT	OR	ON	2017/07/29 17:09
S35	1081387	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2017/07/29 17:09
S36	126	S34 and S35	US-PGPUB; USPAT	OR	ON	2017/07/29 17:09
S37	18967	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2017/07/29 17:09
S38	29	S36 and S37	US-PGPUB; USPAT	OR	ON	2017/07/29 17:09
S39	1	("20100145171").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/01/15 00:07
S40	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/01/15 18:17
S41	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S42	1	(bio-signal biosignal) and motion and S41	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S43	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/01/15 18:17
S44	17001	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD")).AANM.	USPAT	OR	ON	2018/01/15 18:17
S45	41	((("CHOI") near3 ("Chang") near3 ("Mok")).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/01/15 18:17
S46	279	((("KIM") near3 ("Sang") near3 ("Joon")).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/01/15 18:17
S47	41	((("YOON") near3 ("Seung") near3 ("Keun")).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/01/15 18:17
S48	17265	S44 S45 S46 S47	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S49	107	((bio-signal biosignal) and motion).clm.	US-	OR	ON	2018/01/15

			PGPUB; USPAT			18:17
S50	8	S48 and S49	US- PGPUB; USPAT	OR	ON	2018/01/15 18:17
S51	1077971	g06f\$.cpc.	US- PGPUB; USPAT	OR	ON	2018/01/15 18:17
S52	202904	(acceleration motion) near2 (sensor detector)	US- PGPUB; USPAT	OR	ON	2018/01/15 18:17
S53	44597	(watch wearable attachable) and S52	US- PGPUB; USPAT	OR	ON	2018/01/15 18:17
S54	839	(bio-signal biosignal) and S53	US- PGPUB; USPAT	OR	ON	2018/01/15 18:17
S55	1114857	(second two) near2 (signal wave)	US- PGPUB; USPAT	OR	ON	2018/01/15 18:17
S56	152	S54 and S55	US- PGPUB; USPAT	OR	ON	2018/01/15 18:17
S57	21301	(wearable watch) with interfac\$3	US- PGPUB; USPAT	OR	ON	2018/01/15 18:17
S58	37	S56 and S57	US- PGPUB; USPAT	OR	ON	2018/01/15 18:17
S59	1	("20160202759").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2018/01/15 18:17
S60	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US- PGPUB; USPAT	OR	ON	2018/01/15 18:17
S61	1	(bio-signal biosignal) and motion and S60	US- PGPUB; USPAT	OR	ON	2018/01/15 18:17
S62	1	("20160202759").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2018/01/15 18:17
S63	17001	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD")).AANM.	USPAT	OR	ON	2018/01/15 18:17
S64	41	((("CHOI") near3 ("Chang") near3 ("Mok")).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/01/15 18:17
S65	279	((("KIM") near3 ("Sang") near3 ("Joon")).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/01/15 18:17
S66	41	((("YOON") near3 ("Seung") near3 ("Keun")).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/01/15 18:17

S67	17265	S63 S64 S65 S66	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S68	107	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S69	8	S67 and S68	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S70	1077971	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S71	202904	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S72	44597	(watch wearable attachable) and S71	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S73	839	(bio-signal biosignal) and S72	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S74	1114857	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S75	152	S73 and S74	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S76	21301	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S77	37	S75 and S76	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S78	1	("20100145171").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/01/15 18:17
S79	1	("20120172733").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/03/28 15:53
S80	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/03/28 15:55
S81	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S82	1	(bio-signal biosignal) and motion and S81	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S83	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/03/28 15:55
S84	18271	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD")).AANM.	USPAT	OR	ON	2018/03/28 15:55

S85	43	((("CHOI") near3 ("Chang") near3 ("Mok"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/03/28 15:55
S86	288	((("KIM") near3 ("Sang") near3 ("Joon"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/03/28 15:55
S87	41	((("YOON") near3 ("Seung") near3 ("Keun"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/03/28 15:55
S88	18541	S84 S85 S86 S87	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S89	112	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S90	10	S88 and S89	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S91	1089358	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S92	208949	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S93	47101	(watch wearable attachable) and S92	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S94	928	(bio-signal biosignal) and S93	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S95	1128855	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S96	157	S94 and S95	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S97	22240	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S98	41	S96 and S97	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S99	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/03/28 15:55
S100	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S101	1	(bio-signal biosignal) and motion and S100	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S102	1	("20160202759").PN.	US-PGPUB; USPAT;	OR	OFF	2018/03/28 15:55

			USOCR			
S103	18271	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD"))).AANM.	USPAT	OR	ON	2018/03/28 15:55
S104	43	((("CHOI") near3 ("Chang") near3 ("Mok"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/03/28 15:55
S105	288	((("KIM") near3 ("Sang") near3 ("Joon"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/03/28 15:55
S106	41	((("YOON") near3 ("Seung") near3 ("Keun"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/03/28 15:55
S107	18541	S103 S104 S105 S106	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S108	112	((bio-signal biosignal) and motion).clm.	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S109	10	S107 and S108	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S110	1089358	g06f\$.cpc.	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S111	208949	(acceleration motion) near2 (sensor detector)	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S112	47101	(watch wearable attachable) and S111	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S113	928	(bio-signal biosignal) and S112	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S114	1128855	(second two) near2 (signal wave)	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S115	157	S113 and S114	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S116	22240	(wearable watch) with interfac\$3	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S117	41	S115 and S116	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S118	1	("20100145171").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2018/03/28 15:55
S119	1	("20160202759").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2018/03/28 15:55
S120	5	("20060149146" "20060281976"	US-	OR	ON	2018/03/28

		"20100145171" "20120184826" "20120256182").PN.	PGPUB; USPAT			15:55
S121	1	(bio-signal biosignal) and motion and S120	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S122	1	("20160202759").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2018/03/28 15:55
S123	18271	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD")).AANM.	USPAT	OR	ON	2018/03/28 15:55
S124	43	((("CHOI") near3 ("Chang") near3 ("Mok")).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/03/28 15:55
S125	288	((("KIM") near3 ("Sang") near3 ("Joon")).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/03/28 15:55
S126	41	((("YOON") near3 ("Seung") near3 ("Keun")).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/03/28 15:55
S127	18541	S123 S124 S125 S126	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S128	112	((bio-signal biosignal) and motion).clm.	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S129	10	S127 and S128	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S130	1089358	g06f\$.cpc.	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S131	208949	(acceleration motion) near2 (sensor detector)	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S132	47101	(watch wearable attachable) and S131	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S133	928	(bio-signal biosignal) and S132	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S134	1128855	(second two) near2 (signal wave)	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S135	157	S133 and S134	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S136	22240	(wearable watch) with interfac\$3	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S137	41	S135 and S136	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S138	1	("20160202759").PN.	US-	OR	OFF	2018/03/28

			PGPUB; USPAT; USOCR			15:55
S139	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S140	1	(bio-signal biosignal) and motion and S139	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S141	1	("20160202759").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2018/03/28 15:55
S142	18271	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD"))).AANM.	USPAT	OR	ON	2018/03/28 15:55
S143	43	((("CHOI") near3 ("Chang") near3 ("Mok"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/03/28 15:55
S144	288	((("KIM") near3 ("Sang") near3 ("Joon"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/03/28 15:55
S145	41	((("YOON") near3 ("Seung") near3 ("Keun"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/03/28 15:55
S146	18541	S142 S143 S144 S145	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S147	112	((bio-signal biosignal) and motion).clm.	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S148	10	S146 and S147	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S149	1089358	g06f\$.cpc.	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S150	208949	(acceleration motion) near2 (sensor detector)	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S151	47101	(watch wearable attachable) and S150	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S152	928	(bio-signal biosignal) and S151	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S153	1128855	(second two) near2 (signal wave)	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S154	157	S152 and S153	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S155	22240	(wearable watch) with interfac\$3	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55

S156	41	S154 and S155	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S157	1	("20100145171").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/03/28 15:55
S158	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US-PGPUB; USPAT	OR	ON	2018/05/14 17:18
S159	1105417	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2018/05/14 17:19
S160	1105422	S158 S159	US-PGPUB; USPAT	OR	ON	2018/05/14 17:19
S161	100346	sequential near2 (signal process\$3 determin\$5)	US-PGPUB; USPAT	OR	ON	2018/05/14 17:20
S162	1	(sequential near2 (signal process\$3 determin\$5)) with (motion accelerat\$3) with (bio-signal biological)	US-PGPUB; USPAT	OR	ON	2018/05/14 17:21
S163	1	("20150028850").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/14 17:24
S164	154	(sequential near2 (signal process\$3 determin\$5)) with (bio-signal biological)	US-PGPUB; USPAT	OR	ON	2018/05/14 17:25
S165	157	(sequential near2 (signal process\$3 determin\$5)) with (bio-signal biosignal biological)	US-PGPUB; USPAT	OR	ON	2018/05/14 17:25
S166	8	S160 and S165	US-PGPUB; USPAT	OR	ON	2018/05/14 17:26
S167	7	(sequential) with motion with (bio-signal biosignal biological)	US-PGPUB; USPAT	OR	ON	2018/05/14 17:27
S168	1	S160 and S167	US-PGPUB; USPAT	OR	ON	2018/05/14 17:27
S169	2420	(sequential) with (bio-signal biosignal biological)	US-PGPUB; USPAT	OR	ON	2018/05/14 17:27
S170	169	S160 and S169	US-PGPUB; USPAT	OR	ON	2018/05/14 17:28
S171	57	S170 and (motion accelerat\$3)	US-PGPUB; USPAT	OR	ON	2018/05/14 17:28
S172	57	S171 and S169	US-PGPUB; USPAT	OR	ON	2018/05/14 17:31
S173	606	(sequential) with motion with sens\$3	US-PGPUB; USPAT	OR	ON	2018/05/14 17:45
S174	139	S160 and S173	US-	OR	ON	2018/05/14

			PGPUB; USPAT			17:45
S175	38	(watch wearable wear-able) and S174	US- PGPUB; USPAT	OR	ON	2018/05/14 17:46
S176	7	(US-20100145171-\$ or US-20130328761-\$ or US-20140015750-\$ or US-20140258208-\$ or US-20140365979-\$).did. or (US-5954662-\$ or US-5832296-\$).did.	US- PGPUB; USPAT	OR	ON	2018/05/14 17:55
S177	6	(sequential) and S176	US- PGPUB; USPAT	OR	ON	2018/05/14 17:56
S178	1	("20160202759").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S179	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S180	1	(bio-signal biosignal) and motion and S179	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S181	1	("20160202759").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S182	19106	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD"))).AANM.	USPAT	OR	ON	2018/05/15 18:44
S183	45	((("CHOI") near3 ("Chang") near3 ("Mok"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S184	295	((("KIM") near3 ("Sang") near3 ("Joon"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S185	41	((("YOON") near3 ("Seung") near3 ("Keun"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S186	19380	S182 S183 S184 S185	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S187	112	((bio-signal biosignal) and motion).clm.	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S188	10	S186 and S187	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S189	1106227	g06f\$.cpc.	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S190	213039	(acceleration motion) near2 (sensor detector)	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S191	48821	(watch wearable attachable) and S190	US- PGPUB;	OR	ON	2018/05/15 18:44

			USPAT			
S192	1014	(bio-signal biosignal) and S191	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S193	1138450	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S194	167	S192 and S193	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S195	22938	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S196	46	S194 and S195	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S197	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S198	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S199	1	(bio-signal biosignal) and motion and S198	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S200	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S201	19106	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD"))).AANM.	USPAT	OR	ON	2018/05/15 18:44
S202	45	((("CHOI") near3 ("Chang") near3 ("Mok"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S203	295	((("KIM") near3 ("Sang") near3 ("Joon"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S204	41	((("YOON") near3 ("Seung") near3 ("Keun"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S205	19380	S201 S202 S203 S204	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S206	112	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S207	10	S205 and S206	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S208	1106227	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S209	213039	(acceleration motion) near2 (sensor	US-	OR	ON	2018/05/15

		detector)	PGPUB; USPAT			18:44
S210	48821	(watch wearable attachable) and S209	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S211	1014	(bio-signal biosignal) and S210	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S212	1138450	(second two) near2 (signal wave)	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S213	167	S211 and S212	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S214	22938	(wearable watch) with interfac\$3	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S215	46	S213 and S214	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S216	1	("20100145171").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S217	1	("20160202759").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S218	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S219	1	(bio-signal biosignal) and motion and S218	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S220	1	("20160202759").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S221	19106	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD"))).AANM.	USPAT	OR	ON	2018/05/15 18:44
S222	45	((("CHOI") near3 ("Chang") near3 ("Mok"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S223	295	((("KIM") near3 ("Sang") near3 ("Joon"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S224	41	((("YOON") near3 ("Seung") near3 ("Keun"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S225	19380	S221 S222 S223 S224	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S226	112	((bio-signal biosignal) and motion).clm.	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44

S227	10	S225 and S226	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S228	1106227	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S229	213039	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S230	48821	(watch wearable attachable) and S229	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S231	1014	(bio-signal biosignal) and S230	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S232	1138450	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S233	167	S231 and S232	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S234	22938	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S235	46	S233 and S234	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S236	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S237	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S238	1	(bio-signal biosignal) and motion and S237	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S239	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S240	19106	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD"))).AANM.	USPAT	OR	ON	2018/05/15 18:44
S241	45	((("CHOI") near3 ("Chang") near3 ("Mok"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S242	295	((("KIM") near3 ("Sang") near3 ("Joon"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S243	41	((("YOON") near3 ("Seung") near3 ("Keun"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S244	19380	S240 S241 S242 S243	US-PGPUB;	OR	ON	2018/05/15 18:44

			USPAT			
S245	112	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S246	10	S244 and S245	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S247	1106227	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S248	213039	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S249	48821	(watch wearable attachable) and S248	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S250	1014	(bio-signal biosignal) and S249	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S251	1138450	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S252	167	S250 and S251	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S253	22938	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S254	46	S252 and S253	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S255	1	("20100145171").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S256	1	("20120172733").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S257	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S258	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S259	1	(bio-signal biosignal) and motion and S258	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S260	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S261	19106	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD")).AANM.	USPAT	OR	ON	2018/05/15 18:44
S262	45	((("CHOI") near3 ("Chang") near3 ("Mok")).INV.	US-PGPUB;	OR	ON	2018/05/15 18:44

			USPAT; USOCR			
S263	295	((("KIM") near3 ("Sang") near3 ("Joon")).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S264	41	((("YOON") near3 ("Seung") near3 ("Keun")).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S265	19380	S261 S262 S263 S264	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S266	112	((bio-signal biosignal) and motion).clm.	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S267	10	S265 and S266	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S268	1106227	g06f\$.cpc.	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S269	213039	(acceleration motion) near2 (sensor detector)	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S270	48821	(watch wearable attachable) and S269	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S271	1014	(bio-signal biosignal) and S270	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S272	1138450	(second two) near2 (signal wave)	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S273	167	S271 and S272	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S274	22938	(wearable watch) with interfac\$3	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S275	46	S273 and S274	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S276	1	("20160202759").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S277	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S278	1	(bio-signal biosignal) and motion and S277	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S279	1	("20160202759").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44

S280	19106	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD")).AANM.	USPAT	OR	ON	2018/05/15 18:44
S281	45	((("CHOI") near3 ("Chang") near3 ("Mok")).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S282	295	((("KIM") near3 ("Sang") near3 ("Joon")).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S283	41	((("YOON") near3 ("Seung") near3 ("Keun")).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S284	19380	S280 S281 S282 S283	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S285	112	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S286	10	S284 and S285	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S287	1106227	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S288	213039	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S289	48821	(watch wearable attachable) and S288	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S290	1014	(bio-signal biosignal) and S289	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S291	1138450	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S292	167	S290 and S291	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S293	22938	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S294	46	S292 and S293	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S295	1	("20100145171").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S296	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S297	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44

S298	1	(bio-signal biosignal) and motion and S297	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S299	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S300	19106	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD"))).AANM.	USPAT	OR	ON	2018/05/15 18:44
S301	45	((("CHOI") near3 ("Chang") near3 ("Mok"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S302	295	((("KIM") near3 ("Sang") near3 ("Joon"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S303	41	((("YOON") near3 ("Seung") near3 ("Keun"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S304	19380	S300 S301 S302 S303	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S305	112	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S306	10	S304 and S305	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S307	1106227	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S308	213039	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S309	48821	(watch wearable attachable) and S308	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S310	1014	(bio-signal biosignal) and S309	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S311	1138450	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S312	167	S310 and S311	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S313	22938	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S314	46	S312 and S313	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S315	1	("20160202759").PN.	US-PGPUB; USPAT;	OR	OFF	2018/05/15 18:44

			USOCR			
S316	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S317	1	(bio-signal biosignal) and motion and S316	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S318	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S319	19106	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD"))).AANM.	USPAT	OR	ON	2018/05/15 18:44
S320	45	((("CHOI") near3 ("Chang") near3 ("Mok"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S321	295	((("KIM") near3 ("Sang") near3 ("Joon"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S322	41	((("YOON") near3 ("Seung") near3 ("Keun"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S323	19380	S319 S320 S321 S322	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S324	112	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S325	10	S323 and S324	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S326	1106227	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S327	213039	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S328	48821	(watch wearable attachable) and S327	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S329	1014	(bio-signal biosignal) and S328	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S330	1138450	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S331	167	S329 and S330	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S332	22938	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S333	46	S331 and S332	US-PGPUB;	OR	ON	2018/05/15 18:44

			USPAT			
S334	1	("20100145171").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S335	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S336	1106227	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S337	1106232	S335 S336	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S338	100346	sequential near2 (signal process\$3 determin\$5)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S339	1	(sequential near2 (signal process\$3 determin\$5)) with (motion accelerat\$3) with (bio-signal biological)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S340	1	("20150028850").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S341	154	(sequential near2 (signal process\$3 determin\$5)) with (bio-signal biological)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S342	157	(sequential near2 (signal process\$3 determin\$5)) with (bio-signal biosignal biological)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S343	8	S337 and S342	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S344	7	(sequential) with motion with (bio-signal biosignal biological)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S345	1	S337 and S344	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S346	2420	(sequential) with (bio-signal biosignal biological)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S347	169	S337 and S346	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S348	57	S347 and (motion accelerat\$3)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S349	57	S348 and S346	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S350	606	(sequential) with motion with sens\$3	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S351	140	S337 and S350	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44

S352	39	(watch wearable wear-able) and S351	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S353	7	(US-20100145171-\$ or US-20130328761-\$ or US-20140015750-\$ or US-20140258208-\$ or US-20140365979-\$).did. or (US-5954662-\$ or US-5832296-\$).did.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S354	6	(sequential) and S353	US-PGPUB; USPAT	OR	ON	2018/05/15 18:45
S355	7	(US-20100145171-\$ or US-20130328761-\$ or US-20140015750-\$ or US-20140258208-\$ or US-20140365979-\$).did. or (US-5954662-\$ or US-5832296-\$).did.	US-PGPUB; USPAT	OR	ON	2018/11/06 10:52
S356	3	S355 and threshold\$4	US-PGPUB; USPAT	OR	ON	2018/11/06 10:54
S357	1177670	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2018/11/06 11:13
S358	5800	sens\$3 with target with threshold	US-PGPUB; USPAT	OR	ON	2018/11/06 11:14
S359	6123	sens\$3 with target\$4 with threshold\$3	US-PGPUB; USPAT	OR	ON	2018/11/06 11:14
S360	723	S357 and S359	US-PGPUB; USPAT	OR	ON	2018/11/06 11:15
S361	5052	bio-signal\$4 biosignal\$4	US-PGPUB; USPAT	OR	ON	2018/11/06 11:15
S362	1807	S359 and ((second two another other) near2 signal\$4)	US-PGPUB; USPAT	OR	ON	2018/11/06 11:16
S363	94	S359 with ((second two another other) near2 signal\$4)	US-PGPUB; USPAT	OR	ON	2018/11/06 11:17
S364	5	S357 and S363	US-PGPUB; USPAT	OR	ON	2018/11/06 11:17
S365	1603	signal\$4 with sens\$3 with target\$4 with threshold\$3	US-PGPUB; USPAT	OR	ON	2018/11/06 11:19
S366	96	S357 and S365	US-PGPUB; USPAT	OR	ON	2018/11/06 11:19
S367	4	(US-20100145171-\$ or US-20140365979-\$ or US-20040042637-\$).did. or (US-5832296-\$).did.	US-PGPUB; USPAT	OR	ON	2018/11/06 18:21
S368	1603	signal\$4 with sens\$3 with target\$4 with threshold\$3	US-PGPUB; USPAT	OR	ON	2018/11/06 18:22
S369	1	S368 and S367	US-PGPUB; USPAT	OR	ON	2018/11/06 18:22

S370	1	(sens\$3 with target\$4 with threshold\$3) and S369	US-PGPUB; USPAT	OR	ON	2018/11/06 18:23
S371	10	(US-20100145171-\$ or US-20130328761-\$ or US-20140015750-\$ or US-20140258208-\$ or US-20140365979-\$ or US-20040042637-\$ or US-20130151730-\$ or US-20140050354-\$).did. or (US-5954662-\$ or US-5832296-\$).did.	US-PGPUB; USPAT	OR	ON	2019/06/15 13:30
S372	761	(bio-signal\$4 biosignal\$4 (biologic\$5 near2 (signal data))) with period	US-PGPUB; USPAT	OR	ON	2019/06/16 13:32
S373	1219871	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2019/06/16 13:33
S374	125	S372 and S373	US-PGPUB; USPAT	OR	ON	2019/06/16 13:33
S375	73	(wearable watch) and S374	US-PGPUB; USPAT	OR	ON	2019/06/16 13:33

EAST Search History (I nterference)

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Substitute for PTO/SB/08a/b

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	New Application
				Filing Date	Concurrently Herewith
				First Named Inventor	Chang Mok CHOI
				Art Unit	Not Yet Assigned
				Examiner Name	Not Yet Assigned
Sheet	1	of	2	Attorney Docket Number	012052.0799C1

U.S. PATENT DOCUMENTS					
Examiner Initials*	Cite No. ¹	Document Number Number - Kind Code ² (if known)	Patent or Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
	A	US 5,832,296 A	11-03-1998	Wang et al.	
	B	US 2006/0149146 A1	07-06-2006	Schmidt et al.	
	C	US 2006/0281979 A1	12-14-2006	Kim et al.	
	D	US 2010/0145171 A1	06-10-2010	PARK et al.	
	E	US 2012/0172733 A1	07-05-2012	Park	
	F	US 2012/0184826 A1	07-19-2012	KEENAN et al.	
	G	US 2012/0259182 A1	10-11-2012	KIM et al.	

FOREIGN PATENT DOCUMENTS						
Examiner Initials*	Cite No. ¹	Foreign Patent Document Country Code ³ - Number ⁴ - Kind Code ⁵ (if known)	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear	T ⁶
	a	CN 1488115 A	04-07-2004	Bridicum AS	Abstract of WO02059803 A1	X
	b	CN 100440310 C	12-03-2008	Thomson Licensing	Abstract of US 2004218104 A1	X
	c	KR 10-2003-0040316 A	05-22-2003	LEE		X
	d	KR 10-2006-0069985 A	06-23-2006	ELECTRONICS AND TELECOMMUNICATIONS RESEARCH INSTITUTE		X
	e	KR 10-2007-0000848 A	01-03-2007	PANTECH CO., LTD.		X
	f	KR 10-2007-0109044 A	11-15-2007	ELECTRONICS AND TELECOMMUNICATIONS RESEARCH INSTITUTE		X
	g	KR 10-2009-0027390 A	03-17-2009	Lee et al.		X

Examiner Signature		Date Considered	
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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	New Application
				Filing Date	Concurrently Herewith
				First Named Inventor	Chang Mok CHOI
				Art Unit	Not Yet Assigned
				Examiner Name	Not Yet Assigned
Sheet	2	of	2	Attorney Docket Number	012052.0799C1

FOREIGN PATENT DOCUMENTS						
Examiner Initials*	Cite No. ¹	Foreign Patent Document	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear	T ⁶
		Country Code ³ - Number ⁴ - Kind Code ⁵ (if known)				
	h	KR 10-2009-0032537 A	04-01-2009	ELECTRONICS AND TELECOMMUNICATIONS RESEARCH INSTITUTE		X
	i	KR 10-2010-0027375 A	03-11-2010	KOREA INSTITUTE OF ORIENTAL MEDICINE		X
	j	KR 10-2011-0123708 A	11-15-2011	Jawon Medical Co., Ltd.		X
	k	KR 10-2011-0062558 A	06-10-2011	SOONGSIL UNIVERSITY RESEARCH CONSORTIUM TECHNOPARK		X
	l	KR 10-2012-0014233 A	02-16-2012	QUALCOMM INCORPORATED	Abstract of WO/2010/141878	X
	m	KR 10-2012-0111030 A	10-10-2012	LAXTHA		X
	n	KR 10-2012-0133350 A	12-10-2012	NEOFACT CO., LTD.		X
	o	WO 2008/073801 A2	06-19-2008	Emotiv Systems Pty Ltd.		

NON-PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author, title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ⁶
	1	International Search Report issued on August 7, 2014 in counterpart International Application No. PCT/KR2014/004073 (2 pages in English; 2 pages in Korean).	✓
	2	European Search Report issued on July 11, 2017 in Corresponding European Patent Application No. 14838613.9 (11 pages in English).	
	3	Chinese Office Action issued on January 2, 2018 in corresponding Chinese Patent Application No. 201480044084.7 (8 pages in English and 7 pages in Chinese).	✓

Examiner Signature	/Gustavo Polo/ (06/17/2019)	Date Considered	
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16267752	CTNF	06/21/2019	012052.0799C1
	892	06/21/2019	012052.0799C1
	1449	06/21/2019	012052.0799C1

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In re the Application of:

Chang Mok CHOI

Application No. 16/267,752

Art Unit: 2694

Confirmation No. 9126

Filed: February 5, 2019

Examiner: Gustavo D. Polo

For: WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF

RESPONSE PURSUANT TO 37 C.F.R. § 1.111

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

This is in response to the Non-Final Office Action mailed June 21, 2019 having a period for response set to expire on September 23, 2019.

The following amendments and remarks are respectfully submitted. Reconsideration of the claims is respectfully requested.

Amendments to the claims are reflected in the listing of the claims that begins on page 2 of this Paper.

Remarks begin on page 2 of this Paper.

IN THE CLAIMS

This listing of the claims replaces all prior versions and listings of the claims in this application.

The text of all pending claims (including any withdrawn claims) is set forth below. Canceled and not entered claims are indicated with claim number and status only. The claims as listed below show added text with underlining and deleted text with ~~striketrough~~ or in [[double brackets]]. The status of each claim is listed with one of (Original), (Currently amended), (Canceled), (Withdrawn), (Previously presented), (New), and (Not entered).

Please **AMEND** claims 1, 2 and 4-20; **CANCEL** claim 6; and **ADD** claims 21-35 in accordance with the following:

1. (Currently amended) A wearable device comprising:
a motion sensor;
a biological sensor; and
a controller configured to:
obtain a movement signal of an object using the motion sensor, and a biological signal of the object using the biological sensor, ~~and~~
determine, based on an analysis of the movement signal, whether to perform a specified function of the wearable device using the biological signal, and
perform ~~a~~ the specified function using ~~at least a~~ select portion of the biological signal obtained during a ~~specified select period, that of the movement signal, determined to~~ corresponds to a specified condition.
2. (Currently amended) The wearable device of claim 1,
wherein the ~~controller is configured to perform, in response to specified condition~~ comprises the movement signal being determined to be less than or equal to a first threshold value, the wearable device using the biological signal.
3. (Original) The wearable device of claim 2, wherein the controller is configured to determine the first threshold value based on a magnitude of the biological signal.
4. (Currently amended) The wearable device of claim 1, wherein the controller is configured to determine the biological signal to be invalid, irrespective of the movement signal, in response to the magnitude of the biological signal not satisfying a second threshold value.

5. (Currently amended) The wearable device of claim 1, wherein, in response to a determination that the biological signal is valid, the controller is configured to operate in an active mode and to control an external device based on the biological signal.

6. (Canceled)

7. (Currently amended) The wearable device of claim 1, further comprising a transmission unit, wherein the controller is configured to transmit an instruction associated with an external electronic device to the external electronic device using the transmission unit as at least part of the specified function.

8. (Currently amended) A wearable device comprising:
a motion sensor;
a biological sensor; and
a controller configured to:

~~obtain, with respect to an object external to the wearable device, a movement signal of the~~
~~an object external to the wearable device using the motion sensor; and~~

~~obtain a biological signal of the object using the biological sensor;~~

~~identify a period, that of the movement signal, the movement signal is determined to be~~
~~smaller-less than a specified condition; and~~

~~perform a specified function using at least a select portion of the biological signal~~
~~corresponding to the period.~~

9. (Currently amended) The wearable device of claim 8, wherein the biological sensor comprises an optical sensor configured to obtain an optical signal ~~related~~ corresponding to the biological signal.

10. (Currently amended) The wearable device of claim 8, wherein the controller is configured to adjust, prior to the identifying of the period, the specified condition based at least in part on a magnitude of the biological signal ~~prior to the identifying of the period.~~

11. (Currently amended) The wearable device of claim 8, wherein the controller is configured to:

in response to a magnitude of the biological signal being determined to be within a first specified range, determine the biological signal as to be a valid signal if that a magnitude of the biological signal falls into a first specified range; and

in response to the magnitude of the biological signal being determined to be within a second specified range, determine the biological signal as to be an invalid signal if that a magnitude of the biological signal falls into a second specified range.

12. (Currently amended) The wearable device of claim 8, wherein, for the performing of the specified function, the controller is configured to generate an instruction using the ~~at least select~~ portion of the biological signal ~~as at least part of the performing~~.

13. (Currently amended) The wearable device of claim 12, wherein, for the performing of the specified function, the controller is configured to control an external electronic device based at least in part on the instruction as at least part of the specified function.

14. (Currently amended) The wearable device of claim 13, further comprising a transmission unit, wherein the controller is configured to transmit the instruction to the external electronic device using the transmission unit.

15. (Currently amended) A wearable device comprising:

a motion sensor;

a biological sensor; and

a controller configured to:

obtain, with respect to an object external to the wearable device, a movement signal of the an object external to the wearable device using the motion sensor; and

obtain a biological signal of the object using the biological sensor;

determine a select period, of the movement signal, that meets a specified condition;

determine, as a valid signal, at least a portion of the biometric signal that is determined to correspond to the determined corresponding to a select period that the movement is smaller than a specified condition as a valid signal; and

perform a specified function using the valid signal.

16. (Currently amended) The wearable device of claim 15, wherein the biological sensor comprises an optical sensor configured to obtain an optical signal ~~related to~~ as the biological signal.

17. (Currently amended) The wearable device of claim 15, wherein the controller is configured to adjust the specified condition based at least in part on a magnitude of the biological signal prior to the ~~identifying-determining~~ of the select period.

18. (Currently amended) The wearable device of claim 15, wherein for the performing of the specified function, the controller is configured to generate an instruction using ~~the~~ at least a portion of the valid signal ~~as at least part of the performing~~.

19. (Currently amended) The wearable device of claim 18, wherein for the performing of the specified function, the controller is configured to control an external electronic device based at least in part on the instruction as at least part of the specified function.

20. (Currently amended) The wearable device of claim 19, further comprising a transmission unit, wherein the controller is configured to transmit the instruction to the external device using the transmission unit.

21. (New) The wearable device of claim 1, wherein
the controller is further configured to validate the biological signal, and
for the performing of the specified function, the controller is configured to perform the specified function using a portion of the validated biological signal.

22. (New) A wearable device system including one or more wearable devices, the wearable device system comprising:

a motion sensor of the one or more wearable devices;

a biological sensor of the one or more wearable devices; and

a controller of the one or more wearable devices, the controller configured to:

obtain, based on motion information from the motion sensor, a movement signal of an object;

obtain, based on biological information from the biological sensor, a biological signal of the object;

determine whether a period of the movement signal exists that meets a specified motion type indicating condition;

determine whether any portions the biological signal correspond to the period and meet a specified value condition; and

perform, in response to at least one of the one or more portions of the biological signal being determined to correspond to the period, a specified function of at least one of the one or more wearable devices and/or an external device respectively dependent on the movement signal.

23. (New) The wearable device system of claim 22, wherein the determination of whether the period of the movement signal exists includes determining what period of the movement signal meets the specified motion type indication condition that is predetermined to be one or more measures of relatedness of obtained motions to an incidental motion, corresponding to a non-spontaneous or non-intentional motion of the object, and/or an intentional motion of the object.

24. (New) The wearable device system of claim 23, the determination of whether the period of the movement signal exists that meets the specified motion type indicating condition includes determining whether a measured rotational velocity magnitude is below a predetermined velocity magnitude, where the measured rotational velocity magnitude being below the predetermined velocity magnitude is representative of the movement signal being not the incidental motion and/or being the intentional motion of the object.

25. (New) The wearable device system of claim 23, the determination of whether the period of the movement signal exists that meets the specified motion type indicating condition includes determining whether a measured velocity variance magnitude is below a predetermined velocity variance magnitude, where the measured rotational velocity variance magnitude being below the predetermined velocity variance magnitude is representative of the movement signal being not the incidental motion and/or being the intentional motion of the object.

26. (New) The wearable device system of claim 25, wherein the predetermined velocity variance magnitude is a predetermined acceleration-based threshold, a predetermined rotation-based threshold, a predetermined bearing signal-based threshold, or a predetermined coordinate signal-based threshold.

27. (New) The wearable device system of claim 22, wherein the specified value condition is a measure of valuableness of a measured biological signal for intentional controlling of one or more functions

of an external device, the measure being a determination that a magnitude of at least one portion of the biological signal, corresponding to the select period, is greater than a predetermined threshold.

28. (New) The wearable device system of claim 27, wherein the magnitude of the at least one portion of the biological signal is a measure of a relaxation and/or contraction of one or more muscles of the object.

29. (New) The wearable device system of claim 28, wherein the biosensor includes one or more biosensors, and the muscles include one or more of one or more wrist muscles, one or more hand muscles, one or more forearm muscles, one or more face muscles, and/or one or more neck muscles of the object.

30. (New) The wearable device system of claim 29, further comprising determining which one or more of different interaction controls are represented the movement signal, with the performing of the specified function being a performance of the specified function of the determined one or more of the different interaction controls represented by the movement signal.

31. (New) The wearable device system of claim 30, wherein the specified motion type indicating condition is a predetermined condition that, when met, the one or more wearable devices are controlled according to a determined gesture of the object, represented by the movement signal, and the biological signal of the object.

32. (New) The wearable device system of claim 31, wherein the determination of whether the period of the movement signal exists includes determining what period of the movement signal meets the specified motion type indication condition that is predetermined to be one or more measures of relatedness of obtained motions to an incidental motion, corresponding to a non-spontaneous or non-intentional motion of the object, and/or an intentional motion of the object, and

wherein the determining of what period of the movement signal meets the specified motion type indication condition includes determining whether a measured rotational velocity magnitude is below a predetermined velocity magnitude, where the measured rotational velocity magnitude being below the predetermined velocity magnitude is representative of the movement signal being not the incidental motion and/or being the intentional motion of the object, and/or

wherein the determining of what period of the movement signal meets the specified motion type indication condition includes determining whether a measured velocity variance magnitude is below a

predetermined velocity variance magnitude, where the measured rotational velocity variance magnitude being below the predetermined velocity variance magnitude is representative of the movement signal being not the incidental motion and/or being the intentional motion of the object.

33. (New) The wearable device system of claim 32, wherein the predetermined velocity variance magnitude is a predetermined acceleration-based threshold, a predetermined rotation-based threshold, a predetermined bearing signal-based threshold, or a predetermined coordinate signal-based threshold.

34. (New) The wearable device system of claim 22, wherein the specified motion type indicating condition is a predetermined condition that, when met, the one or more wearable devices are controlled to shift modes of operation from an inactive mode of operation to an active mode of operation.

35. (New) The wearable device system of claim 22, wherein the biological signal is a bio-electric/magnetic signal, a bioimpedance signal, and/or a bio-mechanical signal generated in association with skeletal muscle movement signal or skeletal muscle tone of the object.

REMARKS

The Office is thanked for the review and consideration of the pending application. The Non-Final Office Action mailed June 21, 2019 ("Office Action") has been received and its contents reviewed. Reconsideration of the pending application is respectfully requested in view of the amendments and the following remarks.

In accordance with the foregoing, claims 1, 2 and 4-20 have been amended; claim 6 has been canceled; and claims 21-35 have been added. Accordingly, claims 1-35 are pending, with claims 1, 8, 15, and 22 being independent claims.

The amendments address typographical errors, improve readability, and clarify the nature of aspects of the claims. No new matter is presented in the amendments and newly added claims. Applicant has amended claims solely for the purpose of furthering the prosecution of a disclosed embodiment, and not in response to the merits of the outstanding rejections. Applicant reserves the right to pursue other claims, including broader claims or embodiments, in a continuation application.

Claim Rejections Pursuant to 35 U.S.C. § 103

Claims 1-20 stand rejected pursuant to 35 U.S.C. § 103 as allegedly being unpatentable over U.S. Patent App. Pub. No. 2010/0145171 to Park et al. ("Park") in view of U.S. Patent App. Pub. No. 2012/0157789 to Kangas et al. ("Kangas"). These rejections are respectively traversed.

Independent Claims 1, 8 & 15

In rejecting independent claim 1, the Office sets forth that the combination of Park and Kangas discloses all the claimed features. Applicant respectfully disagrees and respectfully submits that neither Park, Kangas, nor any combination thereof discloses or suggests all the claimed features of independent claim 1.

- 1. Perform the specified function using a select portion of the biological signal obtained during a select period, of the movement signal, determined to correspond to a specified condition*

As a non-limiting example, neither Park, Kangas, nor any combination thereof discloses "perform the specified function using a select portion of the biological signal obtained during a select period, of the movement signal, determined to correspond to a specified condition," as recited in independent claim 1.

For example, in its rejection of independent claim 1, the Office asserts:

Park is silent on perform a specified function using at least a portion of the biological signal obtained during a specified period that the movement corresponds to a specified condition. Kangas teaches perform a specified function using at least a portion of the biological signal obtained during a specified period [¶ 96 *bio-signal information which is stored in the bio-signal profile 25 may comprise data which has been collected, over an extended period of time*] that the movement corresponds to a specified condition [¶ 110 *by collecting data over a period of time and storing it in a bio-signal profile 25 the outputs provided are personal to the user of the apparatus and may be optimized to make the apparatus as convenient as possible for the user*].

Office Action at 2, 3.

As shown, it appears that the Office asserts that Kangas's collecting and/or storing of bio-signal information discloses performing a specified function using at least a portion of a biological signal.

However, while Kangas discloses collecting and storing bio-signal information to generate or update a bio-signal profile for a user, the bio-signal information is not collected "using a select portion of the biological signal obtained during a select period, of the movement signal, determined to correspond to a specified condition," as recited in independent claim 1.

For example, Kangas merely discloses "The bio-signal information which is stored in the bio-signal profile 25 may comprise data which has been collected, over an extended period of time, while the user uses the apparatus 1. The data may be collected by using the method illustrated in FIG. 3. In some embodiments of the invention the method illustrated in FIG. 3 may be repeated at regular or predetermined intervals." Kangas at ¶ 0096.

As shown, Kangas merely collects the bio-signal information at predetermined intervals, and this bio-signal information collected at predetermined intervals fails to disclose "a select portion of the biological signal obtained during a select period, of the movement signal, determined to correspond to a specified condition." as recited in independent claim 1.

Moreover, while Kangas discloses that "A bio-mechanical signal may comprise the user of the apparatus 1 moving a part of their body," Kangas at ¶ 0066, Kangas merely discloses that the bio-signal information may be of any one of "a bio-electrical signal, a bio-mechanical signal, an aural signal, a chemical signal or an optical signal." Kangas at ¶ 0063.

Accordingly, an embodiment of Kangas collects and analyzes one of the aforementioned types of bio-signals, and Kangas fails to disclose performing a specified function using a select portion of a bio-electrical, aural, a chemical, or optical signal obtained during a select period, of a bio-mechanical signal, determined to correspond to a specified condition.

Accordingly, Kangas fails to disclose “using a select portion of the biological signal obtained during a select period, of the movement signal, determined to correspond to a specified condition,” as recited in independent claim 1.

II. Determine, based on an analysis of the movement signal, whether to perform a specified function of the wearable device using the biological signal

As a non-limiting example, neither Park, Kangas, nor any combination thereof discloses “determine, based on an analysis of the movement signal, whether to perform a specified function of the wearable device using the biological signal,” as recited in independent claim 1.

As shown, features of previous claim 6 have been incorporated into independent claim 1.

In its rejection of claim 6, the Office appears to rely on Kangas with respect to measured biological signal of Kangas without consideration of the claimed relationship of “the movement” the Office Action has interpreted of Park, and the required relationship between the interpreted “movement” of Park and the pulse wave signal of Park.

For example, independent claim 1 now sets forth “determine, based on an analysis of the movement signal, whether to perform a specified function of the wearable device using the biological signal.”

Here, while Kangas may perform different functions based on different determined biological signals, there is no suggestion that *whether* those functions are performed have any dependence on the acceleration signal of the motion signal of Park or any similarly utilized motion signal.

The motion signal of Park is particularly derived from a motion sensor designed for collecting biosignals particularly related, and using a motion sensor particularly positioned, to/for removing motion artifacts or noise from the measured pulse wave signal.

This motion sensor would not be so similarly placed, or a movement signal determined therefrom, for the purpose of Park and for any related purpose of the different biosignal dependent function implementations of Kangas. Thus, neither Park nor Kangas, nor any combination of the same, could or would disclose “determine, based on an analysis of the movement signal, whether to perform a specified function of the wearable device using the biological signal,” as recited in independent claim 1.

Park fails to cure the aforementioned deficiencies in Kangas with respect to independent claim 1. It is further respectfully submitted that the combination of Park and Kangas also fails to disclose each and every claimed feature of each of independent claims 8 and 15.

III. Obviousness

In addition, it is respectfully submitted that the Office Action’s rejection rationale for independent claims 1, 8, and 15 is improper.

For emphases, in full, the Office Action sets forth:

1. Park discloses a wearable device [Fig. 1] comprising: a motion sensor [Fig. 2, 150]; a biological sensor [130]; and a controller [110] configured to: obtain a movement of an object using the motion sensor [¶ 50], and a biological signal of the object using the biological sensor [¶ 47]. Park is silent on perform a specified function using at least a portion of the biological signal obtained during a specified period that the movement corresponds to a specified condition. Kangas teaches perform a specified function using at least a portion of the biological signal obtained during a specified period [¶ 96 bio-signal information which is stored in the bio-signal profile 25 may comprise data which has been collected, over an extended period of time] that the movement corresponds to a specified condition [¶ 110 by collecting data over a period of time and storing it in a bio-signal profile 25 the outputs provided are personal to the user of the apparatus and may be optimized to make the apparatus as convenient as possible for the user]. *Therefore it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify Park with Kangas as required by this claim, since such a modification improves the performance of the device.*

Here, the Office Action has only argued that Kangas sets forth a feature that Park does not already have, and that the addition of that feature to Park would “improve the performance of the device of Park.”

It is respectfully submitted that a prima facie case of obviousness is not met when the rejection rationale does not include an explanation of why Park would add the particular feature of Kangas, other than the addition would provide the additional feature of the Kangas and ‘improve performance’ of the Park device. The Office Action has not explained why the performance of Park would be increased, or why one skilled in the art would look to the particular additional feature of Kangas and find a reason for modifying Park to include that feature outside of the disclosure of the present application.

Park uses a measured motion for removing motion artifacts from a sensed biosignal.

For example, paragraph [0003] of Park sets forth:

The present invention relates to an apparatus for measuring a motion noise robust pulse wave and a method thereof, and more particularly, to an apparatus for measuring a motion noise robust pulse wave at the back of a wrist, i.e., that part of the wrist which extends into the back of the hand, through an adaptive signal processing technique by estimating a motion component using an acceleration sensor and a method thereof.

Further, independent claim 1 of Park further explains how noise is estimated based on the acceleration signal:

a signal preprocessor that estimates a motion noise model corresponding to the motion of the examinee by predicting the motion of the examinee on the basis of an acceleration signal detected by the acceleration sensor and removes motion noise for the pulse wave signal from the motion noise model.

Thus, Park presents a device directed to producing a more noiseless pulse wave signal by estimating the noise in the measured pulse wave signal based on the acceleration signal.

For example, see paragraph [0006] discussing that the pulse wave signal may be used to measuring a health index of a user. As another example, paragraph [0032] of Park sets forth:

In addition, with an increase in concern about health, the present invention may be used as a health measurement means for adults or old men and an emergency precursor situation detection function and when contents for stress, concentration, sleep quality, etc. through HRV analysis are provided to a host device, the present invention may be used as more various purposes, that is, for a game, education, wellbeing contents and in addition, when the wireless protocol such as Zigbee is used, since data of a plurality of wearable apparatuses in groups such as a hospital, a senior center, and a silver town can be collected and processed, the present invention may be used as a basic apparatus in a ubiquitous-based health care field.

Thus, there is no suggestion in Park to use the measured pulse wave signal for anything other than for more accurate health related contents or information.

Accordingly, the Office is proposing to modify Park to include an additional function, not needed or necessary with the invention of Park, to use the noiseless pulse wave signal of Park for controlling operations of the device. This operation would be independent of the noise removing operation of Park based on the motion sensor of Park, as it must occur after the noise removing operations of Park to achieve the benefit of Park

It is respectfully submitted that the addition of controlling a device based on the noiseless pulse signal of Park does not improve the operation of Park, but only adds a function that Park did not suggest, need, or desire.

For example, in the PTAB decision of *Kranos v Riddel* (Case IPR2016-1649, Feb 7, 2018), where the PTA explained:

The Supreme Court made clear that we apply “an expansive and flexible approach” to the question of obviousness. *KSR*, 550 U.S. at 415. Whether a patent claiming the combination of prior art elements would have been obvious is determined by whether the improvement is more than the predictable use of prior art elements according to their established functions. *Id.* at 417. To reach this conclusion, however, it is not enough to show merely that the prior art includes separate references covering each separate limitation in a challenged claim. *Unigene Labs., Inc. v. Apotex, Inc.*, 655 F.3d 1352, 1360 (Fed. Cir. 2011). Rather, obviousness additionally requires that a person of ordinary skill at the time of the invention “would have selected and combined those prior art elements in the normal course of research and development to yield the claimed invention.” *Id.*

As another example, see *Arendi v Apple*, (Fed. Cir. 2015-2073, August 10, 2016) setting forth:

[T]here are at least three caveats to note in applying “common sense” in an obviousness analysis. First, common sense is typically invoked to provide a known motivation to combine, not to supply a missing claim limitation. In *DyStar*, a pre-*KSR* case, we held a patent

obvious where “all claim limitations [were] found in a number of prior art references,” 464 F.3d at 1360, and a person of ordinary skill in the art would have combined the references to achieve a “cheaper, faster, and more convenient” process. *Id.* at 1371 (quoting *Sandt Tech., Ltd. v. Resco Metal & Plastics Corp.*, 264 F.3d 1344, 1355 (Fed. Cir. 2001)). We

stated that: “an implicit motivation to combine exists not only when a suggestion may be gleaned from the prior art as a whole, but when the ‘improvement’ is technology-independent and the combination of references results in a product or process that is more desirable.” DyStar, 464 F.3d at 1368. Similarly, in Randall, we invoked common sense to vacate the Board’s determination of non-obviousness where “the Board failed to consider a wealth of well-documented knowledge that is highly material to evaluating the motivation to combine references.” Randall, 733 F.3d at 1356. There, the Board’s decision to ignore the existence of a “prevalent, perhaps even predominant method” of achieving the limitation at issue was prejudicial error. *Id.* at 1363.

Second, in Perfect Web, the only case Appellees identifies in which common sense was invoked to supply a limitation that was admittedly missing from the prior art, the limitation in question was unusually simple and the technology particularly straightforward. “The patented invention involves comparing the number of successfully delivered e-mail messages in a delivery against a predetermined desired quantity, and if the delivery does not reach the desired quantity, repeating the process of selecting and e-mailing a group of customers until the desired number of delivered messages has been achieved.” Perfect Web, 587 F.3d at 1326. There, the missing claim limitation—step D of steps A-D—was nothing more than an instruction to repeat steps A, B, and C until a particular quantity of email was sent in accordance with the claim. By contrast, the missing search at issue here “plays a major role in the subject matter claimed” and “affects much more than step (i).” Appellant’s Br. 24. That is—if the search in step (i) is missing, then “the claims would be almost void of content” because the premise of the patent is to use information in a first program to find related information in a second program. *Id.* at 25. Thus, the facts in Perfect Web are distinguishable from the case at bar and ought to be treated as the exception, rather than the rule.

Third, our cases repeatedly warn that references *to “common sense”—whether to supply a motivation to combine or a missing limitation—cannot be used as a wholesale substitute for reasoned analysis and evidentiary support, especially when dealing with a limitation missing from the prior art references specified.* Indeed, we stated that although there is no problem with using common sense “without any specific hint or suggestion in a particular reference,” the Board’s “utter failure to explain the ‘common knowledge and common sense’ on which it relied” is problematic. DyStar, 464 F.3d at 1366 (explaining our reasoning in *In re Lee*, 277 F.3d 1338, 1341, 1344 (Fed. Cir. 2002)). See also *In re Zurko*, 258 F.3d 1379, 1383, 1385 (Fed. Cir. 2001) (reversing Board where it adopted examiner’s unsupported assertion that claim limitation missing from cited references was “basic knowledge” and it “would have been nothing more than good common sense” to combine the references).

For example, in *In re Lee*, the Board adopted the examiner’s statements during prosecution that combining two prior art references to achieve the claimed invention—a method of automatically displaying the functions of a video display device and demonstrating how to select and adjust the functions—would have been obvious to a person of ordinary skill. The examiner had stated that the combination would have been obvious “since the demonstration mode is just a programmable feature which can be used in many different devices for providing automatic introduction by adding the proper programming software,” and that “another motivation would be that the automatic demonstration mode is user friendly and it functions as a tutorial.” *In re Lee*, 277 F.3d 1338, 1341 (Fed. Cir. 2002). We vacated the Board’s decision,

holding that “[c]onclusory statements such as those here provided do not fulfill the agency’s obligation” to explain all material facts relating to a motivation to combine. Id. at 1344.

...

Based on this prior precedent, we conclude that while “common sense” can be invoked, even potentially to supply a limitation missing from the prior art, it must still be supported by evidence and a reasoned explanation. In cases in which “common sense” is used to supply a missing limitation, as distinct from a motivation to combine, moreover, our search for a reasoned basis for resort to common sense must be searching. And, this is particularly true where the missing limitation goes to the heart of an invention.

See also, In re: Marcel Van Os (Fed. Cir. 2015-1975, January 3, 2017), and In re: Nuvasive (Fed. Cir. 2015-1670, December 7, 2016)

Here, as noted above the Office Action’s suggested motivation of improving Park is not related to the invention of purpose of Park, and thus would appear to be merely a conclusion that if the additional functions of Kangas were added after the removing of the noise of the pulse wave signal, then the combination would be an improvement over Park, which would appear to be a conclusion that the addition of the additional features of Kangas would have been common sense for creating a larger combination what includes the features of Park in whole and then the identified features of Kangas. However, as noted above, it is respectfully submitted that this conclusion is improper.

Based at least on the foregoing, the rejections of independent claims 1, 8, and 15 pursuant to 35 U.S.C § 103 based on Park and Kangas are deficient, and Applicant respectfully requests the rejections be withdrawn.

Dependent Claims

Claims 2-7, 9-14, and 16-21 are dependent upon independent claim 1, 8, or 15 and incorporate all the respective features therein. Accordingly, the 35 U.S.C. § 103 rejections of the dependent claims based on Park and Kangas are deficient for at least the same rationale as applied to respective independent claim 1, 8, or 15 above, and Applicant respectfully requests the rejections be withdrawn.

In addition, it is respectfully submitted that the rejections of dependent claims 2-6, 10, 12-13, and 17-19 are improper. As noted above, a substantial evidence rejection rationale must be provided for each rejection.

Rather, the rejections of dependent claims 2-6, 10, 12-13, and 17-19 merely identify additional features of Kangas, without an explanation why the further modification of the combination of Park and Kangas would have been obvious, including an explanation of why one of ordinary skill in the art would make the additional modification.

It is respectfully submitted that these rejections are improper, and respectfully submitted that any next Office Action must be non-Final and clearly present the missing motivation for the Office's proposed further modifications of the combination applied to the independent claims, as it is respectfully submitted that the rejections of the independent claims do not state that or suggest that the entirety of the invention and disclosure of Kangas are modified into the invention of Park, and further provide no substantial evidence reasons for such a substantial change in Park.

Accordingly, withdrawal of the rejections of all dependent claims is further respectfully requested.

Conclusion

There being no further outstanding objections or rejections, it is submitted that the application is in condition for allowance. An early action to that effect is courteously solicited.

In the event this paper is not being timely filed, Applicant respectfully petitions for an appropriate extension of time. Any fees for such an extension together with any additional fees only associated with the processing of this Response and any other documents filed concurrently with this Response may be charged to Counsel's Deposit Account 50-5113. Finally, if there are any formal matters remaining after this response, the Office is requested to telephone the undersigned to attend to these matters.

Respectfully submitted,

Date: September 23, 2019

/Thomas J. Matikainen/
Thomas J. Matikainen
Registration No. 73,291

NSIP Law
1120 Connecticut Avenue, NW, Suite 304
Washington, DC 20036
Telephone (202) 429-0020 x123
Facsimile (202) 315-3758
STB/TJM

Electronic Patent Application Fee Transmittal

Application Number:	16267752			
Filing Date:	05-Feb-2019			
Title of Invention:	WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF			
First Named Inventor/Applicant Name:	Chang Mok CHOI			
Filer:	Thomas Joseph Matikainen			
Attorney Docket Number:	012052.0799C1			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
CLAIMS IN EXCESS OF 20	1202	14	100	1400
INDEPENDENT CLAIMS IN EXCESS OF 3	1201	1	460	460
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1860

Electronic Acknowledgement Receipt

EFS ID:	37237852
Application Number:	16267752
International Application Number:	
Confirmation Number:	9126
Title of Invention:	WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF
First Named Inventor/Applicant Name:	Chang Mok CHOI
Customer Number:	89980
Filer:	Thomas Joseph Matikainen
Filer Authorized By:	
Attorney Docket Number:	012052.0799C1
Receipt Date:	23-SEP-2019
Filing Date:	05-FEB-2019
Time Stamp:	11:04:34
Application Type:	Utility under 35 USC 111(a)

Payment information:

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Payment Type	CARD
Payment was successfully received in RAM	\$ 1860
RAM confirmation Number	E20199MB05409114
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Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		OA20190621_0120520799C1_ResponseAsFiled.pdf	6756848	yes	16
			8a0ff82a030b28f41aea7ba8e74842edb47e55c3		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Amendment/Req. Reconsideration-After Non-Final Reject		1	1	
	Claims		2	8	
	Applicant Arguments/Remarks Made in an Amendment		9	16	
Warnings:					
Information:					
2	Fee Worksheet (SB06)	fee-info.pdf	32214	no	2
			84e4ccd4d15803a01b0c5cb273c5f204cc5b86a8		
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If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

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New International Application Filed with the USPTO as a Receiving Office

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 16/267,752	Filing Date 02/05/2019	☐ To be Mailed
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ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO

APPLICATION AS FILED - PART I

FOR	(Column 1) NUMBER FILED	(Column 2) NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 = *		x \$100 =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 = *		x \$460 =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED - PART II

	(Column 1)	(Column 2)	(Column 3)	RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	☐ Application Size Fee (37 CFR 1.16(s))				
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				
					TOTAL ADD'L FEE
AMENDMENT	09/23/2019				
	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	
	Total (37 CFR 1.16(i))	* 34	Minus	** 20	= 14
	Independent (37 CFR 1.16(h))	* 4	Minus	*** 3	= 1
	☐ Application Size Fee (37 CFR 1.16(s))				
☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
					TOTAL ADD'L FEE
					1860
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.					LIE
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".					/CHANTAE S DESSAU/
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".					
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.					

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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NOTICE OF ALLOWANCE AND FEE(S) DUE

89980 7590 01/09/2020
NSIP LAW
P.O. Box 65745
Washington, DC 20035

EXAMINER

POLO, GUSTAVO D

ART UNIT

PAPER NUMBER

2694

DATE MAILED: 01/09/2020

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/267,752	02/05/2019	Chang Mok CHOI	012052.0799C1	9126

TITLE OF INVENTION: WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0.00	\$0.00	\$1000	04/09/2020

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

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89980 7590 01/09/2020
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(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/267,752	02/05/2019	Chang Mok CHOI	012052.0799C1	9126

TITLE OF INVENTION: WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0.00	\$0.00	\$1000	04/09/2020

EXAMINER	ART UNIT	CLASS-SUBCLASS
POLO, GUSTAVO D	2694	345-156000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-09 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required) ☐ Advance Order - # of Copies _____

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. **Change in Entity Status** (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/267,752	02/05/2019	Chang Mok CHOI	012052.0799C1	9126
89980	7590	01/09/2020	EXAMINER	
NSIP LAW			POLO, GUSTAVO D	
P.O. Box 65745			ART UNIT	
Washington, DC 20035			PAPER NUMBER	
			2694	
DATE MAILED: 01/09/2020				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 16/267,752	Applicant(s) CHOI et al.	
	Examiner GUSTAVO POLO	Art Unit 2694	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 09/23/2019.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 1-5 and 7-35. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☒ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☒ All b) ☐ Some *c) ☐ None of the:

1. ☒ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892) 2. ☐ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date _____. 3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____. 4. ☐ Interview Summary (PTO-413), Paper No./Mail Date. _____.	5. ☐ Examiner's Amendment/Comment 6. ☒ Examiner's Statement of Reasons for Allowance 7. ☐ Other _____.
--	--

/Gustavo Polo/
Primary Examiner, Art Unit 2694

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Allowable Subject Matter

Claims 1-5 and 7-35 are allowed.

The following is an examiner's statement of reasons for allowance: arguments [09/23/19] are persuasive particularly with regards to the "select portion" and related limitations.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to GUSTAVO POLO whose telephone number is (571)270-7613. The examiner can normally be reached on Mon-Fri 9am-5pm PT.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Patrick Edouard can be reached on (571) 272-7603. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Art Unit: 2694

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Gustavo Polo/
Primary Examiner, Art Unit 2694

Issue Classification 	Application/Control No. 16/267,752	Applicant(s)/Patent Under Reexamination CHOI et al.
	Examiner GUSTAVO POLO	Art Unit 2694

CPC						
Symbol					Type	Version
G06F	/	3	/	015	F	2013-01-01
G06F	/	3	/	038	I	2013-01-01
G06F	/	1	/	163	I	2013-01-01
G06F	/	1	/	1694	I	2013-01-01
G06F	/	3	/	017	I	2013-01-01
G06F	/	3	/	0346	I	2013-01-01
A61B	/	5	/	6898	I	2013-01-01
A61B	/	5	/	7475	I	2013-01-01
A61B	/	5	/	1123	I	2013-01-01
A61B	/	5	/	221	I	2013-01-01
G06F	/	3	/	014	I	2013-01-01
A61B	/	5	/	0488	A	2013-01-01
A61B	/	2562	/	0219	A	2013-01-01

CPC Combination Sets							
Symbol				Type	Set	Ranking	Version
	/		/				

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	34	
/Gustavo Polo/ Primary Examiner, Art Unit 2694	17 December 2019	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	1

Issue Classification 	Application/Control No. 16/267,752	Applicant(s)/Patent Under Reexamination CHOI et al.
	Examiner GUSTAVO POLO	Art Unit 2694

INTERNATIONAL CLASSIFICATION				
CLAIMED				
G06F3/01	/	3	/	01
G06F3/038	/	3	/	038
G06F1/16	/	1	/	16
G06F3/0346	/	3	/	0346
A61B5/00	/	5	/	00
A61B5/11	/	5	/	11
A61B5/22	/	5	/	22
A61B5/0488	/	5	/	0488
NON-CLAIMED				
	/		/	

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	34	
/Gustavo Polo/ Primary Examiner, Art Unit 2694	17 December 2019	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	1

Issue Classification 	Application/Control No. 16/267,752	Applicant(s)/Patent Under Reexamination CHOI et al.
	Examiner GUSTAVO POLO	Art Unit 2694

☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 34	
/Gustavo Polo/ Primary Examiner, Art Unit 2694 (Primary Examiner) _____ (Date) 17 December 2019		O.G. Print Claim(s) 1	O.G. Print Figure 1

<i>Search Notes</i> 	Application/Control No. 16/267,752	Applicant(s)/Patent Under Reexamination CHOI et al.
	Examiner GUSTAVO POLO	Art Unit 2694

CPC - Searched*		
Symbol	Date	Examiner
g06f	06/17/2019	gp
g06f, a61b, g06f3/015	12/17/2019	gp

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
inventor/text	06/17/2019	gp
see above	12/17/2019	gp
cpc	12/17/2019	gp

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
	search history	12/17/2019	gp

/Gustavo Polo/ Primary Examiner, Art Unit 2694	
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EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	1783	g06f3/015.cpc.	US-PGPUB; USPAT	OR	ON	2019/12/17 12:18
L2	4011	((bio-signal biosignal) and motion)	US-PGPUB; USPAT	OR	ON	2019/12/17 12:18
L3	140	1 and 2	US-PGPUB; USPAT	OR	ON	2019/12/17 12:19
L4	1282492	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2019/12/17 12:19
L5	457009	a61b\$.cpc.	US-PGPUB; USPAT	OR	ON	2019/12/17 12:20
L6	1713336	4 5	US-PGPUB; USPAT	OR	ON	2019/12/17 12:20
L7	1310395	(part portion\$4) near2 (signal data information)	US-PGPUB; USPAT	OR	ON	2019/12/17 12:21
L8	1257	2 and 6 and 7	US-PGPUB; USPAT	OR	ON	2019/12/17 12:22
L9	143081	((bio-signal biosignal biological) and motion)	US-PGPUB; USPAT	OR	ON	2019/12/17 12:22
L10	2553	((bio-signal biosignal biological) with (motion mov\$4 movement) with (state condition))	US-PGPUB; USPAT	OR	ON	2019/12/17 12:23
L11	400	6 and 7 and 10	US-PGPUB; USPAT	OR	ON	2019/12/17 12:23
L12	255	(watch wear\$6) and 11	US-PGPUB; USPAT	OR	ON	2019/12/17 12:24
S1	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2017/07/27 19:18
S2	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US-PGPUB; USPAT	OR	ON	2017/07/28 12:11
S3	1	(bio-signal biosignal) and motion and S2	US-PGPUB; USPAT	OR	ON	2017/07/28 12:51
S4	1	("20160202759").PN.	US-	OR	OFF	2017/07/28

			PGPUB; USPAT; USOCR			18:22
S5	14137	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD"))).AANM.	USPAT	OR	ON	2017/07/29 17:00
S6	40	((("CHOI") near3 ("Chang") near3 ("Mok"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2017/07/29 17:00
S7	256	((("KIM") near3 ("Sang") near3 ("Joon"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2017/07/29 17:00
S8	37	((("YOON") near3 ("Seung") near3 ("Keun"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2017/07/29 17:00
S9	14382	S5 S6 S7 S8	US- PGPUB; USPAT	OR	ON	2017/07/29 17:03
S10	97	((bio-signal biosignal) and motion).clm.	US- PGPUB; USPAT	OR	ON	2017/07/29 17:03
S11	6	S9 and S10	US- PGPUB; USPAT	OR	ON	2017/07/29 17:04
S12	1029344	g06f\$.cpc.	US- PGPUB; USPAT	OR	ON	2017/07/29 17:04
S13	188878	(acceleration motion) near2 (sensor detector)	US- PGPUB; USPAT	OR	ON	2017/07/29 17:05
S14	38789	(watch wearable attachable) and S13	US- PGPUB; USPAT	OR	ON	2017/07/29 17:06
S15	602	(bio-signal biosignal) and S14	US- PGPUB; USPAT	OR	ON	2017/07/29 17:06
S16	1081387	(second two) near2 (signal wave)	US- PGPUB; USPAT	OR	ON	2017/07/29 17:07
S17	126	S15 and S16	US- PGPUB; USPAT	OR	ON	2017/07/29 17:08
S18	18967	(wearable watch) with interfac\$3	US- PGPUB; USPAT	OR	ON	2017/07/29 17:08
S19	29	S17 and S18	US- PGPUB; USPAT	OR	ON	2017/07/29 17:08
S20	1	("20160202759").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2017/07/29 17:09
S21	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US- PGPUB; USPAT	OR	ON	2017/07/29 17:09

S22	1	(bio-signal biosignal) and motion and S21	US-PGPUB; USPAT	OR	ON	2017/07/29 17:09
S23	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2017/07/29 17:09
S24	14137	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD"))).AANM.	USPAT	OR	ON	2017/07/29 17:09
S25	40	((("CHOI") near3 ("Chang") near3 ("Mok"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2017/07/29 17:09
S26	256	((("KIM") near3 ("Sang") near3 ("Joon"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2017/07/29 17:09
S27	37	((("YOON") near3 ("Seung") near3 ("Keun"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2017/07/29 17:09
S28	14382	S24 S25 S26 S27	US-PGPUB; USPAT	OR	ON	2017/07/29 17:09
S29	97	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2017/07/29 17:09
S30	6	S28 and S29	US-PGPUB; USPAT	OR	ON	2017/07/29 17:09
S31	1029344	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2017/07/29 17:09
S32	188878	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2017/07/29 17:09
S33	38789	(watch wearable attachable) and S32	US-PGPUB; USPAT	OR	ON	2017/07/29 17:09
S34	602	(bio-signal biosignal) and S33	US-PGPUB; USPAT	OR	ON	2017/07/29 17:09
S35	1081387	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2017/07/29 17:09
S36	126	S34 and S35	US-PGPUB; USPAT	OR	ON	2017/07/29 17:09
S37	18967	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2017/07/29 17:09
S38	29	S36 and S37	US-PGPUB; USPAT	OR	ON	2017/07/29 17:09
S39	1	("20100145171").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/01/15 00:07

S40	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/01/15 18:17
S41	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S42	1	(bio-signal biosignal) and motion and S41	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S43	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/01/15 18:17
S44	17001	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD"))).AANM.	USPAT	OR	ON	2018/01/15 18:17
S45	41	((("CHOI") near3 ("Chang") near3 ("Mok"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/01/15 18:17
S46	279	((("KIM") near3 ("Sang") near3 ("Joon"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/01/15 18:17
S47	41	((("YOON") near3 ("Seung") near3 ("Keun"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/01/15 18:17
S48	17265	S44 S45 S46 S47	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S49	107	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S50	8	S48 and S49	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S51	1077971	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S52	202904	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S53	44597	(watch wearable attachable) and S52	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S54	839	(bio-signal biosignal) and S53	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S55	1114857	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S56	152	S54 and S55	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S57	21301	(wearable watch) with interfac\$3	US-PGPUB;	OR	ON	2018/01/15 18:17

			USPAT			
S58	37	S56 and S57	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S59	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/01/15 18:17
S60	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S61	1	(bio-signal biosignal) and motion and S60	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S62	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/01/15 18:17
S63	17001	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD"))).AANM.	USPAT	OR	ON	2018/01/15 18:17
S64	41	((("CHOI") near3 ("Chang") near3 ("Mok"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/01/15 18:17
S65	279	((("KIM") near3 ("Sang") near3 ("Joon"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/01/15 18:17
S66	41	((("YOON") near3 ("Seung") near3 ("Keun"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/01/15 18:17
S67	17265	S63 S64 S65 S66	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S68	107	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S69	8	S67 and S68	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S70	1077971	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S71	202904	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S72	44597	(watch wearable attachable) and S71	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S73	839	(bio-signal biosignal) and S72	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S74	1114857	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2018/01/15 18:17
S75	152	S73 and S74	US-	OR	ON	2018/01/15

			PGPUB; USPAT			18:17
S76	21301	(wearable watch) with interfac\$3	US- PGPUB; USPAT	OR	ON	2018/01/15 18:17
S77	37	S75 and S76	US- PGPUB; USPAT	OR	ON	2018/01/15 18:17
S78	1	("20100145171").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2018/01/15 18:17
S79	1	("20120172733").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2018/03/28 15:53
S80	1	("20160202759").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2018/03/28 15:55
S81	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S82	1	(bio-signal biosignal) and motion and S81	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S83	1	("20160202759").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2018/03/28 15:55
S84	18271	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD"))).AANM.	USPAT	OR	ON	2018/03/28 15:55
S85	43	((("CHOI") near3 ("Chang") near3 ("Mok"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/03/28 15:55
S86	288	((("KIM") near3 ("Sang") near3 ("Joon"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/03/28 15:55
S87	41	((("YOON") near3 ("Seung") near3 ("Keun"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/03/28 15:55
S88	18541	S84 S85 S86 S87	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S89	112	((bio-signal biosignal) and motion).clm.	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S90	10	S88 and S89	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S91	1089358	g06f\$.cpc.	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S92	208949	(acceleration motion) near2 (sensor detector)	US- PGPUB;	OR	ON	2018/03/28 15:55

			USPAT			
S93	47101	(watch wearable attachable) and S92	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S94	928	(bio-signal biosignal) and S93	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S95	1128855	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S96	157	S94 and S95	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S97	22240	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S98	41	S96 and S97	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S99	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/03/28 15:55
S100	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S101	1	(bio-signal biosignal) and motion and S100	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S102	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/03/28 15:55
S103	18271	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD"))).AANM.	USPAT	OR	ON	2018/03/28 15:55
S104	43	((("CHOI") near3 ("Chang") near3 ("Mok"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/03/28 15:55
S105	288	((("KIM") near3 ("Sang") near3 ("Joon"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/03/28 15:55
S106	41	((("YOON") near3 ("Seung") near3 ("Keun"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/03/28 15:55
S107	18541	S103 S104 S105 S106	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S108	112	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S109	10	S107 and S108	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S110	1089358	g06f\$.cpc.	US-	OR	ON	2018/03/28

			PGPUB; USPAT			15:55
S111	208949	(acceleration motion) near2 (sensor detector)	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S112	47101	(watch wearable attachable) and S111	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S113	928	(bio-signal biosignal) and S112	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S114	1128855	(second two) near2 (signal wave)	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S115	157	S113 and S114	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S116	22240	(wearable watch) with interfac\$3	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S117	41	S115 and S116	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S118	1	("20100145171").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2018/03/28 15:55
S119	1	("20160202759").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2018/03/28 15:55
S120	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S121	1	(bio-signal biosignal) and motion and S120	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55
S122	1	("20160202759").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2018/03/28 15:55
S123	18271	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD"))).AANM.	USPAT	OR	ON	2018/03/28 15:55
S124	43	((("CHOI") near3 ("Chang") near3 ("Mok"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/03/28 15:55
S125	288	((("KIM") near3 ("Sang") near3 ("Joon"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/03/28 15:55
S126	41	((("YOON") near3 ("Seung") near3 ("Keun"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/03/28 15:55
S127	18541	S123 S124 S125 S126	US- PGPUB; USPAT	OR	ON	2018/03/28 15:55

S128	112	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S129	10	S127 and S128	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S130	1089358	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S131	208949	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S132	47101	(watch wearable attachable) and S131	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S133	928	(bio-signal biosignal) and S132	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S134	1128855	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S135	157	S133 and S134	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S136	22240	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S137	41	S135 and S136	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S138	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/03/28 15:55
S139	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S140	1	(bio-signal biosignal) and motion and S139	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S141	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/03/28 15:55
S142	18271	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD"))).AANM.	USPAT	OR	ON	2018/03/28 15:55
S143	43	((("CHOI") near3 ("Chang") near3 ("Mok"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/03/28 15:55
S144	288	((("KIM") near3 ("Sang") near3 ("Joon"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/03/28 15:55
S145	41	((("YOON") near3 ("Seung") near3 ("Keun"))).INV.	US-PGPUB; USPAT;	OR	ON	2018/03/28 15:55

			USOCR			
S146	18541	S142 S143 S144 S145	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S147	112	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S148	10	S146 and S147	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S149	1089358	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S150	208949	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S151	47101	(watch wearable attachable) and S150	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S152	928	(bio-signal biosignal) and S151	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S153	1128855	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S154	157	S152 and S153	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S155	22240	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S156	41	S154 and S155	US-PGPUB; USPAT	OR	ON	2018/03/28 15:55
S157	1	("20100145171").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/03/28 15:55
S158	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US-PGPUB; USPAT	OR	ON	2018/05/14 17:18
S159	1105417	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2018/05/14 17:19
S160	1105422	S158 S159	US-PGPUB; USPAT	OR	ON	2018/05/14 17:19
S161	100346	sequential near2 (signal process\$3 determin\$5)	US-PGPUB; USPAT	OR	ON	2018/05/14 17:20
S162	1	(sequential near2 (signal process\$3 determin\$5)) with (motion accelerat\$3) with (bio-signal biological)	US-PGPUB; USPAT	OR	ON	2018/05/14 17:21
S163	1	("20150028850").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/14 17:24

S164	154	(sequential near2 (signal process\$3 determin\$5)) with (bio-signal biological)	US-PGPUB; USPAT	OR	ON	2018/05/14 17:25
S165	157	(sequential near2 (signal process\$3 determin\$5)) with (bio-signal biosignal biological)	US-PGPUB; USPAT	OR	ON	2018/05/14 17:25
S166	8	S160 and S165	US-PGPUB; USPAT	OR	ON	2018/05/14 17:26
S167	7	(sequential) with motion with (bio-signal biosignal biological)	US-PGPUB; USPAT	OR	ON	2018/05/14 17:27
S168	1	S160 and S167	US-PGPUB; USPAT	OR	ON	2018/05/14 17:27
S169	2420	(sequential) with (bio-signal biosignal biological)	US-PGPUB; USPAT	OR	ON	2018/05/14 17:27
S170	169	S160 and S169	US-PGPUB; USPAT	OR	ON	2018/05/14 17:28
S171	57	S170 and (motion accelerat\$3)	US-PGPUB; USPAT	OR	ON	2018/05/14 17:28
S172	57	S171 and S169	US-PGPUB; USPAT	OR	ON	2018/05/14 17:31
S173	606	(sequential) with motion with sens\$3	US-PGPUB; USPAT	OR	ON	2018/05/14 17:45
S174	139	S160 and S173	US-PGPUB; USPAT	OR	ON	2018/05/14 17:45
S175	38	(watch wearable wear-able) and S174	US-PGPUB; USPAT	OR	ON	2018/05/14 17:46
S176	7	(US-20100145171-\$ or US-20130328761-\$ or US-20140015750-\$ or US-20140258208-\$ or US-20140365979-\$).did. or (US-5954662-\$ or US-5832296-\$).did.	US-PGPUB; USPAT	OR	ON	2018/05/14 17:55
S177	6	(sequential) and S176	US-PGPUB; USPAT	OR	ON	2018/05/14 17:56
S178	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S179	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S180	1	(bio-signal biosignal) and motion and S179	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S181	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44

S182	19106	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD"))).AANM.	USPAT	OR	ON	2018/05/15 18:44
S183	45	((("CHOI") near3 ("Chang") near3 ("Mok"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S184	295	((("KIM") near3 ("Sang") near3 ("Joon"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S185	41	((("YOON") near3 ("Seung") near3 ("Keun"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S186	19380	S182 S183 S184 S185	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S187	112	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S188	10	S186 and S187	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S189	1106227	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S190	213039	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S191	48821	(watch wearable attachable) and S190	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S192	1014	(bio-signal biosignal) and S191	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S193	1138450	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S194	167	S192 and S193	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S195	22938	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S196	46	S194 and S195	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S197	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S198	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S199	1	(bio-signal biosignal) and motion and S198	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44

S200	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S201	19106	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD"))).AANM.	USPAT	OR	ON	2018/05/15 18:44
S202	45	((("CHOI") near3 ("Chang") near3 ("Mok"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S203	295	((("KIM") near3 ("Sang") near3 ("Joon"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S204	41	((("YOON") near3 ("Seung") near3 ("Keun"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S205	19380	S201 S202 S203 S204	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S206	112	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S207	10	S205 and S206	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S208	1106227	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S209	213039	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S210	48821	(watch wearable attachable) and S209	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S211	1014	(bio-signal biosignal) and S210	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S212	1138450	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S213	167	S211 and S212	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S214	22938	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S215	46	S213 and S214	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S216	1	("20100145171").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S217	1	("20160202759").PN.	US-PGPUB; USPAT;	OR	OFF	2018/05/15 18:44

			USOCR			
S218	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S219	1	(bio-signal biosignal) and motion and S218	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S220	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S221	19106	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD"))).AANM.	USPAT	OR	ON	2018/05/15 18:44
S222	45	((("CHOI") near3 ("Chang") near3 ("Mok"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S223	295	((("KIM") near3 ("Sang") near3 ("Joon"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S224	41	((("YOON") near3 ("Seung") near3 ("Keun"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S225	19380	S221 S222 S223 S224	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S226	112	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S227	10	S225 and S226	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S228	1106227	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S229	213039	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S230	48821	(watch wearable attachable) and S229	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S231	1014	(bio-signal biosignal) and S230	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S232	1138450	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S233	167	S231 and S232	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S234	22938	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S235	46	S233 and S234	US-PGPUB;	OR	ON	2018/05/15 18:44

			USPAT			
S236	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S237	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S238	1	(bio-signal biosignal) and motion and S237	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S239	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S240	19106	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD"))).AANM.	USPAT	OR	ON	2018/05/15 18:44
S241	45	((("CHOI") near3 ("Chang") near3 ("Mok"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S242	295	((("KIM") near3 ("Sang") near3 ("Joon"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S243	41	((("YOON") near3 ("Seung") near3 ("Keun"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S244	19380	S240 S241 S242 S243	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S245	112	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S246	10	S244 and S245	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S247	1106227	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S248	213039	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S249	48821	(watch wearable attachable) and S248	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S250	1014	(bio-signal biosignal) and S249	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S251	1138450	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S252	167	S250 and S251	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S253	22938	(wearable watch) with interfac\$3	US-	OR	ON	2018/05/15

			PGPUB; USPAT			18:44
S254	46	S252 and S253	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S255	1	("20100145171").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S256	1	("20120172733").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S257	1	("20160202759").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S258	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S259	1	(bio-signal biosignal) and motion and S258	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S260	1	("20160202759").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S261	19106	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD"))).AANM.	USPAT	OR	ON	2018/05/15 18:44
S262	45	((("CHOI") near3 ("Chang") near3 ("Mok"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S263	295	((("KIM") near3 ("Sang") near3 ("Joon"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S264	41	((("YOON") near3 ("Seung") near3 ("Keun"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S265	19380	S261 S262 S263 S264	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S266	112	((bio-signal biosignal) and motion).clm.	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S267	10	S265 and S266	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S268	1106227	g06f\$.cpc.	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S269	213039	(acceleration motion) near2 (sensor detector)	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S270	48821	(watch wearable attachable) and S269	US- PGPUB;	OR	ON	2018/05/15 18:44

			USPAT			
S271	1014	(bio-signal biosignal) and S270	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S272	1138450	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S273	167	S271 and S272	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S274	22938	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S275	46	S273 and S274	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S276	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S277	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S278	1	(bio-signal biosignal) and motion and S277	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S279	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S280	19106	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD"))).AANM.	USPAT	OR	ON	2018/05/15 18:44
S281	45	((("CHOI") near3 ("Chang") near3 ("Mok"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S282	295	((("KIM") near3 ("Sang") near3 ("Joon"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S283	41	((("YOON") near3 ("Seung") near3 ("Keun"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S284	19380	S280 S281 S282 S283	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S285	112	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S286	10	S284 and S285	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S287	1106227	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S288	213039	(acceleration motion) near2 (sensor	US-	OR	ON	2018/05/15

		detector)	PGPUB; USPAT			18:44
S289	48821	(watch wearable attachable) and S288	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S290	1014	(bio-signal biosignal) and S289	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S291	1138450	(second two) near2 (signal wave)	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S292	167	S290 and S291	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S293	22938	(wearable watch) with interfac\$3	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S294	46	S292 and S293	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S295	1	("20100145171").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S296	1	("20160202759").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S297	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S298	1	(bio-signal biosignal) and motion and S297	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S299	1	("20160202759").PN.	US- PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S300	19106	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD"))).AANM.	USPAT	OR	ON	2018/05/15 18:44
S301	45	((("CHOI") near3 ("Chang") near3 ("Mok"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S302	295	((("KIM") near3 ("Sang") near3 ("Joon"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S303	41	((("YOON") near3 ("Seung") near3 ("Keun"))).INV.	US- PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S304	19380	S300 S301 S302 S303	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44
S305	112	((bio-signal biosignal) and motion).clm.	US- PGPUB; USPAT	OR	ON	2018/05/15 18:44

S306	10	S304 and S305	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S307	1106227	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S308	213039	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S309	48821	(watch wearable attachable) and S308	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S310	1014	(bio-signal biosignal) and S309	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S311	1138450	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S312	167	S310 and S311	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S313	22938	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S314	46	S312 and S313	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S315	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S316	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S317	1	(bio-signal biosignal) and motion and S316	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S318	1	("20160202759").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S319	19106	((("SAMSUNG") near3 ("ELECTRONICS") near3 ("CO") near3 ("LTD"))).AANM.	USPAT	OR	ON	2018/05/15 18:44
S320	45	((("CHOI") near3 ("Chang") near3 ("Mok"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S321	295	((("KIM") near3 ("Sang") near3 ("Joon"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S322	41	((("YOON") near3 ("Seung") near3 ("Keun"))).INV.	US-PGPUB; USPAT; USOCR	OR	ON	2018/05/15 18:44
S323	19380	S319 S320 S321 S322	US-PGPUB;	OR	ON	2018/05/15 18:44

			USPAT			
S324	112	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S325	10	S323 and S324	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S326	1106227	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S327	213039	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S328	48821	(watch wearable attachable) and S327	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S329	1014	(bio-signal biosignal) and S328	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S330	1138450	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S331	167	S329 and S330	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S332	22938	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S333	46	S331 and S332	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S334	1	("20100145171").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S335	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S336	1106227	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S337	1106232	S335 S336	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S338	100346	sequential near2 (signal process\$3 determin\$5)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S339	1	(sequential near2 (signal process\$3 determin\$5)) with (motion accelerat\$3) with (bio-signal biological)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S340	1	("20150028850").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2018/05/15 18:44
S341	154	(sequential near2 (signal process\$3 determin\$5)) with (bio-signal biological)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44

S342	157	(sequential near2 (signal process\$3 determin\$5)) with (bio-signal biosignal biological)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S343	8	S337 and S342	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S344	7	(sequential) with motion with (bio-signal biosignal biological)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S345	1	S337 and S344	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S346	2420	(sequential) with (bio-signal biosignal biological)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S347	169	S337 and S346	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S348	57	S347 and (motion accelerat\$3)	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S349	57	S348 and S346	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S350	606	(sequential) with motion with sens\$3	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S351	140	S337 and S350	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S352	39	(watch wearable wear-able) and S351	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S353	7	(US-20100145171-\$ or US-20130328761-\$ or US-20140015750-\$ or US-20140258208-\$ or US-20140365979-\$).did. or (US-5954662-\$ or US-5832296-\$).did.	US-PGPUB; USPAT	OR	ON	2018/05/15 18:44
S354	6	(sequential) and S353	US-PGPUB; USPAT	OR	ON	2018/05/15 18:45
S355	7	(US-20100145171-\$ or US-20130328761-\$ or US-20140015750-\$ or US-20140258208-\$ or US-20140365979-\$).did. or (US-5954662-\$ or US-5832296-\$).did.	US-PGPUB; USPAT	OR	ON	2018/11/06 10:52
S356	3	S355 and threshold\$4	US-PGPUB; USPAT	OR	ON	2018/11/06 10:54
S357	1177670	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2018/11/06 11:13
S358	5800	sens\$3 with target with threshold	US-PGPUB; USPAT	OR	ON	2018/11/06 11:14
S359	6123	sens\$3 with target\$4 with threshold\$3	US-PGPUB; USPAT	OR	ON	2018/11/06 11:14

S360	723	S357 and S359	US-PGPUB; USPAT	OR	ON	2018/11/06 11:15
S361	5052	bio-signal\$4 biosignal\$4	US-PGPUB; USPAT	OR	ON	2018/11/06 11:15
S362	1807	S359 and ((second two another other) near2 signal\$4)	US-PGPUB; USPAT	OR	ON	2018/11/06 11:16
S363	94	S359 with ((second two another other) near2 signal\$4)	US-PGPUB; USPAT	OR	ON	2018/11/06 11:17
S364	5	S357 and S363	US-PGPUB; USPAT	OR	ON	2018/11/06 11:17
S365	1603	signal\$4 with sens\$3 with target\$4 with threshold\$3	US-PGPUB; USPAT	OR	ON	2018/11/06 11:19
S366	96	S357 and S365	US-PGPUB; USPAT	OR	ON	2018/11/06 11:19
S367	4	(US-20100145171-\$ or US-20140365979-\$ or US-20040042637-\$).did. or (US-5832296-\$).did.	US-PGPUB; USPAT	OR	ON	2018/11/06 18:21
S368	1603	signal\$4 with sens\$3 with target\$4 with threshold\$3	US-PGPUB; USPAT	OR	ON	2018/11/06 18:22
S369	1	S368 and S367	US-PGPUB; USPAT	OR	ON	2018/11/06 18:22
S370	1	(sens\$3 with target\$4 with threshold\$3) and S369	US-PGPUB; USPAT	OR	ON	2018/11/06 18:23
S371	10	(US-20100145171-\$ or US-20130328761-\$ or US-20140015750-\$ or US-20140258208-\$ or US-20140365979-\$ or US-20040042637-\$ or US-20130151730-\$ or US-20140050354-\$).did. or (US-5954662-\$ or US-5832296-\$).did.	US-PGPUB; USPAT	OR	ON	2019/06/15 13:30
S372	761	(bio-signal\$4 biosignal\$4 (biologic\$5 near2 (signal data))) with period	US-PGPUB; USPAT	OR	ON	2019/06/16 13:32
S373	1219871	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2019/06/16 13:33
S374	125	S372 and S373	US-PGPUB; USPAT	OR	ON	2019/06/16 13:33
S375	73	(wearable watch) and S374	US-PGPUB; USPAT	OR	ON	2019/06/16 13:33
S376	761	(bio-signal\$4 biosignal\$4 (biologic\$5 near2 (signal data))) with period	US-PGPUB; USPAT	OR	ON	2019/06/17 11:33
S377	1219871	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2019/06/17 11:33

S378	125	S376 and S377	US-PGPUB; USPAT	OR	ON	2019/06/17 11:33
S379	73	(wearable watch) and S378	US-PGPUB; USPAT	OR	ON	2019/06/17 11:33
S380	5	(US-20030078505-\$ or US-20090326398-\$ or US-20120157789-\$ or US-20130096397-\$ or US-20130109997-\$).did.	US-PGPUB	OR	ON	2019/06/17 12:05
S381	5	S379 and S380	US-PGPUB; USPAT	OR	ON	2019/06/17 12:05
S382	15	(US-20100145171-\$ or US-20130328761-\$ or US-20140015750-\$ or US-20140258208-\$ or US-20140365979-\$ or US-20040042637-\$ or US-20130151730-\$ or US-20140050354-\$ or US-20030078505-\$ or US-20090326398-\$ or US-20120157789-\$ or US-20130096397-\$ or US-20130109997-\$).did. or (US-5954662-\$ or US-5832296-\$).did.	US-PGPUB; USPAT	OR	ON	2019/06/17 12:35
S383	10	S382 and threshold	US-PGPUB; USPAT	OR	ON	2019/06/17 12:35
S384	7	((("5832296") or ("20060149146") or ("20060281979") or ("20100145171") or ("20120172733") or ("20120184826") or ("20120259182")).PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2019/06/17 13:23
S393	25913	S389 S390 S391 S392	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S394	139	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S395	21	S393 and S394	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S396	1219871	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S397	249799	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S398	64689	(watch wearable attachable) and S397	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S399	1689	(bio-signal biosignal) and S398	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S400	1218439	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S401	241	S399 and S400	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S402	28769	(wearable watch) with interfac\$3	US-PGPUB;	OR	ON	2019/06/17 13:25

			USPAT			
S412	25913	S408 S409 S410 S411	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S413	139	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S414	21	S412 and S413	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S415	1219871	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S416	249799	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S417	64689	(watch wearable attachable) and S416	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S418	1689	(bio-signal biosignal) and S417	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S419	1218439	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S420	241	S418 and S419	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S421	28769	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S432	25913	S428 S429 S430 S431	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S433	139	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S434	21	S432 and S433	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S435	1219871	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S436	249799	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S437	64689	(watch wearable attachable) and S436	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S438	1689	(bio-signal biosignal) and S437	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S439	1218439	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S440	241	S438 and S439	US-PGPUB;	OR	ON	2019/06/17 13:25

			USPAT			
S441	28769	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S451	25913	S447 S448 S449 S450	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S452	139	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S453	21	S451 and S452	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S454	1219871	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S455	249799	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S456	64689	(watch wearable attachable) and S455	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S457	1689	(bio-signal biosignal) and S456	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S458	1218439	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S459	241	S457 and S458	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S460	28769	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S472	25913	S468 S469 S470 S471	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S473	139	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S474	21	S472 and S473	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S475	1219871	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S476	249799	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S477	64689	(watch wearable attachable) and S476	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S478	1689	(bio-signal biosignal) and S477	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S479	1218439	(second two) near2 (signal wave)	US-PGPUB;	OR	ON	2019/06/17 13:25

			USPAT			
S480	241	S478 and S479	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S481	28769	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S491	25913	S487 S488 S489 S490	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S492	139	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S493	21	S491 and S492	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S494	1219871	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S495	249799	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S496	64689	(watch wearable attachable) and S495	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S497	1689	(bio-signal biosignal) and S496	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S498	1218439	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S499	241	S497 and S498	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S500	28769	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S511	25913	S507 S508 S509 S510	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S512	139	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S513	21	S511 and S512	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S514	1219871	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S515	249799	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S516	64689	(watch wearable attachable) and S515	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S517	1689	(bio-signal biosignal) and S516	US-PGPUB;	OR	ON	2019/06/17 13:25

			USPAT			
S518	1218439	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S519	241	S517 and S518	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S520	28769	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S530	25913	S526 S527 S528 S529	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S531	139	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S532	21	S530 and S531	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S533	1219871	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S534	249799	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S535	64689	(watch wearable attachable) and S534	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S536	1689	(bio-signal biosignal) and S535	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S537	1218439	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S538	241	S536 and S537	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S539	28769	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S542	5	("20060149146" "20060281976" "20100145171" "20120184826" "20120256182").PN.	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S543	1219871	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S544	1219876	S542 S543	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S545	109622	sequential near2 (signal process\$3 determin\$5)	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S548	159	(sequential near2 (signal process\$3 determin\$5)) with (bio-signal biological)	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S549	163	(sequential near2 (signal process\$3 determin\$5)) with (bio-signal biosignal	US-PGPUB;	OR	ON	2019/06/17 13:25

		biological)	USPAT			
S551	7	(sequential) with motion with (bio-signal biosignal biological)	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S553	2625	(sequential) with (bio-signal biosignal biological)	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S554	102	S544 and S553	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S557	676	(sequential) with motion with sens\$3	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S570	25913	S566 S567 S568 S569	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S571	139	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S572	21	S570 and S571	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S573	1219871	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S574	249799	(acceleration motion) near2 (sensor detector)	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S575	64689	(watch wearable attachable) and S574	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S576	1689	(bio-signal biosignal) and S575	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S577	1218439	(second two) near2 (signal wave)	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S578	241	S576 and S577	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S579	28769	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S589	25913	S585 S586 S587 S588	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S590	139	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S591	21	S589 and S590	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S592	1219871	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S593	249799	(acceleration motion) near2 (sensor detector)	US-PGPUB;	OR	ON	2019/06/17 13:25

			USPAT			
S594	64689	(watch wearable attachable) and S593	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S595	1689	(bio-signal biosignal) and S594	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S596	28769	(wearable watch) with interfac\$3	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S606	25913	S602 S603 S604 S605	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S607	139	((bio-signal biosignal) and motion).clm.	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S608	21	S606 and S607	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25
S609	1219871	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2019/06/17 13:25

EAST Search History (I nterference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L13	1282492	g06f\$.cpc.	US-PGPUB; USPAT	OR	ON	2019/12/17 12:24
L14	457009	a61b\$.cpc.	US-PGPUB; USPAT	OR	ON	2019/12/17 12:24
L15	1713336	13 14	US-PGPUB; USPAT	OR	ON	2019/12/17 12:34
L16	379	((bio-signal biosignal biological) with (motion mov\$4 movement) with (state condition)).clm.	US-PGPUB; USPAT	OR	ON	2019/12/17 12:35
L17	135072	(watch wearable attachable).clm.	US-PGPUB; USPAT	OR	ON	2019/12/17 12:35
L18	36	16 and 17	US-PGPUB; USPAT	OR	ON	2019/12/17 12:35

12/ 17/ 2019 12:41:18 PM
C:\ Users\ gpolo\ Documents\ EAST\ Workspaces\ 14913085.wsp

Bibliographic Data

Application No: 16/267,752

Foreign Priority claimed: ☒ Yes ☐ No

35 USC 119 (a-d) conditions met: ☒ Yes ☐ No ☐ Met After Allowance

Verified and Acknowledged:

/Gustavo Polo/

Examiner's Signature

Initials

Title:

WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
02/05/2019	345	2694	012052.0799C1
RULE			

APPLICANTS

Samsung Electronics Co., Ltd., Suwon-si, KOREA, REPUBLIC OF

INVENTORS

Chang Mok CHOI Suwon-si, KOREA, REPUBLIC OF

Sang Joon KIM Suwon-si, KOREA, REPUBLIC OF

Seung Keun YOON Suwon-si, KOREA, REPUBLIC OF

CONTINUING DATA

This application is a CON of 14913085 02/19/2016 PAT 10241574

14913085 is a 371 of PCT/KR2014/004073 05/08/2014

FOREIGN APPLICATIONS

KOREA, REPUBLIC OF 10-2013-0098476 08/20/2013

IF REQUIRED, FOREIGN LICENSE GRANTED**

02/21/2019

STATE OR COUNTRY

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ADDRESS

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Washington, DC 20035

UNITED STATES

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Application	Document	Mailroom Date	Attorney Docket No.
16267752	NOA	01/09/2020	012052.0799C1

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 Commissioner for Patents
 P.O. Box 1450
 Alexandria, VA 22313-1450

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I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO	CONFIRMATION NO
16/267,752	02/05/2019	Chang Mok CHOI	012052.0799C1	9126

TITLE OF INVENTION:

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0	\$0	\$1000	04/09/2020

EXAMINER	ART UNIT	CLASS-SUBCLASS

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363)

- ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
- ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-09 or more recent) attached. **Use of a Customer Number is required.**

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- (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

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2.

3.

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PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

Samsung Electronics Co., Ltd.

(B) RESIDENCE: (CITY and STATE or COUNTRY)

Suwon-si, Republic of Korea

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. Fees Submitted: ☒ Issue Fee ☒ Publication Fee (if required) ☐ Advance Order - # of Copies _____

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- ☐ Applicant asserting small entity status. See 37 CFR 1.27.
- ☐ Applicant changing to regular undiscounted fee status.

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Authorized Signature /Thomas J. Matikainen/

Date April 2, 2020

Typed or printed name Thomas J. Matikainen

Registration No. 73,291

Electronic Patent Application Fee Transmittal

Application Number:	16267752			
Filing Date:	05-Feb-2019			
Title of Invention:	WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF			
First Named Inventor/Applicant Name:	Chang Mok CHOI			
Filer:	Thomas Joseph Matikainen/Mark Gambriel			
Attorney Docket Number:	012052.0799C1			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
PUBL. FEE- EARLY, VOLUNTARY, OR NORMAL	1504	1	0	0
UTILITY APPL ISSUE FEE	1501	1	1000	1000
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1000

Electronic Acknowledgement Receipt

EFS ID:	39040661
Application Number:	16267752
International Application Number:	
Confirmation Number:	9126
Title of Invention:	WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF
First Named Inventor/Applicant Name:	Chang Mok CHOI
Customer Number:	89980
Filer:	Thomas Joseph Matikainen/Mark Gambriel
Filer Authorized By:	Thomas Joseph Matikainen
Attorney Docket Number:	012052.0799C1
Receipt Date:	02-APR-2020
Filing Date:	05-FEB-2019
Time Stamp:	09:27:00
Application Type:	Utility under 35 USC 111(a)

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Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 1000
RAM confirmation Number	E202042027517735
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Authorized User	

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1	Issue Fee Payment (PTO-85B)	NOA20200109_0120520799C1 _IFTransmittalForm.pdf	218466	no	1
			8b4d4f0f3532a4e851900341a67a7182453 d489e		
Warnings:					
Information:					
2	Fee Worksheet (SB06)	fee-info.pdf	32069	no	2
			84e857cdb3d6f65e02911103764850b0990 d6f3c		
Warnings:					
Information:					
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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/267,752	05/05/2020	10642359	012052.0799C1	9126

89980 7590 04/15/2020
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ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

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If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

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APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

Samsung Electronics Co., Ltd., Suwon-si, KOREA, REPUBLIC OF;
Chang Mok CHOI, Suwon-si, KOREA, REPUBLIC OF;
Sang Joon KIM, Suwon-si, KOREA, REPUBLIC OF;
Seung Keun YOON, Suwon-si, KOREA, REPUBLIC OF;

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Subject: Private PAIR Correspondence Notification for Customer Number 89980

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Application	Document	Mailroom Date	Attorney Docket No.
16267752	ISSUE.NTF	04/15/2020	012052.0799C1

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